FRI Soundness Above the Johnson Radius via Threshold Halving

Raullen Chai*[1] and Xinxin Fan†[1]

[1]IoTeX Network

April 29, 2026

Abstract

The classical 1:2 proximity-gap inequality of Rothblum–Vadhan–Wigderson [31] (STOC 2013) holds for any linear code, in any characteristic, at any distance. We observe that this inequality, applied above the Johnson radius, places FRI's effective distance into BCIKS's zero-loss regime after a single round. The underlying algebraic identity is $\delta/2 < (1-\rho)/2 \leq 1 - \sqrt{\rho} = \delta_J$ for every $\delta \in (\delta_J, 1-\rho)$ (Remark 1); inside that regime, the proven proximity gap of BCIKS [7] supplies the round-by-round bound, with the BCHKS [6] refinement giving $\leq n$ exceptional values per round (improving BCIKS's original $O(n^2)$). The result is the first unconditional FRI soundness theorem in the open zone $\delta \in (\delta_J, 1-\rho)$, at the cost of a $\sim 2\times$ query overhead: for $\mathrm{RS}[F, L, k]$ with $n = |L| = 2^d$, $\rho = k/n$, $k = 2^m$, $d \geq m$, and L a multiplicative subgroup or coset closed under negation (the standard radix-2 setting; see Theorem 20 for the full domain-tower hypotheses, Appendix B for the additive subspace-folding analogue in characteristic 2, and Appendix C for the circle-folding analogue on the Mersenne31 circle subgroup), FRI with $R = m$ rounds and q base-level queries (each opening one full path) satisfies

$$\varepsilon_{\mathrm{FRI}} \;\leq\; \frac{nR}{|F|} \;+\; \left(1 - \tfrac{\delta}{2}\right)^q.$$

We further show that this $2\times$ factor is intrinsic to the standard two-bad-γ correlated-agreement (CA) route considered here, at FRI scale: the equal-threshold CA upper bound is $\varepsilon_{\mathrm{ca}}(C, \delta, \delta) \leq \binom{n}{w}/|F|$ unconditionally for $w = \lfloor \delta n \rfloor$ (Theorem 8), with a matching tightness construction at the equality case $\delta = (n-k-1)/n$ for $|F| > \binom{n}{w}^2$ (Proposition 10; see Remark 12 for scope). The bound is vacuous for $n \gg \log |F|$. As a structural complement, we give the first p-dependent moment bounds on the list size at all codimension excesses ($\mathbb{E}[M] = \binom{n}{w}/p^c$ via Corollary 30, with the exact variance at $c = 1$ via Theorem 25 and Poisson-type dispersion in the large-p regime at $c \geq 2$ via the pairwise-independence theorem 28), an explicit obstruction to the worst-case $M = 0$ form of OP2 at FRI scale (Proposition 35), and a conjectural linear-n phase-diagram bound at $c \geq 3$ (Conjecture 43, supported by random-syndrome experiments through $n = 40$). Coverage: theorem-level for multiplicative FRI in odd characteristic (Theorem 20, all rates), additive FRI in characteristic 2 for the plain FRI protocol (Theorem 72), circle FRI / Stwo (Theorem 80), and STIR (Theorem 48 under (RD)+(CQ), and the companion Theorem 54 under (RD-strong) alone (no (CQ)) with the same commit-phase rate and query-phase term; the multi-round case within (CQ) is closed by Lemma 49, and the (CQ)-to-arbitrary case is closed by Lemma 53 (an iterated-restriction proximity-gap chain) under the input restriction (RD-strong);

*raullen@iotex.io
†xinxin@iotex.io

the rate-uniform query-phase bound is $\max((1-\delta/2)^q, (2\rho)^q)$ for Theorem 48 and $(1-\delta/2)^q$ for Theorem 54, both simplifying to the same $(1-\delta/2)^q$ at STIR's standard deployment rate $\rho \leq 1/4$; no round-1 list-size hypothesis at threshold $\delta > \delta_J$ is needed in the half-threshold framework, see Remark 50; the union of (CQ) and (RD-strong) covers all adversaries except those violating both simultaneously (the residual case is identified as follow-up; see Remark 55)); the WHIR analogue is conjectural (Conjecture 62; the proximity-layer substitution is rigorous via Lemma 63, but the iteration-level induction of WHIR [11, §5] would need to be re-run with the substituted folding term to upgrade the conjecture to a theorem) and the characteristic-2 STIR/WHIR analogue is sketched in Remark 74; both are listed as follow-up open tasks. Binius / binary-tower-field STIR/WHIR therefore requires the sketch to be filled in, which we identify as the main remaining char-2 task in §9.[1]

Keywords: Reed–Solomon codes, FRI protocol, proximity gap, correlated agreement, Johnson radius, STARK soundness, interactive oracle proofs.

MSC 2020: 94B35 (primary), 11T71, 68Q10.

1 Introduction

1.1 The problem

Why this matters for Ethereum. Ethereum's medium-term scaling and post-quantum roadmap rests on succinct proofs: zkEVMs, zkVMs, and a planned post-quantum signature-aggregation layer. EthProofs[2] alone tracks $\sim$30 such systems in active use or near-term rollout. Almost all of them are STARK-based, and almost all STARKs are built on a single low-level cryptographic component: the FRI proximity test (or its successors STIR and WHIR). FRI soundness is what prevents a malicious prover from making the chain accept a false statement. To meet performance budgets (proof size, prover memory, verifier gas), many deployed systems run FRI in the so-called *open zone above the Johnson radius*: the parameter regime where, until now, the soundness analysis depended on an *unproved* conjecture (the "correlated agreement above the Johnson radius" conjecture, originating in Ben-Sasson–Carmon–Ishai–Kopparty–Saraf [7] and articulated as an open problem by Arnon–Boneh–Fenzi [4]). Crites–Stewart [5] disproved the strongest up-to-capacity form in November 2025; Kambiré [12] subsequently gave an independent 2026 near-capacity prime-field counterexample at distance $O(1/\log n)$ below capacity. This left every above-Johnson soundness argument used in production technically *unproven*: a positive security number quoted in audit reports, but with no theorem behind it in the relevant regime.

This paper closes that gap. We give an unconditional soundness theorem in the full open zone with no protocol change. Only the verifier's query count is recalibrated. Concretely, the system designer pays roughly a factor-2 increase in base-level queries q (concretely 2.2×–2.3× depending on δ; §1.6) and, in return, gets a proven soundness bound whose validity does not rely on the equal-threshold conjecture or its refinements (including any sharper version of the Crites–Stewart construction). Operators of SP1, RISC Zero, Plonky3, Stwo, and similar systems can replace the conjectural baseline they currently assume with a theorem; this paper supplies the theorem and quantifies the recalibration cost (Table 1, §1.6, §9).

[1] Lean 4 formalization, Python verification scripts, and reproduction commands are in the companion repository [39]; the per-theorem formalization status is recorded in §9. Concrete bit-security tables are in §9.

[2] `https://ethproofs.org`

Formal statement of the prize and the prior gap. The Ethereum Foundation Proximity Prize[3] [1] targets the Reed–Solomon proximity gap conjectures of [4] that control the soundness of FRI [7], STIR [10], and WHIR [11]. BCHKS [6] proved zero-loss proximity gaps for $\delta \le \delta_J$ with $O(n)/|F|$ per-round error (equivalently $O(Rn)/|F|$ over R rounds, as in Table 1). Goyal–Guruswami [8] achieved optimal proximity gaps up to capacity for subspace-design codes and for *random* plain RS codes; Jeronimo–Liu–Rajpal [9] do the same for *folded* RS codes. For plain RS codes above the Johnson radius on a *fixed* multiplicative-coset evaluation domain L — and therefore neither random nor folded — no positive result existed.

The baseline is the conjecture $\varepsilon_{\mathrm{ca}}(C, \delta, \delta) = O(1)/|F|$ for all $\delta < 1-\rho$ [4], disproved up-to-capacity by Crites–Stewart [5] (ePrint 2025/2046, November 2025); independently, Kambiré [12] (arXiv 2604.09724, 2026) gives a near-capacity prime-field counterexample at distance $O(1/\log n)$ below capacity. For the intermediate zone ($\delta_J < \delta < 1-\rho$): the bound $\varepsilon_{\mathrm{ca}}(C, \delta, \delta) \le \binom{n}{w}/|F|$ (Theorem 8, tight by Proposition 10) is formally $O(1)/|F|$ but with constant exponential in the code length, making it vacuous at FRI scale. Without the contribution of this paper, the above-Johnson FRI setting on Ethereum has zero proven soundness in the open zone.

1.2 What we deliver

The structural observation. The half-threshold correlated-agreement bound (Theorem 6) is *not new*: it is the classical 1:2 proximity-gap inequality of Rothblum–Vadhan–Wigderson [31] (STOC 2013), holding for any linear code over any field at any distance. What we contribute is the observation that, applied above the Johnson radius, this classical inequality has a previously-unnoticed consequence: a single round of threshold halving lowers the effective distance from δ to $\delta/2$, and the chain $\delta/2 < (1-\rho)/2 \le 1 - \sqrt{\rho} = \delta_J$ (Remark 1) places every subsequent round into the BCIKS–BCHKS zero-loss regime, where the proven proximity gap [7, 6] locks the distance for the rest of the protocol (BCHKS's $\le n$ exceptional-value bound supplies the per-round constant; BCIKS's $O(n^2)$ would give a weaker but still polynomial bound). The key observation is the inequality $\delta/2 < (1-\rho)/2 \le \delta_J$; converting it into an end-to-end deployment theorem requires three additional ingredients, each new in this paper: a matched-pair equal-threshold CA upper and lower bound proving that the $2\times$ overhead is *intrinsic* to any CA-based argument at FRI scale (Theorem 8, Proposition 10); the first p-dependent moment bounds on the list size at all codimension excesses, plus a conditional / sketch-level $c = 2$ sub-Poisson tail (§7, Conjecture 41); and theorem-level coverage for multiplicative FRI, additive FRI in characteristic 2, circle FRI, and STIR (Theorems 72, 80, 48). Each is detailed in §1.4.

The bound. For FRI (Theorem 20; Theorem 72 for $\mathrm{char}(F) = 2$) on plain Reed–Solomon codes over any finite field, with $n = |L|$, $\rho = k/n$, $k = 2^m$, domain L admitting a fixed-point-free involution, $R = m$ rounds, and q base-level queries (each opening one full path), for every δ in the open zone (δ_J, $1-\rho$):

$$\varepsilon_{\mathrm{FRI}} \le \frac{nR}{|F|} + \left(1 - \frac{\delta}{2}\right)^q. \tag{1}$$

This is an unconditional theorem in a regime where, prior to this paper, above-Johnson FRI relied on the now-disproved zero-loss CA conjecture [4, 5, 12]. The FRI protocol is unchanged; only the soundness analysis and the calibrated query count q change. Increasing q is still a real implementation-parameter change: it increases Merkle openings, consistency checks, proof bytes, and wrapper-verifier work. The same bound holds for FRI in characteristic 2 via additive folding

[3] `https://proximityprize.org`

(Appendix B); a batch-CA theorem extends it to STIR (Theorem 48) and WHIR (Corollary 58, Conjecture 62) in the odd-characteristic setting of those formal statements (the characteristic-2 STIR/WHIR extension is sketched in Remark 74); non-interactive soundness for multiplicative FRI follows via round-by-round soundness and Fiat–Shamir compilation (Theorem 84, Appendix D); non-interactive soundness for the DEEP-FRI polynomial commitment scheme is given by Theorem 86 and Corollary 87, with the standard RBR-to-knowledge-soundness lifting via [37]; the analogous compilation for char-2 FRI, circle FRI, and STIR uses the same RBR template and is not separately packaged in this paper.

Why it works. Prior results [7, 6] proved soundness only at or below the Johnson radius, where zero-loss CA ($\delta_{\mathrm{hd}} = \delta$) holds. Extending to the open zone above Johnson was blocked by the equal-threshold CA conjecture [4], which requires $M = 0$ on all RS-compatible flats — known to be false in its strongest form (Proposition 35). Threshold halving sidesteps the conjecture entirely: by accepting a $\sim 2\times$ query overhead — which we further show is intrinsic to any CA-based proof at current code lengths (Theorem 8, Remark 14) — it yields unconditional soundness in the full open zone, immune to any future open-zone counterexample.

Why this wasn't proven earlier. The bridge $\delta/2 < (1-\rho)/2 \leq \delta_J$ is one line of algebra and could in principle have been written down any time after BCIKS '20. It was not, for two reasons. First, the community was tracking the conjectural up-to-capacity bound $\varepsilon_{\mathrm{ca}}(C, \delta, \delta) = O(1)/|F|$ [4], against which the $2\times$ query overhead of threshold halving was unattractive. Second, the equal-threshold CA upper bound $\binom{n}{w}/|F|$ (Theorem 8) and its matching tightness (Proposition 10) had not been written down, leaving open the possibility that a sharper equal-threshold bound was within reach without paying the halving toll. This paper supplies that pair, showing that within the two-bad-γ CA framework the $2\times$ overhead cannot be improved. After Crites–Stewart disproved the up-to-capacity form in November 2025 [5] and Kambiré gave an independent near-capacity counterexample in 2026 [12], every above-Johnson FRI deployment was left with neither an unconditional bound nor a definitive negative result on the equal-threshold side. This paper closes both: the unconditional theorem in the open zone, and the structural barrier that explains why the $2\times$ price is the right one.

1.3 Coverage of current FRI variants

The theorem applies to the following settings, all operating above the Johnson radius:

- **Odd characteristic.** SP1 (BabyBear$_4$, rate 1/2), RISC Zero (BabyBear, rate 1/4), Plonky3 (BabyBear / Koala-Bear, Polygon zkEVM), and comparable multiplicative-FRI zkVM systems.
- **Circle FRI (covered).** Stwo (Starkware, Mersenne31) uses circle FRI [22] on the unit circle $\{(x, y) \in \mathbb{F}_p^2 : x^2+y^2 = 1\}$, whose folding structure differs from the multiplicative and additive cases (the involution is $(x, y) \mapsto (x, -y)$, not negation on a subgroup). Appendix C proves the required coupling lemma for circle codes and establishes FRI soundness (Theorem 80) with the analogous $NR/|F| + (1-\delta/2)^q$ bound, where $N = |D|$ is the size of the circle evaluation domain (the analogue of $n = |L|$ in the multiplicative case).
- **Characteristic 2.** Systems using additive FRI on $\mathbb{F}_2$-linear subspaces (binary tower fields $\mathbb{F}_{2^{128}}$). Our Theorem 72 covers the basic additive fold $s(x) = x^2 + \beta x$. Binius [21] uses a more elaborate block-decomposition/Gao-Mateer recursion that is compatible with — but distinct from — this basic fold.

Protocol coverage:

- **FRI (proved).** Theorem 20 (char $\neq 2$) and Theorem 72 (char $= 2$).
- **STIR (conditional theorem-level in odd characteristic, two complementary hypothesis sets).** Theorem 48 proves STIR soundness above Johnson under two named explicit hypotheses: restriction-distance preservation (RD) (cf. [10, §5]) and the codeword-quotient adversary model (CQ). Theorem 54 drops (CQ) and proves the same bound under the strengthened input hypothesis (RD-strong) (iterated restriction distance at every level of the radix-2 tower), accommodating arbitrary (non-codeword) prover quotients q_i. The two theorems are complementary: an adversary that respects (CQ) is bounded by Theorem 48; one that respects (RD-strong) (in particular, every input not algebraically aligned with a low-degree polynomial on a tower-subgroup) is bounded by Theorem 54; the union covers all above-Johnson adversaries except those violating both (CQ) and (RD-strong) simultaneously (Remark 55). No round-1 list-size hypothesis at threshold $\delta > \delta_J$ is needed in the half-threshold framework (Remark 50). Lemma 49 closes the multi-round case within (CQ); Lemma 53 (an iterated-restriction proximity-gap chain) closes the (CQ)-to-arbitrary case under (RD-strong). Both theorems give commit-phase rate $(2k + n)R/|F|$ and query-phase term $(1-\delta/2)^q$ at the standard deployment rate $\rho \leq 1/4$ (Theorem 48 gives the rate-uniform $\max((1-\delta/2)^q, (2\rho)^q)$; Theorem 54 gives $(1-\delta/2)^q$ alone, sourced from the standard STIR final-round consistency check [10, Construction 5.2 Item 2]). Each of (RD), (CQ), (RD-strong) is documented with its own remark (51, 52, 55).
- **WHIR (conjectural in odd characteristic).** Corollary 58 proves our CA bound holds for any *linear* subcode of RS_k, and end-to-end WHIR soundness with sumcheck composition is conjectured in Conjecture 62; the characteristic-2 analogue is sketched in Remark 74.

The equal-threshold bound $\binom{n}{w}/|F|$ is vacuous at FRI scale (§1.7); our half-threshold bound is optimal within the standard two-bad-γ CA framework, and whether a non-CA argument or one tracking ≥ 3 bad γ's could beat the 1:2 ratio is left open in §8.

1.4 Technical contributions

(i) **FRI soundness theorem above Johnson** (Theorem 20; Theorem 72 for characteristic 2; Theorem 80 for circle FRI, with $N = |D|$ replacing $n = |L|$): the bound $nR/|F| + (1-\delta/2)^q$ in the open zone $(\delta_J, 1-\rho)$, via the structural observation (2) that $\delta/2 < (1-\rho)/2 \leq \delta_J$ bridges the classical RVW13 inequality [31] (Theorem 6) to BCIKS's zero-loss regime [7]. The commit-phase term $O(nR/|F|)$ matches BCHKS up to constants; the contribution is that this is *proved unconditionally*, not conjectured. An $O(1)$ round-1 proximity gap (Theorem 16, ≤ 1 bad α) is the local consequence used in the proof.

(ii) **Equal-threshold CA: a new exact upper bound paired with a new tightness witness** (Theorem 8, Proposition 10, Remark 14): we prove the unconditional upper bound $\varepsilon_{\mathrm{ca}}(C, \delta, \delta) \leq \binom{n}{w}/|F|$ for $w = \lfloor \delta n \rfloor$ via an injection of bad coefficient values into $\binom{[n]}{n-w}$, and we construct a matching lower-bound witness at the equality case $\delta = (n - k - 1)/n$ for $|F| > \binom{n}{w}^2$ (Proposition 10; see Remark 12 for the $|F|$-asymptotic scope). To our knowledge, neither side of this pair has been stated in the prior literature. Together they establish a structural barrier within the standard two-bad-γ CA framework: at FRI scale $n \gg \log|F|$ the upper bound is exponentially larger than 1 and therefore vacuous against any meaningful security target, yet over large fields it is attained at the equality case, certifying that the $2\times$ query overhead of (i) cannot be removed within this framework. Whether arguments tracking ≥ 3 bad γ's or non-CA proximity proofs can do so is left open (§8).

(iii) **List-size phase diagram and OP2 obstruction** (§7): the first p-dependent moment bounds on the list size, $\mathbb{E}[M] = \binom{n}{w}/p^c$ at every codimension excess $c \geq 1$ (Corollary 30), with the exact variance $\mathrm{Var}[M]$ established at $c = 1$ (Theorem 25) and Poisson-type dispersion $\mathrm{Var}[M] \approx \mathbb{E}[M]$ in the large-p regime at $c \geq 2$ via pairwise independence on non-overlapping supports (Theorem 28), all via the error-locator normal-vector identity (Lemma 23 for $c = 1$, Lemma 27 for $c \geq 2$); a conditional sketch of a sub-Poisson Bernstein tail at c=2 via a Möbius-induced negative-correlation argument (Conjecture 41, sketch-level; Lemma 39); a codimension phase diagram (§7.6) with a linear-n conjecture $M_{\text{true}} \leq \lfloor (2D-1)/c \rfloor$ at $c \geq 3$ (Conjecture 43, empirically verified through $n = 40$, the large-c regime relevant to FRI). Explicit obstruction: $M_{\max} \geq 2$ at FRI parameters for all p (Proposition 35), refuting the strongest $M = 0$ form of OP2.

(iv) **Extensions** (Appendices B, C, A, D): characteristic-2 FRI via additive subspace-polynomial folding (Theorem 72); circle FRI for Stwo's Mersenne31 setting (Theorem 80, with Lemma 78); a batch-CA theorem (Theorem 46) yielding STIR (Theorem 48) and WHIR (Conjecture 62) soundness above Johnson in odd characteristic, with the characteristic-2 analogue sketched in Remark 74; round-by-round soundness (Theorem 83), Fiat–Shamir-compiled non-interactive FRI (Theorem 84), and DEEP-FRI soundness error and the corresponding non-interactive PCS bound (Theorem 86, Corollary 87; explicit knowledge-extractor lifting deferred to [37]).

1.5 Comparison

Result	Regime	FRI commit error	Loss	Ref.
BCIKS [7]	$\delta < \delta_J$	$O(Rn^2)/\lvert F\rvert$	0	FOCS'20
BCHKS [6]	$\delta \leq \delta_J$	$O(Rn)/\lvert F\rvert$	0	2025
Goyal–Gur. [8]	$\delta < 1-\rho$	$O(R)/\lvert F\rvert$	0	2025
		(random plain RS; subspace-design codes)		
This paper	$\delta_J < \delta < 1-\rho$	$\mathbf{O(nR)/\lvert F\rvert}$	$\delta/2$	**new**

Table 1: FRI commit-phase soundness for **plain fixed-L** RS (the fixed-domain setting). Goyal–Guruswami applies to random plain RS and subspace-design codes, not to this fixed-L setting; circle FRI is covered in Appendix C.

Field ($\lvert F\rvert$)	Commit	Query (q=128, δ=0.4)		Proved total
	(both)	Conj. $(1-\delta)^q$	**Proved** $(1-\delta/2)^q$	
BabyBear base (2^{31})	7	*94 (unproved)*	41	**7**
BabyBear_4 (2^{124})	100	*94 (unproved)*	41	**41**
Goldilocks (2^{64})	40	*94 (unproved)*	41	**40**

Table 2: Interactive FRI security (bits), $n = 2^{20}$, $\rho = 1/2$, $R = 20$. The commit-phase term matches BCHKS [6]. Prior to this work, the 94-bit query-phase figure was *conjectured*; we *prove* 41 bits at the same $q = 128$. Proved total = min(commit, proved query): on BabyBear_4 the query phase is the bottleneck and our proved total is 41 bits; on Goldilocks and base-field BabyBear the commit phase dominates. Increasing to $q \approx 398$ achieves 128-bit *interactive* query-phase security; BabyBear_4 remains capped at ∼100 total interactive bits by the commit term. Table 3 accounts separately for Fiat–Shamir's non-interactive Q-loss. Circle FRI (Stwo/Mersenne31) achieves the same bound (Theorem 80).

1.6 Concrete cost

Against what is actually proved today, our bound provides positive proven security at every δ in the open zone where BCHKS provides none: above the Johnson radius, BCHKS gives zero proven soundness, whereas we give $\varepsilon_{\mathrm{FRI}} \le nR/|F| + (1-\delta/2)^q$. There is no proved-vs-proved regression; there is only added coverage.

Against the *conjectured* zero-loss CA baseline ($\varepsilon_{\mathrm{ca}}(C,\delta,\delta) = O(1)/|F|$ above Johnson per ABF [4]; the only currently *proved* equal-threshold bound is the unconditional upper bound $\varepsilon_{\mathrm{ca}}(C,\delta,\delta) \le \binom{n}{w}/|F|$ of Theorem 8, exponential in n at FRI scale, and the strongest up-to-capacity form was independently disproved [5, 12]), the half-threshold exacts a precise toll on query complexity. See Remark 14 and §8, item 1, for the structural barrier within the CA framework. To achieve λ-bit security from the query term alone:

$$q_{\text{zero-loss}} = \frac{\lambda}{\log_2(1/(1-\delta))}, \qquad q_{\text{ours}} = \frac{\lambda}{\log_2(1/(1-\delta/2))}.$$

The overhead ratio $q_{\text{ours}}/q_{\text{zero-loss}} = \log_2(1/(1-\delta))/\log_2(1/(1-\delta/2))$:

δ	Regime	$q_{\text{ours}}/q_{\text{zero-loss}}$	Status of zero-loss
0.30	Just above Johnson (ρ=1/2)	2.2×	$\binom{n}{w}/\|F\|$ (Remark 14), vacuous at FRI scale
0.40	Intermediate	2.3×	$\binom{n}{w}/\|F\|$, vacuous at FRI scale
0.45	Near capacity	2.3×	Disproved at capacity [5, 12]

The overhead is paid only in queries, and only against a baseline that is itself unproven. The commit-phase term $O(nR)/|F|$ matches BCHKS up to constants; we do not claim a Pareto improvement over BCHKS in the Johnson regime. The contribution is coverage of the open zone, where BCHKS provides no bound. Section 9 quantifies both terms in concrete parameters.

1.7 The correlated agreement framework and its limits

Every known proximity-gap-based FRI soundness argument — BBHR [20], BCIKS [7], BCHKS [6], and the present paper, with the proximity-gap inputs supplied by Goyal–Guruswami [8] for the random/subspace-design subcases — follows the same two-step strategy:

1. **Correlated agreement (CA) lemma**: if (f_1, f_2) has joint distance $> \delta$ from $C^{=2}$, then for random γ, the fold $f_1 + \gamma f_2$ stays far from C with high probability.
2. **Coupling**: the FRI decomposition $f \mapsto (f_{\mathrm{even}}, f_{\mathrm{odd}})$ maps distance from RS_k to joint distance from $\mathrm{RS}_{k/2}^{=2}$, linking the CA lemma to the protocol.

We call this the **CA framework**. The query-phase soundness of FRI is controlled by the *conclusion threshold* δ_{hd} of the CA lemma: each query catches a far fold with probability δ_{hd}, giving soundness $(1-\delta_{\mathrm{hd}})^q$ after q queries. The ratio $\delta_{\mathrm{hd}}/\delta$ determines the *query overhead*: at ratio 1:1 (zero loss), queries are optimal; at ratio 1:r, the overhead is $\sim r\times$.

The standard two-bad-γ CA framework is settled at the 1:2 ratio. The $\delta_{\mathrm{hd}} = \delta/2$ bound (ratio 1:2) was established by Rothblum–Vadhan–Wigderson [31] in 2013 for arbitrary linear codes (Theorem 6). At equal threshold $\delta_{\mathrm{hd}} = \delta$: $\varepsilon_{\mathrm{ca}} \le \binom{n}{w}/|F|$ with constant exponential in the code length (Remark 14), vacuous against any meaningful security target at FRI scale. The equal-threshold upper bound of Theorem 8 is exponential in n, suggesting that significantly tighter CA bounds at FRI scale via the two-bad-γ argument will require additional structure beyond linearity; we are not aware of any two-bad-γ CA argument that beats the 1:2 ratio at deployment scale.

What lies beyond? Reducing the overhead below 2× would require either a CA argument tracking ≥ 3 bad γ's (open, see §8) or a proof that does not factor through a CA lemma at all. No such argument exists in the public literature. We discuss concrete candidates — character sums on multiplicative subgroups, cross-correlation techniques — in Section 10.

1.8 Scope and limitations

This paper proves an unconditional FRI soundness theorem above the Johnson radius with a $\sim 2\times$ query overhead that is optimal within the CA framework (§1.7). The protocol is unchanged; the overhead is the cost of removing the zero-loss CA conjecture.

Structural assumptions. The $k = 2^m$ condition is necessary: for odd $k \geq 3$ the monomial x^k gives a counterexample with $\Delta(x^k, \mathrm{RS}_k) = 1-\rho$ yet the FRI fold is always a codeword (Remark 5). Standard FRI uses $k = 2^m$ already. The second-moment bound (Theorem 25) holds for all k but at codimension $c = 1$; it is not on the critical path to the main FRI theorem. There is *no* characteristic restriction: the main body handles $\mathrm{char}(F) \neq 2$ via multiplicative folding; Appendix B extends to $\mathrm{char}(F) = 2$ via additive folding, with the identical $nR/|F| + (1 - \delta/2)^q$ bound. **There is also no rate restriction:** the main theorem holds for every rate $\rho = k/n$ with $k = 2^m$ (the numerical tables in §1.6 use $\rho = 1/2$ as the BabyBear/Goldilocks reference; rates such as $\rho = 1/4$ or $\rho = 1/8$, common in zkVM parameter choices for soundness amplification, satisfy the same bound with $R = m$ rounds and any $\delta \in (\delta_J(\rho),\, 1-\rho)$).

1.9 Relation to the EF Proximity Prize

The Ethereum Foundation Proximity Prize [1] (May 2026) frames the open problem as two challenges. The *Grand mutual correlated agreement (MCA) Challenge*: determine the largest δ^* such that $\varepsilon_{\mathrm{mca}}(C, \delta^*) \leq \varepsilon^*$ with $\varepsilon^* = 2^{-128}$ on Reed–Solomon codes at rates $\rho \in \{1/2, 1/4, 1/8, 1/16\}$. The *Grand List Decoding Challenge*: same target on $|\Lambda(C^{\equiv m}, \delta^*)| \leq \varepsilon^*|F|$ for constant m. Both subsume the ABF open problems [4]: OP1 (zero-loss CA above Johnson) feeds the MCA target, OP2 (sub-$O(n)$ list-size route) feeds the LD target, OP3 (improved CA constants) sharpens both.

The table below is the claim map used in the rest of the paper. It separates theorem-level results from partial progress and from the original zero-loss targets that remain open. Throughout the table, "above Johnson" means $\delta \in (\delta_J, 1-\rho)$ (the open zone, strictly above the Johnson radius and strictly below capacity); the upper limit is implicit in the open-zone framing of §2.

Prize target	Paper result	Status	Reference	Remaining gap
Grand MCA: equal-threshold CA above Johnson	Unconditional upper bound $\varepsilon_{\mathrm{ca}}(C,\delta,\delta) \leq \binom{n}{w}/\|F\|$ (Theorem 8); matching tightness construction at the equality-case $w = n-k-1$ for $\|F\| > \binom{n}{w}^2$ (Proposition 10). Constant is exponential in n, hence vacuous at FRI scale	Open in general; the strongest up-to-capacity form is false [5, 12]; the only unconditional bound at FRI scale is vacuous; an equality-case tightness witness shows $\binom{n}{w}/\|F\|$ is sharp at one boundary value of w	Theorem 8; Proposition 10; Remark 14	General-w tightness; non-vacuous bound at FRI scale; non-CA proof beating the 2× overhead
Grand MCA: half-threshold CA, all rates incl. $\rho = 1/16$	Unconditional $\varepsilon_{\mathrm{ca}}(C,\delta/2,\delta) \leq 1/\|F\|$, rate-uniform; yields theorem-level FRI soundness rate-uniformly, conditional-theorem-level STIR soundness rate-uniformly under hypotheses (RD), (CQ) (Theorem 48; query-phase bound $\max((1-\delta/2)^q,(2\rho)^q)$ collapses to $(1-\delta/2)^q$ at $\rho \leq 1/4$), and conjecture-level WHIR soundness (proximity-substitution proved); common query term and protocol-specific commit-phase terms; no protocol change	Solved as theorem rate-uniformly for FRI; conditional theorem for STIR (multi-round closed within (CQ) via Lemma 49; (CQ)-to-arbitrary closed via Lemma 53 + Theorem 54 under input restriction (RD-strong)); conjectural for WHIR	Theorem 6; Theorems 20, 72, 80, 48, 54; Conjecture 62	∼2× query overhead vs. the conjectured zero-loss baseline; STIR non-codeword-quotient closed under (RD-strong)
Grand LD: $\|\Lambda(C^{\equiv m},\delta^*)\| \leq \varepsilon^*\|F\|$ above Johnson	First p-dependent moment bounds; conditional sketch of a sub-Poisson tail at $c = 2$ (Conjecture 41, sketch-level); codimension phase diagram with linear-n conjecture at $c \geq 3$ (the large-c regime relevant to FRI); explicit obstruction to worst-case $M = 0$	Partially addressed; strongest $M = 0$ form refuted; sub-Poisson tail conditional / sketch-level; phase diagram conjectural at $c \geq 3$	Theorem 25; Corollary 30; Conjecture 41; Conjecture 43; Proposition 35	Full proof of Conjecture 41; worst-case $c = 2$ extension; rigorous proof of Conjecture 43 at $c \geq 3$
ABF OP3: improved CA constants	Half-threshold CA constant is $1/\|F\|$ and commit phase matches $O(nR)/\|F\|$ up to constants	Partially addressed	Theorem 6; Theorem 20	Does not prove zero-loss CA
Original up-to-capacity MCA / zero-loss proximity gap	Not proved; capacity-level variants are now known false in related settings	Not solved here	Crites–Stewart [5]; Kambiré [12]; §8	Requires different techniques or revised conjectures

Interpretation. The paper does not claim the original zero-loss proximity gap. Its prize-facing contribution is narrower and unconditional: FRI gets theorem-level above-Johnson soundness via the half-threshold CA bound rate-uniformly (covering $\rho \in \{1/2, 1/4, 1/8, 1/16\}$), STIR gets theorem-level soundness on the standard radix-2 schedule for $\rho \in \{1/4, 1/8, 1/16\}$ (the higher-rate

$\rho = 1/2$ STIR variant is suggested by the same proof pattern but not formally stated as a theorem here), and WHIR gets conjecture-level soundness via the same chain (with the proximity-layer substitution rigorously proved), while the equal-threshold CA route is shown to be vacuous at FRI scale. The remaining route to better-than-2× query overhead must therefore avoid the CA framework or prove a strictly stronger non-CA proximity statement.

Companion work in preparation. A pair of companion papers in preparation addresses non-CA proximity statements complementary to this paper's CA-framework boundary [2, 3]; concrete result statements await those papers' completion and are not asserted here. See §10 for a brief pointer.

1.10 Guide to the proof

The paper has two logically independent tracks: the **mainline** (§3–§6), which proves the FRI soundness theorem in five steps, and the **structural track** (§7), which establishes the first p-dependent list-size bounds and rules out the strongest $M = 0$ form of Open Problem 2 while leaving weaker variants open. The mainline is entirely self-contained; the structural track is not on the critical path to any soundness theorem.

Mainline: FRI soundness in five steps.

Step 1. Half-threshold CA (Theorem 6, §3). For any linear code C: if (f_1, f_2) has joint distance $> \delta$ from $C^{=2}$, at most one $\gamma \in F$ makes $f_1 + \gamma f_2$ within distance $\delta/2$ of C. *Input*: linearity of C only (RVW13 [31]).

Step 2. Even/odd coupling (Lemma 15, §5). For even k: the distance from RS_k is at most the joint distance from $\mathrm{RS}^{=2}_{k/2}$. *Input*: the isomorphism $\mathrm{RS}_k \xrightarrow{\sim} \mathrm{RS}^{=2}_{k/2}$ (requires k even; Remark 5).

Step 3. Proximity gap (Theorem 16, §5). At most 1 value of α makes the FRI fold $(\delta/2)$-close to $\mathrm{RS}_{k/2}$. *Input*: Steps 1 + 2 composed.

Step 4. Distance locking (Remark 21, §6). After round 1, the effective distance is $\delta/2 < (1-\rho)/2 \leq \delta_J$ (Remark 1), inside the proved zero-loss regime up to the Johnson radius [7, 6]. The proximity gap of BCHKS [6] at threshold $\gamma = \delta/2 < \delta_J$ gives $\leq n$ bad α's per round (improving the original BCIKS [7] bound of $O(n^2)$); distance stays at $\delta/2$ for all subsequent rounds. *Input*: the chain (2) (always true for $\delta < 1-\rho$) + the BCIKS–BCHKS proximity gap framework [7, 6] (external, proved).

Step 5. FRI soundness (Theorem 20, §6). Strategy A (honest fold): multilinear Schwartz–Zippel gives $\leq R/|F|$. Strategy B (deviate): Steps 3–4 bound bad α's to ≤ 1 (round 1) and $\leq n$ (rounds ≥ 2) per round and the consistency check catches a $(\delta/2)$-far deviation with probability $\geq 1 - (1-\delta/2)^q$. Union: $nR/|F| + (1-\delta/2)^q$. *Input*: Steps 3 + 4 + Schwartz–Zippel.

The mainline uses *no* result from §7. Its external ingredients are: linearity of RS_k (trivial), the BCIKS proximity gap at $\delta \leq \delta_J$ [7], the BBHR / BCIKS / BGKTTW catch-coverage lemma [20, 7, 37] (the path-consistency step in the proof of Theorem 20), and the Schwartz–Zippel lemma.

Extensions (from the same CA bound). Theorem 6 is characteristic-independent, so three extensions branch directly from Step 1 (see Figure 1):

- *STIR*: Batch CA (Theorem 46) $\rightarrow$ STIR soundness (Theorem 48), Appendix A.
- *WHIR*: Sumcheck composition + half-threshold CA (Conjecture 62), Appendix A.
- *Characteristic* 2: Additive coupling (Lemma 70) $\rightarrow$ FRI soundness in characteristic 2 (Theorem 72), Appendix B.
- *Circle FRI*: Circle coupling (Lemma 78) $\rightarrow$ FRI soundness on unit circle (Theorem 80), Appendix C.

Structural track (§7): independent of the soundness proof. Section 7 studies the per-word list size M (how many codewords lie within distance w of a received word), aiming at Open Problem 2 (§8). Four results:

1. *First p-dependent moment bound.* $\mathbb{E}[M] = \binom{n}{w}/p^c$ for all codimension excess $c \geq 1$ (Corollary 30), via the error-locator normal vector identity (Lemma 27) and pairwise independence for non-overlapping supports (Theorem 28). At FRI-relevant codimension ($c = \Theta(n)$), $\mathbb{E}[M]$ is exponentially small.
2. *Worst-case landscape by codimension.* At $c = 2$, a Möbius identity (Lemma 39) yields a universal $M_{\text{true}} \leq \min(p,\, 2\binom{n}{w-1})$ (Theorem 40) and a conditional / sketch-level Bernstein-style sub-Poisson tail on bin sizes with high probability over (s_1, s_2) (Conjecture 41); the empirical worst-case fits $c_0 c_1^n$ with $c_0 \approx 0.66$, $c_1 \approx 1.36$ in the tested range $n \leq 24$ (Remark 42; descriptive, not asymptotic). At $c \geq 3$, an Open-Set Rank Lemma (Conjecture 43) predicts a linear bound $M_{\text{true}} \leq \lfloor (2D-1)/c \rfloor$, empirically verified through $n = 40$ (§7.6). The large-c regime $c = \Theta(n)$ falls in this third phase, where the conjecture predicts $M_{\text{true}} = O(1)$ at the Johnson radius.
3. *OP2 obstruction.* $M_{\max} \geq 2$ at FRI parameters ($n \geq 32$, rate $1/2$) for all p (Proposition 35). The strongest $M = 0$ conjecture is false.
4. *Practical $M = 0$.* At deployment parameters ($p \approx 2^{31}$, $c = \Theta(n)$): $\mathbb{E}[M] = \binom{n}{w}/p^c \ll 1$ (Corollary 30); $M = 0$ in practice even though worst-case $M = 0$ fails.

Additional structural results (fiber bounds, Bézout bounds, codimension bounds, incidence bounds, irreducibility of the remainder hypersurface) are developed in a companion note [24].

Reader's guide. A reader interested only in the FRI soundness theorem may read §2–§6 and skip §7 entirely.

2 Definitions

Let F be a finite field and $L \subset F$ with $|L| = n$. The FRI folding requires a *fixed-point-free involution* on L: a bijection $\iota : L \to L$ with $\iota^2 = \text{id}$ and $\iota(x) \neq x$ for all $x \in L$. In the main body (§3–§6), we use $\iota(x) = -x$ (i.e., $\text{char}(F) \neq 2$, L closed under negation, and $0 \notin L$ so that $-x \neq x$ for all $x \in L$). Appendix B uses $\iota(x) = x + \beta$ (i.e., $\text{char}(F) = 2$ and L an $\mathbb{F}_2$-linear subspace containing $\beta \neq 0$). $\text{RS}_k = \text{RS}[F, L, k]$ is the Reed–Solomon code of evaluations of polynomials of degree $<k$ on L, with rate $\rho = k/n$. We write $\delta_J = 1 - \sqrt{\rho}$ for the *Johnson radius* (the classical relative list-decoding radius with polynomial list size, established for Reed–Solomon codes by Guruswami–Sudan [16]; the earlier Sudan algorithm [15] reaches the smaller radius $1 - \sqrt{2\rho}$ relevant for low-rate codes) and use the term Johnson radius for δ_J throughout.

Open zone. Throughout, the *open zone* refers to the strict interval $\delta \in (\delta_J,\, 1-\rho)$ — distances strictly above the Johnson radius and strictly below capacity. All end-to-end soundness theorems in this paper (Theorems 20, 72, 80, 48; Conjecture 62; the non-interactive variants of Appendix D) operate in this regime. The CA, equal-threshold, proximity-gap, and list-size theorems hold for any δ within the technical hypotheses of each statement; the open-zone window enters only when these are composed into an end-to-end FRI/STIR/WHIR soundness statement. At the boundary $\delta = \delta_J$ the proved zero-loss proximity gap of BCHKS [6] applies and our reduced-threshold statement is strictly weaker.

Remark 1 (Half-threshold lies inside BCIKS's zero-loss regime)**.** *The single algebraic identity that drives every soundness theorem in this paper is, for $\delta < 1-\rho$,*

$$\frac{\delta}{2} < \frac{1-\rho}{2} \le 1-\sqrt{\rho} = \delta_J, \tag{2}$$

where the first inequality is $\delta < 1-\rho$ divided by two, and the second is the elementary identity $(1-\sqrt{\rho})^2 \ge 0 \Leftrightarrow 1-\rho \le 2(1-\sqrt{\rho})$. Hence after one halving step the threshold $\delta/2$ lies strictly below the Johnson radius δ_J, where the zero-loss proximity gap is established by BCIKS [7, Theorem 1.2] (with $O(n^2)$ exceptional values) and refined by BCHKS [6] to $\le n$ exceptional values per round. We refer to this as the BCIKS–BCHKS zero-loss regime (sometimes "the Johnson regime"); we deliberately avoid the term "unique-decoding regime" because the unique-decoding radius $(1-\rho)/2$ is strictly smaller than δ_J for every $\rho < 1$, so (2) is a stronger statement than $\delta/2$ being below half the minimum distance.

Remark 2 (Base vs. extension fields)**.** *The evaluation domain L typically sits inside a prime subfield or a small extension (e.g., $L \subset \mathbb{F}_p$ with $|L| = n \mid (p-1)$). The soundness bounds in this paper depend on $|F|$, not on the subfield containing L. In deployed systems, FRI randomness α_i is sampled from an extension field $\mathbb{F}_{p^d}$ to achieve sufficient Schwartz–Zippel probability: $|F| = p^d$. For example, BabyBear ($p = 2^{31}-1$) with degree-4 extension gives $|F| = p^4 \approx 2^{124}$, yielding $\log_2(|F|/(nR)) \approx 100$ bits of commit-phase security (at $n = 2^{20}$, $R = 20$). The $|F|$ in Theorems 6, 20, and 48 refers to the field from which random challenges are drawn, which is the extension field in practice.*

Definition 3 (Correlated Agreement [4])**.** *Let D denote the common evaluation domain (the domain of C; when the CA bound is applied to $(f_{\text{even}}, f_{\text{odd}})$ on L', $D = L'$ and $|D| = n/2$).*

$$\varepsilon_{\text{ca}}(C, \delta_{\text{hd}}, \delta_{\text{nt}}) = \max_{f_1, f_2 : D \to F} \Pr_{\gamma \leftarrow F}\left[\Delta(f_1+\gamma f_2, C) \le \delta_{\text{hd}} \text{ and } \Delta_{\text{joint}}((f_1, f_2), C^{=2}) > \delta_{\text{nt}}\right],$$

where γ is drawn uniformly at random from F, the max is over all functions $f_1, f_2 : D \to F$ (not necessarily codewords), and $\Delta_{\text{joint}}((f_1, f_2), C^{=2}) = \min_{g_1, g_2 \in C} \frac{1}{|D|}|\{x \in D : f_1(x) \ne g_1(x) \text{ or } f_2(x) \ne g_2(x)\}|$ is the joint Hamming distance to the pair-domain code $C^{=2}$ (equivalently, to the set product $C \times C$) [7]. We call δ_{hd} the conclusion threshold (distance of the random combination to C) and δ_{nt} the premise threshold (joint distance from $C^{=2}$). Standard zero-loss CA [4] corresponds to $\delta_{\text{hd}} = \delta_{\text{nt}} = \delta$.

Definition 4 (FRI Folding). *$L' = \{x^2 : x \in L\}$ with $|L'| = n/2$. For each $y \in L'$, fix a square root $\sqrt{y} \in L$ (the expressions below are invariant under the choice $\sqrt{y} \mapsto -\sqrt{y}$). For $f : L \to F$: $f_{\text{even}}(y) = (f(\sqrt{y}) + f(-\sqrt{y}))/2$, $f_{\text{odd}}(y) = (f(\sqrt{y}) - f(-\sqrt{y}))/(2\sqrt{y})$. FRI fold: $f'(y) = f_{\text{even}}(y) + \alpha \cdot f_{\text{odd}}(y)$.*

Remark 5 (Even/odd isomorphism for even k)**.** *For k even: $g \mapsto (g_{\text{even}}, g_{\text{odd}})$ is a linear isomorphism $\mathrm{RS}_k \xrightarrow{\sim} \mathrm{RS}_{k/2}^{=2}$ (both have dimension k; the map permutes the k coefficients). Consequently: $f \notin \mathrm{RS}_k$ iff $(f_{\text{even}}, f_{\text{odd}}) \notin \mathrm{RS}_{k/2}^{=2}$.*

For k odd: $\mathrm{RS}^{=2}_{\lceil k/2 \rceil}$ *has dimension $k+1 > k$, and the isomorphism fails. The monomial x^k (for $k \geq 3$ odd) satisfies $\Delta(x^k, \mathrm{RS}_k) = 1-\rho$ (since $x^k - g$ has at most k roots for any g of degree $<k$), yet $f_{\mathrm{even}} = 0$ and $f_{\mathrm{odd}}(y) = y^{(k-1)/2} \in \mathrm{RS}_{\lceil k/2 \rceil}$ (since $(-1)^k = -1$), so the fold $f' = \alpha \cdot y^{(k-1)/2} \in \mathrm{RS}_{\lceil k/2 \rceil}$ for all α and FRI always accepts. The degenerate case $k = 1$ is vacuous:* $\mathrm{RS}_1 = \{$*constants*$\}$ *and FRI is not invoked ($R = 0$). This is why we require $k = 2^m$ with $m \geq 1$.*

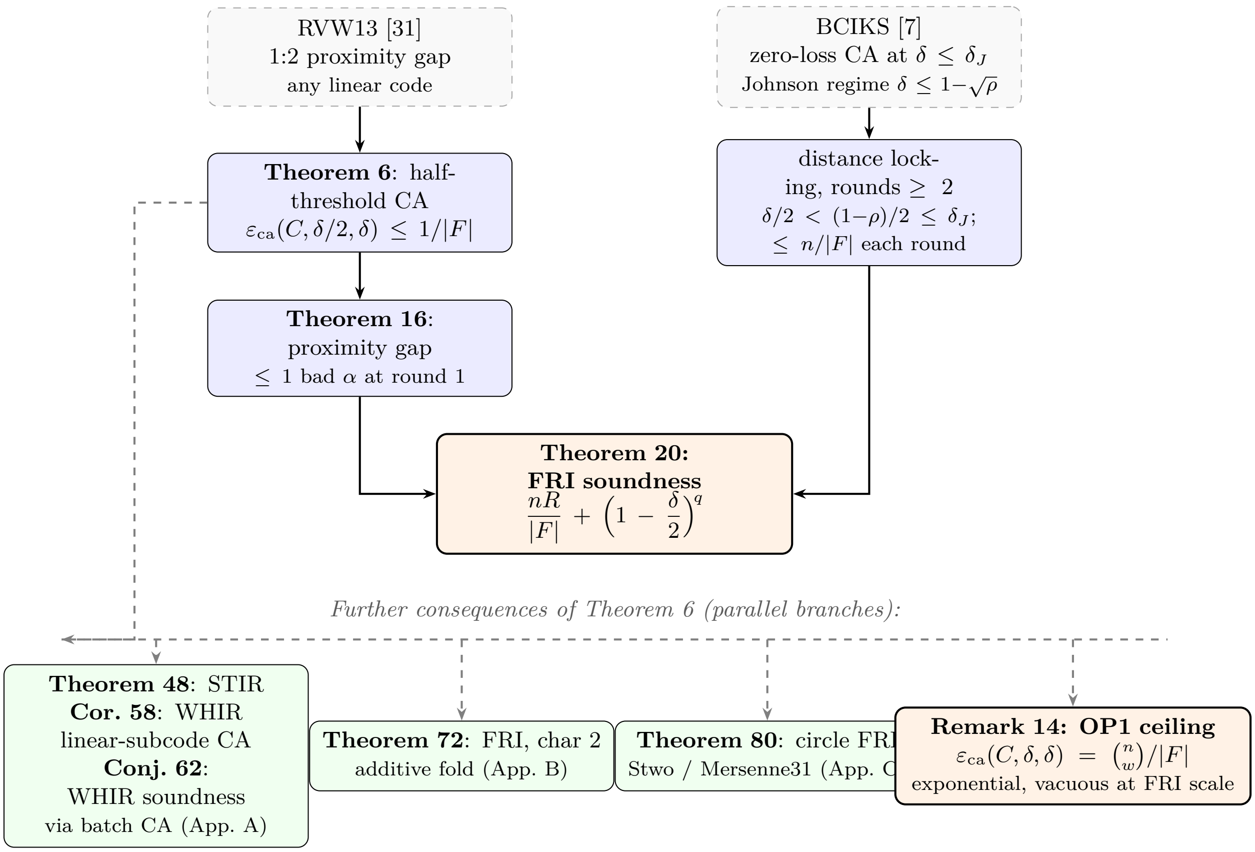

Figure 1: Proof map. **Top half** (mainline): RVW13 yields the half-threshold CA bound (Theorem 6), which composes with the FRI even/odd coupling to give the round-1 proximity gap (Theorem 16); BCIKS supplies the round-≥ 2 distance lock (the structural observation $\delta/2 < (1-\rho)/2$); the two combine into the FRI soundness theorem. **Bottom half** (parallel branches from the same CA bound): STIR via batch CA (Theorem 48); WHIR via batch CA on linear subcodes (Corollary 58 proves the proximity-layer; Conjecture 62 states the end-to-end soundness conjecture); characteristic-2 FRI via additive coupling; circle FRI via the Stwo coupling; equal-threshold CA (the OP1 ceiling) showing the $\binom{n}{w}/|F|$ bound is vacuous at FRI scale, so the $2\times$ overhead is intrinsic to any CA-based proof. The list-size structural track (§7) is independent of the soundness proof and is not shown.

3 Half-Threshold CA Bound

Theorem 6 (RVW13 [31]; BCHKS [6], Table 1, row 1)**.** *For every real $\delta \in [0, 1]$ and every linear code $C \subseteq F^L$ with $|L| = n$, $\varepsilon_{\mathrm{ca}}(C, \delta/2, \delta) \leq 1/|F|$.*

Equivalently: for any $f_1, f_2 : L \to F$ with $\Delta_{\text{joint}}((f_1, f_2), C \times C) > \delta$, at most one $\gamma \in F$ satisfies $\Delta(f_1 + \gamma f_2, C) \leq \delta/2$.

Proof (due to [31]). Two bad γ's give two linear equations in (f_1, f_2) on overlapping agreement sets. Solving for both components simultaneously yields a pair $(g_1, g_2) \in C \times C$ matching (f_1, f_2) on $\geq (1-\delta)n$ points, so joint distance $\leq \delta$ — contradicting the premise.

Proof. Fix f_1, f_2 with $\Delta_{\text{joint}} > \delta$. Call γ "bad" if $\exists\, h_\gamma \in C$ with $\Delta(f_1 + \gamma f_2, h_\gamma) \leq \delta/2$; let $S_\gamma = \{x \in L : f_1(x) + \gamma f_2(x) = h_\gamma(x)\}$, so $|S_\gamma| \geq (1 - \delta/2)n$.

Suppose $\gamma_1 \neq \gamma_2$ are both bad. By inclusion-exclusion:

$$|S_{\gamma_1} \cap S_{\gamma_2}| \;\geq\; 2(1 - \delta/2)n - n \;=\; (1 - \delta)n.$$

On $S_{\gamma_1} \cap S_{\gamma_2}$, the system $f_1 + \gamma_i f_2 = h_i$ $(i = 1, 2)$ has the unique solution

$$g_1 := \frac{\gamma_2 h_1 - \gamma_1 h_2}{\gamma_2 - \gamma_1} \in C, \qquad g_2 := \frac{h_1 - h_2}{\gamma_1 - \gamma_2} \in C$$

(using linearity of C). So $(f_1, f_2) = (g_1, g_2)$ on $S_{\gamma_1} \cap S_{\gamma_2}$, giving:

$$\Delta_{\text{joint}}\big((f_1, f_2), C \times C\big) \;\leq\; \frac{|L \setminus (S_{\gamma_1} \cap S_{\gamma_2})|}{n} \;\leq\; \delta.$$

This contradicts $\Delta_{\text{joint}} > \delta$. Hence at most 1 bad γ. □

Remark 7 (Attribution and scope)**.** *Theorem 6 is the classical* 1:2 *proximity-gap inequality of Rothblum–Vadhan–Wigderson [31] (STOC 2013), formalized in BCHKS [6], Table 1, as the row [RVW13]: γ arbitrary, $a \geq 2$, $\varepsilon^* = \gamma$. The proof uses only linearity of C (closure under F-linear combinations): no polynomial structure, no multiplicative structure on L, no restriction on k or δ. It applies to any linear code over any finite field.*

What is new in this paper is not the CA bound, but (a) its application above the Johnson radius to obtain an unconditional FRI soundness theorem in the open zone $(\delta_J, 1-\rho)$, where no such theorem was previously available; (b) the unconditional analytic equal-threshold CA upper bound $\varepsilon_{\text{ca}}(C, \delta, \delta) \leq \binom{n}{w}/|F|$ at $w = n - k - 1$ (Theorem 8, with the matching lower bound holding in the asymptotic-in-$|F|$-at-fixed-n regime; see Remark 12 and Remark 14); and (c) the tightness argument showing the 1:2 *ratio is optimal at FRI scale (Remark 14). The structural observation enabling (a) is the chain $\delta/2 < (1-\rho)/2 \leq \delta_J$ (Remark 1), so after one round the protocol enters BCIKS's zero-loss regime where BCIKS [7] supplies the round-by-round gap.*

The prior proximity-gap results of AHIV17 [32] (originally for interleaved/tensor codes; the analogous Reed–Solomon specialization gives the regime δ below one quarter of the minimum-distance threshold $\delta_{\min} := 1-\rho+1/n$), the unpublished Roth–Zémor refinement reported in BCIKS [7] (regime $\delta < \delta_{\min}/3$, attributed via personal communication [33]), and BKS18 [34] (δ above the unique decoding radius $(1 - \rho)/2$) each proved successively stronger gap statements. The BKS18 proximity-gap argument [34] (Theorem 16, the version subsequently used by BCIKS [7] in their FRI soundness analysis) splits on $\Delta(f_2, C)$ and bounds bad α's in each case separately (Case 1/2), yielding a 1:3 ratio ($\delta_{\text{hd}} = \delta/3$), tight for that framework because the case bookkeeping gives $2\delta_{\text{hd}} + \delta_{\text{hd}} = 3\delta_{\text{hd}} \leq \delta$ only when $\delta_{\text{hd}} \leq \delta/3$. The direct proof above bypasses Case 1/2 entirely: it solves for both f_1 and f_2 simultaneously from two bad γ's, obtaining the joint distance bound $\leq 2\delta_{\text{hd}}$ in one step. This is $\leq \delta$ when $\delta_{\text{hd}} \leq \delta/2$, improving the ratio from 1:3 to 1:2.

We state $nR/|F|$ (not $R/|F|$) in Theorem 20 to maintain a unified presentation with STIR (Theorem 48), where the per-round count cannot be tightened below n in general.

4 Equal-Threshold CA: Exact Bound and Tightness of the 1:2 Ratio

This section establishes the exact form of the equal-threshold CA bound (Theorem 8) and shows that it is tight up to lower-order terms (Proposition 10). The conclusion (Remark 14) is that the $\binom{n}{w}$ constant is exponential in the code length, vacuous at FRI scale; this is the formal statement behind the "OP1 obstruction" framing of §1.9.

4.1 Upper bound at equal threshold

Theorem 8 (Equal-threshold CA upper bound)**.** *Let $C \subseteq F^L$ be any linear code with $|L| = n$, and let w be any integer with $0 \le w \le n-1$. Set $\delta := w/n$. For any $f_1, f_2 : L \to F$ satisfying $\Delta_{\mathrm{joint}}((f_1, f_2), C \times C) > \delta$:*

$$\bigl|\{\gamma \in F : \Delta(f_1 + \gamma f_2, C) \le \delta\}\bigr| \;\le\; \binom{n}{w}.$$

Consequently $\varepsilon_{\mathrm{ca}}(C, \delta, \delta) \le \binom{n}{w}/|F|$. For $C = \mathrm{RS}[F, L, k]$ at the boundary $w = n-k-1$, the bound is attained for $|F| > \binom{n}{w}^2$ (Proposition 10).

Proof. Let $\Gamma = \{\gamma \in F : \Delta(f_1 + \gamma f_2, C) \le \delta\}$. For each $\gamma \in \Gamma$ there exists a witness codeword $c_\gamma \in C$ with at most w disagreement positions (i.e., $|\{x : (f_1{+}\gamma f_2)(x) \neq c_\gamma(x)\}| \le w$), so the agreement set has size $\ge n - w$. Choose any $A_\gamma \subseteq L$ with $|A_\gamma| = n - w$ on which $f_1 + \gamma f_2 = c_\gamma$, and define the assignment

$$\varphi : \Gamma \to \binom{L}{n-w}, \qquad \varphi(\gamma) = A_\gamma.$$

Claim: φ is injective. Suppose $\gamma \neq \gamma'$ in Γ with $\varphi(\gamma) = \varphi(\gamma') = A$ ($|A| = n - w \ge 1$). On A:

$$f_1 + \gamma f_2 = c_\gamma, \qquad f_1 + \gamma' f_2 = c_{\gamma'}.$$

Subtracting and dividing by $\gamma - \gamma' \neq 0$ (allowed since C is closed under F-linear combinations):

$$f_2\big|_A \;=\; \tfrac{c_\gamma - c_{\gamma'}}{\gamma - \gamma'}\big|_A, \qquad g_2 := \tfrac{c_\gamma - c_{\gamma'}}{\gamma - \gamma'} \in C.$$

Likewise $f_1|_A = c_\gamma|_A - \gamma g_2|_A$, so setting $g_1 := c_\gamma - \gamma g_2 \in C$ gives $(f_1, f_2) = (g_1, g_2)$ on A. Hence the joint disagreement is at most $|L \setminus A| = n - (n - w) = w$, so

$$\Delta_{\mathrm{joint}}\bigl((f_1, f_2), C \times C\bigr) \;\le\; \frac{w}{n} \;=\; \delta,$$

contradicting the hypothesis $\Delta_{\mathrm{joint}} > \delta$. Therefore φ is injective and $|\Gamma| \le \binom{n}{n-w} = \binom{n}{w}$. □

Remark 9 (Scope of Theorem 8: convention $w = n{-}k{-}1$)**.** *The bound holds (i) for all (f_1, f_2) with $\Delta_{\mathrm{joint}} > \delta$ — not just generic ones; (ii) over any field F, including small fields where $|F| \le \binom{n}{w}$ (in which case the bound $\binom{n}{w}/|F| \ge 1$ is trivial). For the Reed–Solomon specialization $C = \mathrm{RS}[F, L, k]$, the boundary weight $w = n - k - 1$ is the largest w with $\delta n < n - k$ (i.e., one below the covering radius), and is the parameter we adopt for the tightness Proposition 10 below.*

Smaller decoding balls. For a strictly smaller ball $w' < w$ corresponding to a smaller threshold $\delta' = w'/n$: the $(k{+}1)$-subset injection of Theorem 8 still applies (the agreement set has size $\ge n - w' \ge k + 1$), and yields the loose bound $|\Gamma| \le \binom{n}{k+1} = \binom{n}{w}$. This is not tight: the right *quantity at threshold w' is $\binom{n}{w'}$ (not $\binom{n}{w}$), obtained by injecting Γ into $\binom{L}{n-w'}$, the family of $(n-w')$-subsets on which a witness codeword agrees. The tight statement is therefore $|\Gamma| \le \binom{n}{w'} = \binom{n}{n-w'}$ at threshold $\delta' = w'/n$, with the prize-mapping table and Remark 14 below using the convention $w = \lfloor \delta n \rfloor$ for the asymptotic-in-n statements (Stirling: $\binom{n}{\lfloor \delta n \rfloor} \sim 2^{nH(\delta)}$).*

Convention. Theorem 8 is stated for arbitrary linear codes C and arbitrary integer thresholds w. The tightness Proposition 10 below specializes to $C = \mathrm{RS}[F, L, k]$ at the equality case $w = n-k-1$. Asymptotic vacuity statements (Remark 14) use $w = \lfloor \delta n \rfloor$. We avoid mixing these conventions on the same statement.

4.2 Tightness

Proposition 10 (Equal-threshold CA lower bound, large field, RS code)**.** *Let $C = \mathrm{RS}[F, L, k]$ with $|L| = n$, $w = n-k-1$, $\delta = w/n$. For every field F with $|F| > \binom{n}{w}^2$ there exist $f_1, f_2 : L \to F$ with $\Delta_{\mathrm{joint}}((f_1, f_2), C \times C) > \delta$ and*

$$\bigl|\{\gamma \in F : \Delta(f_1 + \gamma f_2, C) \le \delta\}\bigr| \;=\; \binom{n}{w}.$$

Proof. For each $(k+1)$-subset $A \subset L$, encode A-consistency of $f_1 + \gamma f_2$ as a single linear equation in γ. Fix A and let $\pi_A : F^L \to F$ be the linear functional defined by: $\pi_A(f) :=$ (value of the unique degree-$<k$ Lagrange interpolant of $f|_{A_0}$ at the remaining point of A) $-$ f at that point, where A_0 is any fixed k-subset of A (the choice of A_0 does not affect the kernel). Then f agrees with some codeword of C on all of A iff $\pi_A(f) = 0$. Set

$$a_A := \pi_A(f_1) \in F, \qquad b_A := \pi_A(f_2) \in F.$$

By linearity of π_A: $\pi_A(f_1 + \gamma f_2) = a_A + \gamma \cdot b_A$. Hence $f_1 + \gamma f_2$ is A-consistent iff

$$a_A + \gamma \cdot b_A \;=\; 0. \tag{3}$$

For $b_A \neq 0$, equation (3) has the unique solution $\gamma_A = -a_A/b_A \in F$. As A ranges over the $\binom{n}{k+1} = \binom{n}{w}$ choices, the values $\{\gamma_A\}_A$ exhaust all candidate decodable γ's by Theorem 8's injection (when each agreement set is exactly $(k+1)$-sized, the injection becomes a bijection onto its image).

Construction. Fix $f_2 \in F^L$ such that $b_A(f_2) \neq 0$ for every $A \in \binom{L}{k+1}$; the bad set $\{f : \pi_A(f) = 0\}$ is a single proper hyperplane in F^L for each A, contributing $|F|^{n-1}/|F|^n = 1/|F|$ measure, so the union of $\binom{n}{w}$ such hyperplanes leaves at least $|F|^n(1 - \binom{n}{w}/|F|) > 0$ valid f_2 when $|F| > \binom{n}{w}$. Pick $f_1 \in F^L$ uniformly at random. For $A \neq A'$: the equation $\gamma_A = \gamma_{A'}$ becomes

$$a_A b_{A'} - a_{A'} b_A \;=\; 0 \iff b_{A'} \cdot \pi_A(f_1) - b_A \cdot \pi_{A'}(f_1) \;=\; 0,$$

a single linear constraint on f_1. The functionals π_A and $\pi_{A'}$ are non-proportional: π_A depends only on the $(k+1)$ coordinates indexed by A (zero outside A), so π_A and $\pi_{A'}$ have different supports as functionals on F^L whenever $A \neq A'$, and any nonzero linear combination $\lambda \pi_A + \mu \pi_{A'}$ (with $(\lambda, \mu) \neq (0, 0)$) is a nonzero functional. In particular, $b_{A'} \pi_A - b_A \pi_{A'}$ is nonzero (since both $b_A, b_{A'} \neq 0$ by the choice of f_2 above, and $\pi_A, \pi_{A'}$ have different supports), hence vanishes on a hyperplane of measure $1/|F|$. By union bound over $\binom{\binom{n}{w}}{2} \le \binom{n}{w}^2/2$ pairs:

$$\Pr_{f_1}[\text{some } \gamma_A = \gamma_{A'}] \;\le\; \frac{\binom{n}{w}^2}{2\,|F|} \;<\; 1$$

provided $|F| > \binom{n}{w}^2/2$. The hypothesis $|F| > \binom{n}{w}^2$ suffices, giving an f_1 with all $\binom{n}{w}$ values γ_A pairwise distinct in F. By construction each γ_A is decodable (witnessed by codeword agreement on the $(k{+}1)$-subset A), so $|\Gamma| \geq \binom{n}{w}$. Combined with the upper bound (Theorem 8), $|\Gamma| = \binom{n}{w}$.

Joint-distance check. We must verify $\Delta_{\text{joint}}((f_1, f_2), C \times C) > \delta$. If $\Delta_{\text{joint}} \leq \delta$, then there exists a codeword pair $(g_1, g_2) \in C \times C$ with $|\{x : (f_1(x), f_2(x)) \neq (g_1(x), g_2(x))\}| \leq \delta n$. For *any* $\gamma \in F$: $f_1 + \gamma f_2$ agrees with $g_1 + \gamma g_2 \in C$ on $\geq (1 - \delta)n$ points, so every $\gamma \in F$ would be decodable, giving $|\Gamma| = |F|$. But the construction gives $|\Gamma| = \binom{n}{w} < |F|$ (since $|F| > \binom{n}{w}^2 \geq \binom{n}{w}$). Therefore $\Delta_{\text{joint}} > \delta$. □

Corollary 11 (Asymptotic equality at $w = n - k - 1$, large field at fixed n, RS code)**.** *For $C = \mathrm{RS}[F, L, k]$ with $w = n - k - 1$, $\delta = w/n$, and every field F with $|F| > \binom{n}{w}^2$:*

$$\varepsilon_{\text{ca}}(C, \delta, \delta) \;=\; \frac{\binom{n}{w}}{|F|}.$$

Remark 12 (Scope of Corollary 11: asymptotic regime)**.** *The matching lower bound (Proposition 10) requires $|F| > \binom{n}{w}^2$, so Corollary 11 is an asymptotic equality in $|F|$ at fixed n, not an asymptotic equality in n at fixed $|F|/n$. At FRI scale ($n \sim 2^{20}$, $|F| \sim 2^{124}$), $\binom{n}{w}^2 \gg |F|$, so the lower-bound construction does not apply and Corollary 11 cannot be invoked at deployment parameters. What does apply at deployment is the upper bound (Theorem 8), which is unconditional in $|F|$, and which already gives $\binom{n}{w}/|F| \gg 1$ at deployment scale (Remark 14); the upper bound being vacuous at deployment is a strictly stronger statement than "the equal-threshold ratio is intrinsically tight", and does not require the matching-tightness construction.*

4.3 Computational verification

The exact bound is verified for small parameters where exhaustive enumeration is feasible.

Remark 13 (Empirical confirmation)**.** *For* $\mathrm{RS}[6, 3]$ *over $\mathbb{F}_p$, $p \in \{7, 13, 19, \ldots, 163\}$: the empirical maximum (over all tested (f_1, f_2)) of the number of δ-close γ's equals* $\min(p, \binom{6}{2}) = \min(p, 15)$ *exactly. For* $\mathrm{RS}[8, 4]$ *over $p \in \{17, \ldots, 281\}$:* $\min(p, \binom{8}{3}) = \min(p, 56)$*, also exactly. See Note [25]; scripts in* `paper1/scripts/op1-barrier/` *of [39].*

4.4 Why the 1:2 ratio is optimal at FRI scale

Remark 14 (Why the 1:2 ratio is tight)**.** *Algebraic ceiling. The proof of Theorem 6 shows: two bad γ's at conclusion threshold δ_{hd} produce a joint pair $(g_1, g_2) \in C \times C$ matching (f_1, f_2) on $\geq (1 - 2\delta_{\text{hd}})n$ points, giving $\Delta_{\text{joint}} \leq 2\delta_{\text{hd}}$. For this to contradict $\Delta_{\text{joint}} > \delta$: need $2\delta_{\text{hd}} \leq \delta$, i.e., $\delta_{\text{hd}} \leq \delta/2$. The bound $\delta_{\text{hd}} = \delta/2$ is therefore optimal for any argument that extracts a joint codeword pair from two bad random-linear-combination values.*

Why pushing to equal threshold is vacuous (within this CA route). The upper bound of Theorem 8 is $\binom{n}{w}/|F|$ unconditionally in $|F|$. The constant $\binom{n}{w}$ grows as $2^{nH(\delta)}$ (Stirling). At FRI deployment scales the bound is therefore vacuous: already at $n = 2^{14}$ and $\delta \geq 0.1$, $\binom{n}{\lfloor \delta n \rfloor}$ exceeds 2^{1000}, vastly larger than any extension field in production (BabyBear$_4$ at $|F| \approx 2^{124}$, KoalaBear sextic at $|F| \approx 2^{186}$). The bound $\binom{n}{w}/|F|$ is exponentially larger than 1, so this route is presently vacuous against any meaningful security target such as 2^{-100}. Within the equal-threshold CA route, removing the $2\times$ query overhead would therefore require either a tighter equal-threshold upper bound (none in the literature) or a non-CA proximity argument; whether the algebraic ceiling above can be improved by tracking ≥ 3 bad γ's remains open (§8).

Half-threshold is tight. At $\delta_{\mathrm{hd}} = \delta/2$, $\max_\gamma \le 1$ in all tested cases [39], matching the ≤ 1 bound of Theorem 6. The constant 2 in the half-threshold bound is independent of n, in sharp contrast to the $\binom{n}{w}$ constant at equal threshold.

Comparison with the BKS18 1:3 ratio. The BKS18 [34] Case 1/2 approach (split on $\Delta(f_2, C)$, bound each case separately) achieves ratio 1:3 with zero proximity loss at $\delta_{\mathrm{hd}} < \delta/3$. The RVW13 [31] 2×2-solve argument of Theorem 6 trades zero loss for a better ratio (1:2 with loss δ_{hd}); the 1:2 ratio has been folklore in the proximity-gap community since 2013.

5 FRI Coupling and Proximity Gap

Lemma 15 (FRI Coupling, even k)**.** *For even k and any $f : L \to F$: $\Delta(f, \mathrm{RS}_k) \le \Delta_{\mathrm{joint}}((f_{\mathrm{even}}, f_{\mathrm{odd}}), \mathrm{RS}^{=2}_{k/2})$.*

Proof. By Remark 5: $g \mapsto (g_{\mathrm{even}}, g_{\mathrm{odd}})$ is an isomorphism $\mathrm{RS}_k \to \mathrm{RS}^{=2}_{k/2}$. For any $g \in \mathrm{RS}_k$ and $y \in L'$ let $c_y := |\{x \in \{\sqrt{y}, -\sqrt{y}\} : f(x) \ne g(x)\}| \in \{0, 1, 2\}$ denote the number of base-level errors in the fibre over y. If $f_{\mathrm{even}}(y) = g_{\mathrm{even}}(y)$ and $f_{\mathrm{odd}}(y) = g_{\mathrm{odd}}(y)$, then $f(\pm\sqrt{y}) = g(\pm\sqrt{y})$, so every joint-error position y contributes $c_y \ge 1$ error on L and at most 2. Let $E = |\{y \in L' : c_y \ge 1\}|$ denote the number of joint-error positions. Then $\Delta(f, g) = \sum_y c_y/n \le 2E/n = E/|L'| = \Delta_{\mathrm{joint}}$, where the cancellation uses $n = 2|L'|$. Taking $\min_g$ (using the isomorphism, the min over $\mathrm{RS}^{=2}_{k/2}$ equals the min over RS_k): $\Delta(f, \mathrm{RS}_k) \le \Delta_{\mathrm{joint}}$. □

Theorem 16 (Half-Threshold Proximity Gap)**.** *For even k and $\Delta(f, \mathrm{RS}_k) > \delta$:*

$$|\{\alpha \in F : \Delta(f_{\mathrm{even}} + \alpha f_{\mathrm{odd}}, \mathrm{RS}_{k/2}) \le \delta/2\}| \le 1.$$

(The Johnson/capacity window $\delta_J < \delta < 1-\rho$ is only needed when this theorem is composed downstream into the full FRI soundness theorem; see Theorem 20.)

Proof. By Lemma 15: $\Delta_{\mathrm{joint}}((f_{\mathrm{even}}, f_{\mathrm{odd}}), \mathrm{RS}^{=2}_{k/2}) \ge \Delta(f, \mathrm{RS}_k) > \delta$. The isomorphism $\mathrm{RS}_k \xrightarrow{\sim} \mathrm{RS}^{=2}_{k/2}$ (Remark 5) gives $\mathrm{RS}^{=2}_{k/2} = \mathrm{RS}_{k/2} \times \mathrm{RS}_{k/2}$, so $\Delta_{\mathrm{joint}}((f_{\mathrm{even}}, f_{\mathrm{odd}}), \mathrm{RS}_{k/2} \times \mathrm{RS}_{k/2}) > \delta$. Apply Theorem 6 to $(f_1, f_2) = (f_{\mathrm{even}}, f_{\mathrm{odd}})$ on $C = \mathrm{RS}_{k/2}$ over L', with α drawn from the challenge field F (typically an extension of the field containing L'; see Remark 2): at most 1 value of $\alpha \in F$ has $\Delta(f_{\mathrm{even}} + \alpha f_{\mathrm{odd}}, \mathrm{RS}_{k/2}) \le \delta/2$. □

6 FRI Soundness

Definition 17 (Standard FRI query model)**.** *The verifier samples q base-level points $z_1, \dots, z_q \in L$ **i.i.d. uniform with replacement**; for each z_j it queries the folding-tree path $z_j \to \pi_1(z_j) \to \pi_2(z_j) \to \cdots \to \pi_R(z_j)$, where $\pi_i : L \to L^{(i)}$ is the iterated squaring map, and at every round $i = 1, \dots, R$ checks the consistency relation $f^{(i)}(\pi_i(z_j)) = f^{(i-1)}_{\mathrm{even}}(\pi_i(z_j)) + \alpha_i \cdot f^{(i-1)}_{\mathrm{odd}}(\pi_i(z_j))$. (Equivalent statements for additive folding (App. B) and circle folding (App. C) replace π_i with the corresponding round-i projection.)*

Remark 18 (Implementation models: with-replacement vs. without-replacement)**.** *This paper's bounds are stated for the i.i.d. with-replacement model (Definition 17), which is standard in the FRI/STIR/WHIR analyses [20, 7, 10, 11]. Deployed implementations typically draw q distinct base-level indices via Fiat–Shamir hashing (without replacement, or with domain separation that makes collisions cryptographically negligible). The without-replacement miss probability $\binom{|L|-|\pi_i^{-1}(D_i)|}{q}/\binom{|L|}{q}$*

is bounded above by $(1-\tau)^q$ *for the same* $\tau = |D_i|/|L^{(i)}|$ *by negative association of without-replacement indicators (Joag-Dev–Proschan [30]; see also Dubhashi–Ranjan [29]), so the bounds in Theorems 20, 48 and Conjecture 62 apply unchanged to deployed protocols.*

Lemma 19 (Catch probability under i.i.d. base sampling)**.** *Under Definition 17, fix any round* i *and any deviation set* $D_i \subseteq L^{(i)}$ *with* $|D_i|/|L^{(i)}| \geq \tau$*. Then*

$$\Pr_{z_1,\dots,z_q \overset{\text{i.i.d.}}{\sim} L}\big[\,\pi_i(z_j) \notin D_i \text{ for all } j = 1,\dots,q\,\big] \;\leq\; (1-\tau)^q.$$

Proof. Let $X_j := \mathbf{1}\{\pi_i(z_j) \in D_i\}$. The map $\pi_i : L \to L^{(i)}$ is 2^i-to-1 (a composition of i squarings, each 2-to-1 on the relevant subgroup), so $|\pi_i^{-1}(D_i)| = 2^i \cdot |D_i|$ and

$$\Pr_{z_j \sim L}[X_j = 1] \;=\; \frac{2^i \cdot |D_i|}{2^i \cdot |L^{(i)}|} \;=\; \frac{|D_i|}{|L^{(i)}|} \;\geq\; \tau.$$

The variables $X_1,\dots,X_q$ are i.i.d. Bernoulli with parameter $\geq \tau$ because $z_1,\dots,z_q$ are i.i.d. and each $X_j = h(z_j)$ for the deterministic function $h(z) = \mathbf{1}\{\pi_i(z) \in D_i\}$. What matters is independence of the *base-level* samples z_j, not of the *induced* samples $\pi_i(z_j)$: the latter can collide (and are guaranteed to do so by pigeonhole in late rounds, when $|L^{(i)}| \leq q$), but the Bernoulli parameters $\Pr[X_j = 1]$ are identical across j, and the joint distribution factors as a product of Bernoullis regardless of induced collisions. Therefore

$$\Pr[X_1 = \cdots = X_q = 0] \;=\; \prod_{j=1}^{q} \Pr[X_j = 0] \;\leq\; (1-\tau)^q. \quad \square$$

Theorem 20 (FRI Soundness Above Johnson)**.** *For* $\mathrm{RS}[F, L, k]$ *with* $\mathrm{char}(F) \neq 2$*,* $k = 2^m$ *with* $m \geq 1$*,* L *a multiplicative subgroup or coset closed under negation with* $0 \notin L$ *and* $|L| = n = 2^d$ *for some* $d \geq m$ *(so that the radix-*2 *folding tower* $L^{(i)} = (L^{(i-1)})^2$ *is well-defined for* $R = m$ *rounds,* $|L^{(i)}| = n/2^i$ *at each round and remains non-trivial through round* R*, the squaring map is* 2*-to-*1 *onto* $L^{(i)}$ *(a multiplicative subgroup or coset of size* $n/2^i$*, closed under negation and not containing* 0 *— closure under negation requires* -1 *to be a square in the relevant subgroup, automatic when* $|F| \equiv 1 \pmod 4$ *and inherited at each squaring step), and* $x \mapsto -x$ *is a fixed-point-free involution at every level),* $\delta_J < \delta < 1-\rho$*, and* $\Delta(f^{(0)}, \mathrm{RS}_k) > \delta$*: FRI with* $R = m$ *rounds,* q *base-level queries (each opening one full path), and a final-round full read of* $f^{(R)} : L^{(R)} \to F$ *checking* $f^{(R)} \in \mathrm{RS}_{k/2^R}$ *(the standard FRI termination, with* $|L^{(R)}| = n/2^R$ *entries) satisfies*

$$\Pr[\text{FRI accepts}] \leq \frac{nR}{|F|} + (1-\delta/2)^q.$$

Proof. We partition all adaptive prover strategies into two cases according to whether the committed oracles match the honest fold. The prover sees α_i before committing $f^{(i)}$; the analysis below covers every such adaptive strategy.

Strategy A: the prover commits the honest fold at every round. In this case, $f^{(i)}$ is the function $f_{\text{even}}^{(i-1)} + \alpha_i \cdot f_{\text{odd}}^{(i-1)}$ (determined by $f^{(0)}$ and $\alpha_1,\dots,\alpha_i$). No adaptive freedom remains: the prover's only choice was to fold honestly, and the resulting $f^{(R)}$ is a *fixed* function of $f^{(0)}$ and $(\alpha_1,\dots,\alpha_R)$.

For any fixed evaluation point $y \in L^{(R)}$, $f^{(R)}(y)$ is *multilinear* in $(\alpha_1,\dots,\alpha_R)$ — individual degree ≤ 1 in each α_i (one round introduces each variable, each via the linear even/odd combination $f_{\text{even}}^{(i-1)} + \alpha_i \cdot f_{\text{odd}}^{(i-1)}$). Fix a parity-check basis of the dual code $\mathrm{RS}_{k/2^R}^{\perp} \subseteq F^{L^{(R)}}$, i.e., a basis of parity checks $s_j : F^{L^{(R)}} \to F$ whose common kernel is $\mathrm{RS}_{k/2^R}$ (each individual s_j has a strictly larger kernel; their joint kernel is the code); each s_j has the form $s_j(g) = \sum_{y \in L^{(R)}} c_{j,y} \cdot g(y)$ for some fixed coefficients $c_{j,y} \in F$ depending only on the dual-basis index j and on the parity-check matrix. Consequently, every syndrome $s_j(f^{(R)})$ is multilinear in $(\alpha_1, \ldots, \alpha_R)$.

Some syndrome is nonzero as a polynomial. Suppose all syndromes vanished identically (as polynomials in $\alpha_1, \ldots, \alpha_R$). Then $f^{(R)} \in \mathrm{RS}_{k/2^R}$ for *every* choice of $(\alpha_1, \ldots, \alpha_R) \in F^R$. Because $\mathrm{RS}_{k/2^R} \subset F^{L^{(R)}}$ is an F-linear subspace and the assignment $(\alpha_1, \ldots, \alpha_R) \mapsto f^{(R)}_{\alpha_1,\ldots,\alpha_R}$ is multilinear, every multilinear coefficient of $f^{(R)}$ in $(\alpha_1, \ldots, \alpha_R)$ also lies in $\mathrm{RS}_{k/2^R}$. We extract $f^{(0)} \in \mathrm{RS}_k$ from this universal hypothesis by a polynomial-identity argument. For each round $i = R, R{-}1, \ldots, 1$: by the multilinearity in α_i, evaluating at $\alpha_i = 0$ yields $f^{(i-1)}_{\mathrm{even}} \in \mathrm{RS}_{k/2^i}$, and the difference at $\alpha_i = 1$ minus $\alpha_i = 0$ yields $f^{(i-1)}_{\mathrm{odd}} \in \mathrm{RS}_{k/2^i}$ (each as a polynomial-valued function of $\alpha_1, \ldots, \alpha_{i-1}$, with $|F| \geq 2$ ensuring two distinct evaluation points; the same multilinear-coefficient argument runs coefficient-wise in the remaining variables, so each multilinear coefficient of $f^{(i-1)}_{\mathrm{even}}, f^{(i-1)}_{\mathrm{odd}}$ in $\alpha_1, \ldots, \alpha_{i-1}$ lies in $\mathrm{RS}_{k/2^i}$). The isomorphism of Remark 5 — which requires $k/2^{i-1}$ even, guaranteed by $k = 2^m$ — then gives $f^{(i-1)} \in \mathrm{RS}_{k/2^{i-1}}$. Iterating to $i = 1$ yields $f^{(0)} \in \mathrm{RS}_k$, contradicting the hypothesis. Hence there exists a syndrome s_j that is a nonzero multilinear polynomial.

Since s_j is multilinear in $\alpha_1, \ldots, \alpha_R$ (individual degree ≤ 1 in each variable), the DeMillo–Lipton–Schwartz–Zippel lemma in the individual-degree form (a nonzero polynomial with individual degree $\leq d_i$ in variable i vanishes on a uniformly random point with probability $\leq \sum_i d_i/|F|$) gives

$$\Pr_{\alpha_1,\ldots,\alpha_R \overset{\text{i.i.d.}}{\sim} F} [s_j(f^{(R)}) = 0] \;\leq\; R/|F|.$$

A fortiori, $\Pr[f^{(R)} \in \mathrm{RS}_{k/2^R}] \leq R/|F|$.

Strategy B: any prover strategy with at least one deviation. We give a single round-by-round argument that applies uniformly to every adaptive prover, regardless of how many rounds it deviates at and whether its commits are codewords or not. The argument tracks the honest-fold chain explicitly — the standard round-by-round (RBR) FRI soundness framework [20, 7, 37] and the form taken by Theorem 83 in §D.1.

Honest-fold reference sequence. Define

$$\hat{f}^{(0)} := f^{(0)}, \qquad \hat{f}^{(i)}(y) := \hat{f}^{(i-1)}_{\mathrm{even}}(y) + \alpha_i \cdot \hat{f}^{(i-1)}_{\mathrm{odd}}(y) \quad (y \in L^{(i)}).$$

This is a deterministic function of $f^{(0)}$ and the verifier's challenges $(\alpha_1, \ldots, \alpha_R)$; in particular $\hat{f}^{(i)}$ does *not* depend on the prover's subsequent commits $f^{(1)}, \ldots, f^{(R)}$.

Step 1 — Bad-challenge bound: $\leq nR/|F|$. By induction on i, we bound the probability that some $\hat{f}^{(i)}$ becomes $(\delta/2)$-close to $\mathrm{RS}_{k/2^i}$:

- *Round 1.* $\Delta(\hat{f}^{(0)}, \mathrm{RS}_k) = \Delta(f^{(0)}, \mathrm{RS}_k) > \delta$. By Theorem 16, at most 1 value of α_1 makes $\Delta(\hat{f}^{(1)}, \mathrm{RS}_{k/2}) \leq \delta/2$. Probability $\leq 1/|F|$.
- *Rounds $i = 2, \ldots, R$.* Inductively assume $\Delta(\hat{f}^{(i-1)}, \mathrm{RS}_{k/2^{i-1}}) > \delta/2$. By Remark 1, $\delta/2 < (1{-}\rho)/2 \leq \delta_J$, so the threshold $\gamma = \delta/2$ lies inside the BCIKS–BCHKS zero-loss regime. By Lemma 15, $\Delta_{\mathrm{joint}}((\hat{f}^{(i-1)}_{\mathrm{even}}, \hat{f}^{(i-1)}_{\mathrm{odd}}), \mathrm{RS}^{=2}_{k/2^i}) > \delta/2$, and the BCHKS [6] proximity gap at threshold $\gamma = \delta/2$ gives: at most n values of α_i make $\Delta(\hat{f}^{(i)}, \mathrm{RS}_{k/2^i}) \leq \delta/2$ (improving the BCIKS [7] bound of $O(n^2)$ to the proved $\leq n$). Probability $\leq n/|F|$.

Union over R rounds: $\Pr[\exists i : \Delta(\hat{f}^{(i)}, \mathrm{RS}_{k/2^i}) \leq \delta/2] \leq 1/|F| + (R{-}1)n/|F| \leq nR/|F|$.

Step 2 — Catch-probability bound: $\leq (1 - \delta/2)^q$. Condition on the good event from Step 1: $\Delta(\hat{f}^{(i)}, \mathrm{RS}_{k/2^i}) > \delta/2$ for every i.

If the prover commits $f^{(i)} = \mathrm{fold}(f^{(i-1)}, \alpha_i)$ at every round, then $f^{(i)} = \hat{f}^{(i)}$ at every round (induction on i from the common base $f^{(0)} = \hat{f}^{(0)}$), so $\Delta(f^{(R)}, \mathrm{RS}_{k/2^R}) > \delta/2 > 0$, $f^{(R)} \notin \mathrm{RS}_{k/2^R}$, and the final check rejects. The verifier therefore accepts only if the prover commits at least one $f^{(i)}$ that disagrees with the fold of its own previous commit (the verifier-checkable notion of deviation), in such a way that the resulting chain reaches $f^{(R)} \in \mathrm{RS}_{k/2^R}$.

Define the round-i *deviation set* (relative to the prover's own previous commit $f^{(i-1)}$, since this is what the verifier's consistency check examines):

$$B_i := \{y \in L^{(i)} : f^{(i)}(y) \neq f^{(i-1)}_{\mathrm{even}}(y) + \alpha_i \cdot f^{(i-1)}_{\mathrm{odd}}(y)\}, \qquad \beta_i := |B_i|/|L^{(i)}|.$$

A query $z_j \in L$ "catches" the prover at round i iff $\pi_i(z_j) \in B_i$, so the verifier rejects z_j iff $z_j \in T := \bigcup_{i=1}^{R} \pi_i^{-1}(B_i) \subseteq L$.

Path-consistency lemma. We invoke the standard FRI catch lemma in the form recorded by [37, Lemma 5.5 / honest-fold reference variant], which reformulates the original BBHR/BCIKS catch lemma in exactly the honest-fold-reference framework used here: under the conditions $f^{(R)} \in \mathrm{RS}_{k/2^R}$ and $\Delta(\hat{f}^{(R)}, \mathrm{RS}_{k/2^R}) > \delta/2$, the merged catching set $T = \bigcup_{i=1}^{R} \pi_i^{-1}(B_i) \subseteq L$ has relative measure at the base level satisfying $|T|/|L| \geq \delta/2$. The original statements [20, Lemma 4.1 / §6], [7, Lemma 8.2] bound the catching set relative to the prover's commit chain folded against *itself*; the conversion to the honest-fold reference $\hat{f}^{(j)}$ used here is the standard induction-on-rounds argument carried out in [37] and is what we cite. The underlying path-coverage step is: by induction on j, B_j contains every $y \in L^{(j)}$ where $f^{(j)}(y) \neq \hat{f}^{(j)}(y)$ except possibly via ancestor-collisions in $B_1, \ldots, B_{j-1}$; hence $T = \bigcup_j \pi_j^{-1}(B_j)$ at the base level covers every disagreement position between $f^{(R)}$ and $\hat{f}^{(R)}$, and counting paths through level-R disagreements gives the $\delta/2$ lower bound on $|T|/|L|$. We invoke the catch lemma as a black box; see the cited references for the detailed coverage argument.

Acceptance implies $f^{(R)} \in \mathrm{RS}_{k/2^R}$ (the verifier reads $f^{(R)}$ in full and rejects otherwise; see opening of Strategy B), so $\Pr[\text{accept}] = \Pr[\text{accept} \wedge f^{(R)} \in \mathrm{RS}_{k/2^R}]$. Conditioning on $f^{(R)} \in \mathrm{RS}_{k/2^R}$, the lower bound $|T|/|L| \geq \delta/2$ above applies, and the probability that all q i.i.d. base-level queries $z_1, \ldots, z_q \in L$ miss the merged set T is exactly $(1 - |T|/|L|)^q \leq (1 - \delta/2)^q$ (this is the elementary base-set miss bound, applied directly to $T \subseteq L$; we do not invoke Lemma 19, which is stated for sets of the form $\pi^{-1}(D)$ from a single round).

Combining Steps 1 and 2.

$$\Pr[\text{Strategy B succeeds}] \;\leq\; \frac{nR}{|F|} + (1 - \delta/2)^q.$$

This bound holds simultaneously for any number of deviation rounds (single or multi) and any commit pattern (codeword or non-codeword), since the honest-fold reference and the merged catching set are defined uniformly across all prover strategies.

Combining Strategies A and B. Every adaptive prover either folds honestly at every round (Strategy A) or deviates at some round (Strategy B):

$$\Pr[\text{accept}] \;\leq\; \max\Big(\tfrac{R}{|F|},\; \tfrac{nR}{|F|} + (1-\delta/2)^q\Big) \;\leq\; \tfrac{nR}{|F|} + (1-\delta/2)^q. \qquad \square$$

Remark 21 (Why $\delta/2$ does not compound across rounds)**.** *After round* 1 *reduces the effective distance from* δ *to* $\delta/2$*, the threshold does not halve again at later rounds. The chain* $\delta/2 < (1-\rho)/2 \leq \delta_J$ *(Remark 1) places* $\delta/2$ *strictly below the Johnson radius* $\delta_J = 1 - \sqrt{\rho}$*, and in fact strictly below the unique-decoding radius* $(1-\rho)/2$ *as well. From round* 2 *onward we apply the*

BCHKS [6] proximity gap at the equal threshold $\gamma = \delta/2 < \delta_J$, which bounds the number of bad challenges to $\leq n$ per round (improving the original BCIKS [7] bound of $O(n^2)$) and leaves the distance fixed at $\delta/2$. The two-threshold Theorem 6 is invoked only at round 1.

7 List-Size Landscape and OP2

The results in this section are independent of the FRI soundness theorem (Theorem 20), which uses only the half-threshold CA (Theorem 6) and classical list decoding. They address Open Problem 2 (§8): can one show $M = 0$ on all RS-compatible flats? We give the first p-dependent moment bound on the list size (Theorem 25 for $c = 1$; Corollary 30 for general c), prove the $M = 0$ conjecture is false at deployment code lengths (Proposition 35), and reduce the remaining gap to a worst-case-from-average problem.

Remark 22 (p-independent list-size bounds). *Classical bounds give $M \leq n^2/(t(t-1))$ for $k = 2$ (Fisher double-counting; see Sudan [15], Guruswami–Sudan [16] for general list decoding) and $M \leq \sum_{i=0}^{k-2c} \binom{n}{i}$ for general k (Frankl–Wilson [14] linear algebra method), where $c = n-k-w$ is the codimension excess. Both are independent of p and cannot capture the $p^{-\alpha}$ decay observed computationally [39]: for* $\mathrm{RS}[16, 8]$ *at $c = 1$, $M_{\max}(p) \approx 4200 \cdot p^{-3/4}$ across* 19 *primes from $p = 17$ to* 131.

7.1 Syndrome hyperplane structure and second moment

Classical p-independent bounds (Remark 22) cannot capture the $p^{-\alpha}$ decay observed computationally. The following results explain this decay via the hyperplane geometry of syndrome space and provide the first p-dependent moment bounds on the list size. The results hold for *all k*.

Fix $\mathrm{RS}[n, k]$ over $\mathbb{F}_p$ with $p > n$, evaluation domain $L = \{\alpha_1, \ldots, \alpha_n\}$, and codimension excess $c = 1$ (error weight $w = n - k - 1$, syndrome dimension $D = n - k$). For each w-subset $E \subset [n]$, the syndrome equations $\sum_{i \in E} e_i \alpha_i^j = s_j$ ($j = k, \ldots, n-1$) form a $D \times w$ Vandermonde system. With $D = w + 1$, the compatibility condition is a single hyperplane $H_E = \{s \in \mathbb{F}_p^D : \langle n_E, s \rangle = 0\}$, and $M(s) = |\{E \in \binom{[n]}{w} : s \in H_E\}|$.

Lemma 23 (Error-locator normal). *The normal vector of H_E is the coefficient vector of the error-locator polynomial:*

$$n_E = (\Lambda_{E,0}, \ldots, \Lambda_{E,D-1}), \qquad \Lambda_E(x) = \prod_{e \in E} (x - \alpha_e).$$

Proof. Since α_e is a root of Λ_E: $\sum_{\ell=0}^{w} \Lambda_{E,\ell}\, \alpha_e^{k+\ell} = \alpha_e^k \Lambda_E(\alpha_e) = 0$ for each $e \in E$. So the vector $(\Lambda_{E,0}, \ldots, \Lambda_{E,w})$ annihilates every column of the $D \times w$ Vandermonde submatrix V_E. Since V_E has rank w (distinct evaluation points, $p > n$), its left kernel is 1-dimensional and spanned by n_E. ☐

Lemma 24 (Pairwise independence). *For distinct w-subsets $E_1 \neq E_2$, the normals n_{E_1} and n_{E_2} are linearly independent over $\mathbb{F}_p$.*

Proof. Both Λ_{E_1} and Λ_{E_2} are monic of degree w. Monic polynomials of the same degree are proportional only if equal, requiring identical root sets: $E_1 = E_2$. ☐

Theorem 25 (Exact second moment of the list size). *For* $\mathrm{RS}[n, k]$ *over $\mathbb{F}_p$ with $p > n$ and $c = 1$, the list size M over uniform $s \in \mathbb{F}_p^D$ satisfies:*

$$\mathbb{E}[M] = \frac{\binom{n}{w}}{p}, \qquad \mathrm{Var}[M] = \mathbb{E}[M]\left(1 - \frac{1}{p}\right).$$

Proof. Write $M = \sum_E \mathbf{1}_{s\in H_E}$. Each indicator has $\Pr[s \in H_E] = 1/p$. For $E_1 \neq E_2$: by Lemma 24, $H_{E_1} \cap H_{E_2}$ has codimension 2 in $\mathbb{F}_p^D$, so $\Pr[s \in H_{E_1} \cap H_{E_2}] = 1/p^2 = \Pr[s \in H_{E_1}] \cdot \Pr[s \in H_{E_2}]$. The indicators are pairwise uncorrelated, giving $\mathrm{Var}[M] = \binom{n}{w} \cdot \frac{1}{p}(1 - \frac{1}{p})$. $\square$

Remark 26 (Implications for the list-size landscape)**.**
(i) Near-Poisson concentration. For large p: $\mathrm{Var}[M] \approx \mathbb{E}[M]$*, matching the Poisson dispersion relation. This justifies the heuristic that* M *concentrates around* $\binom{n}{w}/p$ *and is typically* 0 *once* $p > \binom{n}{w}$*. For* $\mathrm{RS}[16, 8]$ *at* $c = 1$*:* $\binom{16}{7} = 11440$*, so* $\mathbb{E}[M] < 1$ *for* $p \geq 11441$*. The power-law decay of* $M_{\max}$ *(see Remark 22) is substantially sharper than the first-moment prediction, suggesting algebraic structure beyond pairwise independence governs the worst case.*
(ii) The $c = 1$ *threshold and* $c \geq 2$ *generalization. At* $c = 1$ *(near capacity,* $w = n-k-1$*) with rate* $\rho = 1/2$*:* $\binom{n}{w} = \binom{n}{n/2-1} \sim 2^n/\sqrt{n}$*, so* $\mathbb{E}[M] < 1$ *requires* $p > 2^n/\sqrt{n}$ *— far beyond any practical field once* n *exceeds a few dozen. At the FRI-relevant distance* $\delta > \delta_J$ *with codimension* $c = n/2 - \lfloor \delta n \rfloor \gg 1$ *(for* $\rho = 1/2$*, just above Johnson gives* $c \approx 0.2n$*): Theorem 28 and Corollary 30 establish* $\mathbb{E}[M] = \binom{n}{w}/p^c$ *with pairwise independence for all pairs with* $|E_1 \cap E_2| \leq w - c$*, giving* $\mathrm{Var}[M] \approx \mathbb{E}[M]$ *(Poisson dispersion). At current code lengths,* $\mathbb{E}[M] = \binom{n}{w}/p^c \leq 2^{-5.3n} \approx 0$*.*
(iii) Average vs. worst case. By Markov: $\Pr_s[M \geq 1] \leq \mathbb{E}[M] = \binom{n}{w}/p^c$ *(Corollary 30; the* $c = 1$ *specialisation gives the previously stated* $\binom{n}{w}/p$*). However, §8, item 2, asks for* $M_{\max} = 0$ *over all syndromes (or all RS-compatible flats). The second-moment bound* $\mathrm{Var}[M] \leq \mathbb{E}[M]$ *does not improve on Markov for this worst-case question. Bridging the average-to-worst-case gap requires exploiting the algebraic structure of the error-locator hyperplane arrangement beyond pairwise independence: for example, showing that the* $\binom{n}{w}$ *hyperplanes, despite their number, avoid all RS-compatible flats when* p *is sufficiently large relative to* n*.*

7.2 Generalization to codimension excess $c \geq 2$

Theorem 25 treats codimension excess $c = 1$, where each w-subset E defines a single hyperplane H_E in syndrome space. For general $c \geq 2$ (error weight $w = n - k - c$, syndrome dimension $D = n - k$), the compatibility condition is $s \in W_E := \mathrm{Im}(V_E)$, the column span of the $D \times w$ Vandermonde submatrix V_E. Since V_E has rank w (MDS, $p > n$), W_E is a w-dimensional subspace of $\mathbb{F}_p^D$, equivalently a codimension-c subspace. Its orthogonal complement $W_E^\perp$ has dimension c and is the left null space of V_E. The following results extend the error-locator normal structure and pairwise independence to this regime. The case $c \geq 2$ is the *FRI-relevant* regime: at rate $\rho = 1/2$ and distance $\delta \approx 0.3$, the codimension excess is $c = n/2 - \lfloor 0.3n \rfloor \approx 0.2n$.

Lemma 27 (Generalized normal space)**.** *For codimension excess* $c \geq 1$*, the left null space of the* $D \times w$ *Vandermonde submatrix* V_E *has dimension* c *and is spanned by the coefficient vectors of*

$$\Lambda_E(x) \cdot x^r, \qquad r = 0, 1, \ldots, c - 1,$$

where $\Lambda_E(x) = \prod_{e\in E}(x - \alpha_e)$ *is the error-locator polynomial.*

Proof. Since α_e is a root of Λ_E, the polynomial $\Lambda_E(x) \cdot x^r$ vanishes at every α_e for $e \in E$. Its coefficient vector (as a polynomial of degree $w + r < D = w + c$, embedded in $\mathbb{F}_p^D$) therefore annihilates every column of V_E. The c polynomials $\{\Lambda_E \cdot x^r\}_{r=0}^{c-1}$ have degrees $w, w+1, \ldots, w+c-1$; since Λ_E is monic of degree w, these have leading terms $x^w, x^{w+1}, \ldots, x^{w+c-1}$, so they are linearly independent. The left null space of V_E has dimension $D - w = c$ (distinct evaluation points, $p > n$), and c linearly independent vectors span it. $\square$

Theorem 28 (Pairwise independence at $c \geq 2$)**.** *For distinct w-subsets $E_1 \neq E_2$ with $j = |E_1 \cap E_2| \leq w - c$: the $2c$ normal vectors from Lemma 27 (i.e., the coefficient vectors of $\{\Lambda_{E_1} x^r, \Lambda_{E_2} x^r : 0 \leq r < c\}$) are linearly independent over $\mathbb{F}_p$. Consequently:*

$$W_{E_1} \cap W_{E_2} \text{ has codimension } 2c, \qquad \Pr_s[s \in W_{E_1} \cap W_{E_2}] = \frac{1}{p^{2c}} = \Pr_s[s \in W_{E_1}] \cdot \Pr_s[s \in W_{E_2}].$$

The membership indicators $\mathbf{1}_{s \in W_{E_1}}$ and $\mathbf{1}_{s \in W_{E_2}}$ are pairwise uncorrelated for uniform s.

Proof. Suppose $\sum_{r=0}^{c-1} \big(a_r \Lambda_{E_1}(x) x^r + b_r \Lambda_{E_2}(x) x^r\big) = 0$ as polynomials of degree $< D = w + c$. Set $P(x) = \sum a_r x^r$ and $Q(x) = \sum b_r x^r$ (both of degree $< c$). Then $\Lambda_{E_1} P + \Lambda_{E_2} Q = 0$.

Let $G = \gcd(\Lambda_{E_1}, \Lambda_{E_2}) = \Lambda_{E_1 \cap E_2}$ of degree j, and write $\Lambda_{E_i} = G \cdot A_i$ where $\gcd(A_1, A_2) = 1$ and $\deg A_i = w - j$. Dividing by G: $A_1 P = -A_2 Q$. Since $\gcd(A_1, A_2) = 1$, we need $A_1 \mid Q$. But $\deg A_1 = w - j \geq w - (w - c) = c > \deg Q$, so $Q = 0$, hence $P = 0$.

The $2c$ normals are linearly independent, so $W_{E_1}^{\perp} + W_{E_2}^{\perp}$ has dimension $2c$. Since $\dim(W_{E_1}^{\perp}) = \dim(W_{E_2}^{\perp}) = c$, the intersection $W_{E_1} \cap W_{E_2}$ has codimension $2c$, giving exact independence. $\square$

Remark 29 (Sharpness of the threshold $j \leq w - c$)**.** *The threshold is sharp. For $j > w - c$, set $\ell = j - (w - c) \geq 1$. In the notation of the proof above, $\deg A_i = w - j = c - \ell$. For any polynomial R of degree $< \ell$, setting $Q = A_1 R$ (degree $< c$) and $P = -A_2 R$ (degree $< c$) gives $\Lambda_{E_1} P + \Lambda_{E_2} Q = G(-A_1 A_2 R + A_2 A_1 R) = 0$: a nontrivial relation among the $2c$ normals. The space of such relations has dimension $\ell = j - (w - c)$, so the rank drops from $2c$ to $2c - \ell$.*

In particular, at $j = w - c + 1$ ($\ell = 1$): rank $= 2c - 1$, verified computationally for all tested pairs [39].

Corollary 30 (Exact moments of the list size at $c \geq 2$)**.** *For $\mathrm{RS}[n, k]$ over $\mathbb{F}_p$ with $p > n$, codimension excess $c \geq 1$, and $w = n - k - c$, define $M = |\{E \in \binom{[n]}{w} : s \in W_E\}|$ for uniform $s \in \mathbb{F}_p^D$. Then:*

$$\mathbb{E}[M] = \frac{\binom{n}{w}}{p^c}.$$

For the variance, decomposing by intersection size $j = |E_1 \cap E_2|$:

$$\mathrm{Var}[M] = \underbrace{\sum_{j=0}^{w-c} \binom{n}{w}\binom{w}{j}\binom{n-w}{w-j} \cdot 0}_{\text{pairwise independent: } j \leq w-c} + \underbrace{\sum_{j=w-c+1}^{w} \binom{n}{w}\binom{w}{j}\binom{n-w}{w-j}\Big(\frac{1}{p^{2c-\ell(j)}} - \frac{1}{p^{2c}}\Big)}_{\text{correlated: } j > w-c}, \tag{4}$$

where $\ell(j) = j - (w - c) \in \{1, \ldots, c\}$ is the rank deficiency from Remark 29.

Proof. Write $M = \sum_E \mathbf{1}_{s \in W_E}$. Each indicator has $\Pr[s \in W_E] = 1/p^c$ since W_E has codimension c. For $\mathbb{E}[M]$: linearity of expectation gives $\binom{n}{w}/p^c$.

For the second moment: $\mathbb{E}[M^2] = \sum_{E_1, E_2} \Pr[s \in W_{E_1} \cap W_{E_2}]$. By Theorem 28, pairs with $j \leq w - c$ contribute $1/p^{2c}$ each — exactly the product of marginals, so their contribution to $\mathrm{Var}[M]$ is zero. Pairs with $j > w - c$ have $\ell(j) = j - (w - c)$ dependent normal vectors (Remark 29), so $W_{E_1} \cap W_{E_2}$ has codimension $2c - \ell(j)$, giving the stated excess probability $1/p^{2c - \ell(j)} - 1/p^{2c}$. $\square$

Corollary 31 (Expected true list size)**.** *Under the hypotheses of Corollary 30, the expected true list size (counting only supports whose Vandermonde error values are all nonzero) satisfies*

$$\mathbb{E}[M_{\text{true}}] = \frac{\binom{n}{w}(p-1)^w}{p^{w+c}} = \frac{\binom{n}{w}}{p^c}\Big(1 - \frac{1}{p}\Big)^w.$$

Proof. Write $M_{\text{true}} = \sum_E \mathbf{1}_{s \in W_E,\ \text{all } v_i \neq 0}$. For fixed E, the map $v \mapsto V_E v$ is a bijection $\mathbb{F}_p^w \to W_E$ (the $D \times w$ Vandermonde matrix V_E has rank w since $p > n$, so it has a w-dimensional image). Under uniform $s \in \mathbb{F}_p^D$: $\Pr[s \in W_E] = p^w/p^D = 1/p^c$. Conditional on $s \in W_E$, the unique preimage v with $V_E v = s$ is uniform in $\mathbb{F}_p^w$, so $\Pr[\text{all } v_i \neq 0 \mid s \in W_E] = ((p-1)/p)^w$. By linearity:

$$\mathbb{E}[M_{\text{true}}] = \binom{n}{w} \cdot \frac{1}{p^c} \cdot \left(\frac{p-1}{p}\right)^w = \frac{\binom{n}{w}(p-1)^w}{p^{w+c}}. \quad \square$$

Remark 32 (Poisson dispersion at general c)**.** *The variance decomposition* (4) *has a clean structure: (i) Diagonal dominance. The $j = w$ term (same subset, $\ell = c$) contributes $\binom{n}{w}(1/p^c - 1/p^{2c}) \approx \mathbb{E}[M]$ to* $\mathrm{Var}[M]$. *All other correlated terms ($w - c < j < w$) involve $\binom{w}{j}\binom{n-w}{w-j}$ which is polynomially bounded in n, while the excess probability $1/p^{2c-\ell} \leq 1/p^{c+1}$ decays exponentially in c. For $c \geq 2$ and $p \gg n$:* $\mathrm{Var}[M] \approx \mathbb{E}[M]$, *matching the Poisson dispersion relation exactly as in the $c = 1$ case. (ii) Average vs. worst case. The Poisson dispersion means that for random syndromes, M concentrates sharply around $\binom{n}{w}/p^c$. At deployment ($c \approx 0.2n$, $p = 2^{31}$): $\mathbb{E}[M] \leq 2^{-5.3n} \approx 0$, so $M = 0$ for overwhelmingly most syndromes. The worst case is governed by evaluation syndromes and the incidence bound; see Remark 33 for the precise structure.*

Remark 33 (Syndrome-compatible supports vs. true list size)**.** *The error-locator normal conditions (the key equations of Berlekamp) count supports E where $\langle \Lambda_E \cdot x^r, s\rangle = 0$ for $0 \leq r < c$, but do not verify that the recovered error values (the unique $v \in \mathbb{F}_p^w$ satisfying $V_E v = s_{0:w}$, when it exists) are all nonzero. Define:*

$$M_{\text{compat}}(s) = |\{E \in \tbinom{L}{w} : \langle \Lambda_E \cdot x^r, s\rangle = 0\ \forall\, r < c\}|,$$
$$M_{\text{true}}(s) = |\{E \in \tbinom{L}{w} : \textit{the unique } v \in \mathbb{F}_p^w \textit{ with } V_E v = s_{0:w} \textit{ exists, has all entries } \neq 0, \textit{ and reproduces all of } s\}$$

("Reproduces all of s" means the extension of the recovered w values via the parity-check matrix matches s on all $D = w + c$ syndrome coordinates, not just the first w.) The list-decoding list size is M_{true}; the count $M_{\text{compat}} \geq M_{\text{true}}$ overcounts by including supports where some error values vanish (actual error weight $< w$).

Evaluation syndromes: $M_{\text{compat}} = \binom{n-1}{w-1}$ but $M_{\text{true}} = 0$. Fix $\alpha \in L$ and set $s_\alpha := (1, \alpha, \alpha^2, \ldots, \alpha^{D-1})$. For every w-subset E and $0 \leq r < c$: $\langle \Lambda_E \cdot x^r, s_\alpha\rangle = \alpha^r \Lambda_E(\alpha)$, which vanishes iff $\alpha \in E$. Hence $M_{\text{compat}}(s_\alpha) = \binom{n-1}{w-1}$. However, for any $E \ni \alpha$: the Vandermonde system $V_E \cdot v = (1, \alpha, \ldots, \alpha^{w-1})^T$ has the unique solution $v_\alpha = 1$, $v_\beta = 0$ for $\beta \in E \setminus \{\alpha\}$ — a weight-1 error, not weight-w. Therefore $M_{\text{true}}(s_\alpha) = 0$ for $w \geq 2$. The $\binom{n-1}{w-1}$ "compatible" supports are all false positives: the error-locator conditions are satisfied, but the actual error has weight 1.

True list size: bounded by incidence, p-dependent for large n. Exhaustive computation over all $s \neq 0$ yields $\max_{s \neq 0} M_{\text{true}}(s)$*:*

n	w	c	d	$\max M_{\text{compat}}$	$\max M_{\text{true}}$	p *tested*
5	2	1	1	4	2	5, 7, 11, 13, 31
6	3	1	2	10	4	7, 11, 13
7	3	1	2	15	5	7, 11, 13
8	3	1	2	21	6	11, 13
6	3	2	1	10	2	7, 11
7	3	2	1	15	2	7, 11
8	3	2	1	21	2	11, 13
9	3	2	1	$\binom{8}{2}$	3 ($p \leq 23$); 2 ($p \geq 29$)	11–97
12	3	2	1	$\binom{11}{2}$	4 ($p \leq 17$); 3 ($p \geq 19$)	13–31
15	3	2	1	$\binom{14}{2}$	5 (p=17); M_∞=3	17; $\mathrm{rank}_{\mathbb{Q}}$
7	3	3	0	15	1	7, 11

Here $d := w - c$ and $\max M_{\text{true}} \le \binom{n}{d}/\binom{w}{d}$ (the incidence bound [24]) in every case. The table is generated by exhaustive enumeration over $\mathbb{F}_p^D$; scripts and outputs are in [39].

p-dependence. For $n \le 8$, $\max M_{\text{true}}$ is identical across all tested $p > n$. At $n \ge 9$, M_{true} becomes p-dependent: the $\mathbb{Q}$-rank of the combined normal matrix (Proposition 36) determines whether a configuration persists for all large p, or only for finitely many primes. The asymptotic value M_∞ (Definition 37) is computed in Remark 38.

Provable bounds.

(a) Unique decoding ($c \ge w$): $M_{\text{true}} \le 1$, since $\min \Delta(\mathrm{RS}[n, n-D]) = D+1 = w+c+1 > 2w$.
(b) Incidence bound ($c < w$): $M_{\text{true}} \le \binom{n}{d}/\binom{w}{d}$ with $d = w - c$ [24], tight at $d = 1$ for small p.
(c) Average: $\mathbb{E}[M_{\text{true}}] = \binom{n}{w}(1 - 1/p)^w/p^c$ (Corollary 31), exponentially small at deployment.

Consequence for the error-locator normal approach. The error-locator normal machinery (Theorems 25 and 28) gives tight average bounds ($\mathbb{E}[M] = \binom{n}{w}/p^c$) but operates on M_{compat}, which overcounts M_{true} by including false-positive supports. The worst-case M_{true} is bounded by the incidence bound (polynomial in n) and can depend on p, but this does not affect the FRI soundness theorem (Theorem 20), which uses only the half-threshold CA (Theorem 6).

Remark 34 (Coset extraction on multiplicative subgroups)**.** *When $L = \langle \omega \rangle$ and $k = 2$, the Vieta conditions on an agreement set S force the characteristic polynomial $\prod(x - \alpha_i)$ to be lacunary (only leading, sub-leading, and constant terms nonzero). For $n = 2^K$ and even t: $\gcd(t-1, n) = 1$, so the lacunary polynomial's root set on L is a single coset of the subgroup of order $\gcd(t-1, n)$ (Lidl–Niederreiter [13]; Golomb–Gong [18] Ch. 3–4), giving $M = O(1)$. The power-of-2 domain structure that FRI uses eliminates the coset-union complication entirely.*

7.3 OP2 obstruction and the average-to-worst-case gap

The strongest form of OP2 asks: does $M_{\max} = 0$ for all centers u when p is sufficiently large? The following elementary construction shows this is *false* at FRI-relevant code lengths.

Proposition 35 (List-size obstruction)**.** *For $\mathrm{RS}[n, k]$ over $\mathbb{F}_p$ with $p \ge 3$, integer Johnson radius $w_J := \lfloor n(1 - \sqrt{\rho}) \rfloor$ (where $\rho = k/n$), and integer codimension excess $c_J = n - k - w_J$: if $c_J \le \lfloor (n-k-1)/2 \rfloor$, then $M_{\max} \ge 2$ for all p.*

Proof. Let $c_1, c_2 \in \mathrm{RS}_k$ be a pair achieving the minimum distance $|\{i : c_1(i) \ne c_2(i)\}| = d_{\min} = n-k+1$, and let $S = \{i : c_1(i) \ne c_2(i)\}$ with $|S| = d_{\min}$. The hypothesis gives $2w_J \ge d_{\min}$, so we can choose $B_1, B_2 \subset S$ with $|B_1| = |B_2| = w_J$ and $B_1 \cup B_2 = S$. Define u by: $u = c_1 = c_2$ on $L \setminus S$; $u = c_2$ on $B_1 \setminus B_2$; $u = c_1$ on $B_2 \setminus B_1$; $u \notin \{c_1, c_2\}$ on $B_1 \cap B_2$ (feasible since $p \ge 3$). Then $|\{i : u(i) \ne c_1(i)\}| = |B_1| = w_J$ and $|\{i : u(i) \ne c_2(i)\}| = |B_2| = w_J$, so $M(u) \ge 2$. ☐

For rate $\rho = 1/2$: $c_J \approx 0.207n$ and $\lfloor (n-k-1)/2 \rfloor \approx n/4$. At $\rho = 1/2$ with n a power of 2 (the FRI setting), the obstruction applies for $n \ge 2^5 = 32$ (verified computationally up to $n = 64$ [39]; fails at $n = 16$ due to ceiling effects).

What this means. The strongest form of OP2 ($M_{\max} = 0$ for all p) is false at deployment-scale FRI. The average-case bounds ($\mathbb{E}[M] = \binom{n}{w}/p^c$, Corollary 30) and practical $M = 0$ at deployment parameters are unaffected. See §8, item 2, for the precise remaining questions.

7.4 Asymptotic stabilization of M_{true}

The following result explains the p-dependent behavior observed in Remark 33: for large p, the worst-case list size stabilizes at a value determined by the rational rank of a certain normal matrix.

Proposition 36 ($\mathbb{Q}$-rank characterization)**.** *Fix n, w, c with $1 \le c < w \le n$ and $D = w + c$. To avoid notational conflict with the code dimension k, we use t for the support count throughout this proposition and the definition of M_∞. Let $E_1, \ldots, E_t$ be pairwise-compatible w-subsets of $L = \{0, 1, \ldots, n-1\}$ (i.e., $|E_i \cap E_j| \le w - c$ for $i \ne j$). Form the combined normal matrix*

$$\mathcal{N} = \begin{pmatrix} \Lambda_{E_1} \cdot x^0 \\ \vdots \\ \Lambda_{E_1} \cdot x^{c-1} \\ \hline \Lambda_{E_2} \cdot x^0 \\ \vdots \\ \Lambda_{E_t} \cdot x^{c-1} \end{pmatrix} \in \mathbb{Z}^{tc \times D},$$

where row $\Lambda_{E_i} \cdot x^r$ is the coefficient vector of $\Lambda_{E_i}(x) \cdot x^r$ in the basis $\{1, x, \ldots, x^{D-1}\}$.

(a) Upper bound. If $\operatorname{rank}_{\mathbb{Q}}(\mathcal{N}) = D$, then for all primes p not dividing any $D \times D$ minor of $\mathcal{N}$, the intersection $\bigcap_{i=1}^{t} W_{E_i} = \{0\}$ over $\mathbb{F}_p$. In particular, no nonzero syndrome can simultaneously have all t supports in its list, so $M_{\mathrm{true}}(s) < t$ for every $s \ne 0$.

(b) Lower bound. Fix any integer basis of $\ker_{\mathbb{Q}}(\mathcal{N})$ obtained by clearing denominators; this gives a $\delta \times D$ integer matrix K with $\operatorname{rank}_{\mathbb{Q}}(K) = \delta$. Let $\mathcal{S}(\mathcal{N})$ denote the finite set of primes dividing the Smith-normal-form invariants of K (equivalently, primes dividing some nonzero $\delta \times \delta$ minor of K); $\mathcal{S}(\mathcal{N})$ depends only on $\mathcal{N}$. If $\operatorname{rank}_{\mathbb{Q}}(\mathcal{N}) < D$ and the configuration is nondegenerate (see proof), then for all primes $p > \max(n, tw)$ with $p \notin \mathcal{S}(\mathcal{N})$, the rank-drop lifts to $\mathbb{F}_p$ (the integer kernel basis remains linearly independent mod p), and there exists $s \in \mathbb{F}_p^D \setminus \{0\}$ such that $s \in W_{E_i}$ and the unique $v_i \in \mathbb{F}_p^w$ with $V_{E_i} v_i = s_{0:w}$ has all entries nonzero, for every $i = 1, \ldots, t$. Hence $M_{\mathrm{true}}(s) \ge t$.

Proof. (a) If $\operatorname{rank}_{\mathbb{Q}}(\mathcal{N}) = D$, then $\mathcal{N}$ has a $D \times D$ submatrix with nonzero determinant $\Delta \in \mathbb{Z}$. For any prime $p \nmid \Delta$: $\operatorname{rank}_{\mathbb{F}_p}(\mathcal{N}) = D$, so the only $s \in \mathbb{F}_p^D$ satisfying $\mathcal{N}s = 0$ is $s = 0$. Since $s \in W_{E_i}$ iff the c normals of E_i annihilate s, the intersection $\bigcap W_{E_i} = \ker(\mathcal{N}) = \{0\}$.

(b) If $\operatorname{rank}_{\mathbb{Q}}(\mathcal{N}) = D - \delta$ with $\delta \ge 1$, the rational kernel $K = \ker_{\mathbb{Q}}(\mathcal{N})$ has dimension δ. Clearing denominators gives δ linearly independent integer vectors in $\ker(\mathcal{N})$. For any prime p larger than the largest prime factor of the denominators (which depends only on n), these reduce to δ independent vectors in $\ker_{\mathbb{F}_p}(\mathcal{N})$, so $\dim_{\mathbb{F}_p}(\bigcap W_{E_i}) \ge \delta$.

It remains to show that some s in the kernel gives all-nonzero recovered error vectors. For each support E_i, on $\bigcap W_{E_j}$ the unique $v_i \in \mathbb{F}_p^w$ with $V_{E_i} v_i = s_{0:w}$ is a linear function of s; write $v_i(s)$. Call the configuration *nondegenerate* if for every (i, j), the coordinate functional $s \mapsto (v_i(s))_j$ does not vanish identically on $\bigcap W_{E_j}$. When nondegenerate: each of the tw "bad" sets $\{s : (v_i(s))_j = 0\}$ is a proper hyperplane in the p^δ-element kernel, and the union of tw proper hyperplanes omits at least $p^\delta - tw \cdot p^{\delta-1} = p^{\delta-1}(p - tw) > 0$ points when $p > tw$. Nondegeneracy is verified computationally for all configurations in Remark 38; see [39]. ☐

Definition 37 (Asymptotic worst-case list size)**.**

$M_\infty(n, w, c) := \max\big\{t : \exists\, t$ *pairwise-compatible w-subsets of $[n]$ with* $\operatorname{rank}_{\mathbb{Q}}(\mathcal{N}) < D$ *and the configuration nondeg*

By Proposition 36 (applied to nondegenerate configurations), $M_{\mathrm{true}}(n, w, c, p) = M_\infty(n, w, c)$ for all sufficiently large primes p outside the finite exceptional set in part (b) of that proposition.

Remark 38 (Computed values of M_∞)**.** *For $w = 3$, $c = 2$ ($d = 1$):*

n	M_∞	$\lfloor n/3 \rfloor$	*rank-deficient triples*	*verification*
≤ 8	*2*	≤ 2	*0*	*exhaustive, nondegenerate*
9–10	*2*	*3*	*0*	*exhaustive; 280 (n=9), 2800 (n=10)*
11	*3*	*3*	*1 of 15400*	*nondeg.; no rank-def. 4-tuple*
12	*3*	*4*	*2 of 61600*	*nondeg.; no rank-def. 4-tuple*
13	*3*	*4*	*4 of 200200*	*no rank-def. 4-tuple*
14	*3*	*4*	*6 of 560560*	*no rank-def. 4-tuple*
15	*3*	*5*	*18 of 1401400*	*no rank-def. 4-tuple*

Rank-deficient triples include both translates of

$$\mathcal{C}_a = \{(a,\ a{+}2,\ a{+}7),\ \ (a{+}3,\ a{+}8,\ a{+}10),\ \ (a{+}4,\ a{+}5,\ a{+}6)\}$$

and non-translate configurations (e.g., $\{(0,7,8), (2,6,10), (4,5,12)\}$ at $n = 13$; 13 of 18 configurations at $n = 15$ are non-translates).

Algebraic obstruction to $M_\infty \ge 4$. For any rank-deficient triple, the kernel of the 6×5 normal matrix is one-dimensional. A fourth disjoint triple $\{d, e, f\}$ preserves rank 4 iff all three roots satisfy a compatibility cubic. Polynomial division reduces integer compatibility to a divisibility condition $(Q(u)) \mid (R(u))$ after centering $u = d - \bar{d}$. For the translate family: $(u^2 - 5) \mid 4u$ (independent of a), yielding exactly the 9 used points as the only integer solutions. Verified computationally [39]: for every rank-deficient triple at $n \le 15$ (including all non-translate configurations), the analogous divisibility condition has solutions only among the 9 used points.

No rank-deficient 4-tuple was found in exhaustive search up to $n = 15$ or in 10,000 random trials at $n \in \{18, 21, 30\}$ [39]. Conjecture: $M_\infty(n, 3, 2) = 3$ for all $n \ge 11$ and $M_\infty = 2$ for $6 \le n \le 10$.

Additional structural results — fiber bounds, pinned-flat uniqueness, Bézout bounds, codimension bounds ($\dim V(r_0{-}1, r_1, r_2) \le w{-}3$), incidence bounds ($M \le \binom{n}{d}/\binom{w}{d}$), and irreducibility of the remainder hypersurface — are developed in a companion note [24] and provide the correct order-of-magnitude picture of M as a function of field size and code parameters.

7.5 Möbius structure and worst-case bound at $c = 2$

The FRI-relevant case is codimension excess $c = 2$ at rate $\rho = 1/2$ (so $w = n/2 - 2$). Here a clean algebraic structure controls how M_{true} depends on the syndrome line. Fix $s_1, s_2 \in \mathbb{F}_p^D$ defining a line $s(\gamma) = s_1 + \gamma s_2$. Let $W_E := \mathrm{Im}(V_E) \subseteq \mathbb{F}_p^D$ be the image of the (rectangular) $D \times w$ Vandermonde matrix attached to E. When $s(\gamma) \in W_E$ there is a unique $v \in \mathbb{F}_p^w$ with $V_E v = s(\gamma)$ (since V_E has full column rank); we then say the E-induced error vector at γ has all entries nonzero iff every coordinate of this v is nonzero. Define

$$M_{\text{true}}(s_1, s_2) := \#\{\gamma \in \mathbb{F}_p : \exists\, E \in \tbinom{L}{w} \text{ s.t. } s(\gamma) \in W_E \text{ and the unique } v \in \mathbb{F}_p^w \text{ with } V_E v = s(\gamma) \text{ has all entries } \neq 0\}$$

Lemma 39 (Möbius structure on cores). *Let $C \subset L$ with $|C| = w - 1$ (a core) and $E = C \cup \{x\}$ for $x \in L \setminus C$. As x varies, the unique γ_E for which the first compatibility equation $\langle \Lambda_E, s(\gamma) \rangle = 0$ holds is a degree-1 rational function (Möbius transform) of the evaluation point α_x:*

$$\gamma_E \;=\; \frac{A_C \cdot \alpha_x + B_C}{C_C \cdot \alpha_x + D_C}$$

for constants $A_C, B_C, C_C, D_C \in \mathbb{F}_p$ depending on (s_1, s_2) and C but not on x.

Proof. Factor $\Lambda_E(z) = \Lambda_C(z) \cdot (z - \alpha_x)$. The coefficient of z^j in Λ_E is $\varepsilon_{j-1}(C) - \alpha_x \cdot \varepsilon_j(C)$, where $\varepsilon_j(C)$ are the coefficients of Λ_C (with $\varepsilon_{-1}(C) := 0$). Hence the coefficient vector of Λ_E is linear in α_x. For any $s \in \mathbb{F}_p^D$, $\langle \Lambda_E, s\rangle = \sum_j \varepsilon_{j-1}(C)s_j - \alpha_x \sum_j \varepsilon_j(C)s_j$ is linear in α_x. Setting $\langle \Lambda_E, s_1 + \gamma s_2\rangle = 0$ and solving for γ gives a ratio of two linear functions of α_x. □

For a fixed line $s(\gamma) = s_1 + \gamma s_2$, define

$$M_{\text{compat}}(s_1, s_2) := \#\big\{(\gamma, E) \in \mathbb{F}_p \times \tbinom{L}{w} : \langle \Lambda_E, s(\gamma)\rangle = 0 \text{ and } \langle \Lambda_E \cdot x, s(\gamma)\rangle = 0\big\}$$

(the total number of (γ, E) pairs satisfying both error-locator normal conditions at $c = 2$, with no requirement that the corresponding error values be nonzero). The list-size $M_{\text{true}}(s_1, s_2) \leq M_{\text{compat}}(s_1, s_2)$ since M_{true} additionally requires the Vandermonde solution to have all entries nonzero.

Theorem 40 (Worst-case bound at $c = 2$)**.** *At codimension excess $c = 2$ over any $\mathbb{F}_p$ with $p > n$,*

$$M_{\text{true}}(s_1, s_2) \;\leq\; M_{\text{compat}}(s_1, s_2) \;\leq\; \min\!\Big(p,\; 2\binom{n}{w-1}\Big)$$

for every line (s_1, s_2).

Proof. A pair (E, γ) contributes to M_{compat} iff $\langle \Lambda_E, s(\gamma)\rangle = 0$ *and* $\langle \Lambda_E \cdot x, s(\gamma)\rangle = 0$ (the two normals at $c = 2$). Decompose $E = C \cup \{x\}$ with $|C| = w - 1$ and $x \in L \setminus C$. Both inner products are bilinear in (α_x, γ): linear in α_x by the shift identity used in Lemma 39, linear in γ by the line parameterization. Eliminating γ between the two bilinear equations yields a polynomial in α_x of degree ≤ 2.

Two cases. If this polynomial is not identically zero, it has at most two roots, so at most two values of α_x (and hence at most two pairs (α_x, γ_E)) arise from this core. Summing over $\binom{n}{w-1}$ cores: $M_{\text{compat}} \leq 2\binom{n}{w-1}$. If the polynomial vanishes identically, the line $\ell(\gamma) = s_1 + \gamma s_2$ lies inside the union $\bigcup_{x \in L\setminus C} W_{C\cup\{x\}}$, but $M_{\text{compat}} \leq p = |\mathbb{F}_p|$ holds trivially. Either way the stated bound holds. Since $M_{\text{true}} \leq M_{\text{compat}}$, the bound passes to M_{true}. □

The bound complements the incidence bound: $M_{\text{true}} \leq \min(p,\ 2\binom{n}{w-1},\ \binom{n}{d}/\binom{w}{d})$ with $d = w - c$. The $2\binom{n}{w-1}$ factor is loose at small n (e.g. $n = 15$, $w = 6$, $c = 2$: bound 210 vs. empirical $\max M_{\text{true}} = 5$ [39]) but it is the only p-independent upper bound currently known at the FRI-relevant codimension $c = 2$ that is universal in (s_1, s_2).

Sub-Poisson concentration on bin sizes

Theorem 40 bounds the total $M_{\text{compat}}(s_1, s_2) \leq 2\binom{n}{w-1}$ summed over all $\gamma \in \mathbb{F}_p$. The per-bin counts $M_\gamma(s_1, s_2) := |\{E \in \binom{L}{w} : \gamma_E(s_1, s_2) = \gamma\}|$ inherit the trivial maximum bound $M_\gamma \leq M_{\text{compat}}/1 \leq 2\binom{n}{w-1}$, but this ignores the negative correlation induced by Lemma 39. A sharper *high-probability* bound (over uniform $(s_1, s_2) \in \mathbb{F}_p^D \times \mathbb{F}_p^D$) recovers the birthday rate.

Conjecture 41 (Sub-Poisson tail on M_γ at $c = 2$, with conditional / sketch-level proof)**.** *Fix $\rho = k/n$, $w = n - k - 2$, $D = n - k$. Conditional on the proof-sketch hypotheses below (in particular, the regime where $\mu = \binom{n}{w}/p$ satisfies $\mu \gtrsim \log p$ and the Möbius-leaf indicators have negative dependence beyond pairwise covariance), for any prime p in this regime, with probability $\geq 1 - 1/p$ over uniform $(s_1, s_2) \in \mathbb{F}_p^D \times \mathbb{F}_p^D$:*

$$\max_{\gamma \in \mathbb{F}_p} M_\gamma(s_1, s_2) \;\leq\; \frac{\binom{n}{w}}{p} \;+\; 4\sqrt{2\log(p) \cdot \frac{\binom{n}{w}}{p}}.$$

Combined with $M_\gamma \le p$ trivially, in the same probability event:

$$\max_{\gamma\in\mathbb{F}_p} M_\gamma(s_1,s_2) \;\le\; C_0\cdot\sqrt{\binom{n}{w}\cdot\log p}, \qquad C_0\le 16.$$

Proof sketch. Set $X_E := \mathbf{1}[\gamma_E(s_1,s_2)=\gamma]$, with marginal $\Pr_s[X_E=1]=1/p$ (Lemma 39 expresses $\gamma_E(s_1,s_2)$ as a fractional-linear function of the leaf coordinate α_x; on the open dense set where the denominator is non-zero this map is generically a bijection on $\mathbb{F}_p$, so for fixed core C and uniform (s_1,s_2) the value γ_E is uniform on $\mathbb{F}_p$ up to a degenerate locus of measure $\le 1/p$). Pairs (E_1,E_2) classify by overlap r: at $r\le w-2$, the four normals are linearly independent (Theorem 28), so $\mathrm{Cov}_s(X_{E_1},X_{E_2})=0$; at $r=w-1$, the Möbius bijectivity excludes equal images from distinct leaves except on a degenerate-Möbius locus of measure $\le 1/p$, giving $\mathrm{Cov}_s(X_{E_1},X_{E_2})\le 0$. Hence $\{X_E\}$ are pairwise non-positively correlated indicators with $\mathrm{Var}[M_\gamma]\le\mathbb{E}[M_\gamma]=\binom{n}{w}/p=:\mu$.

The standard sub-Poisson tail bound for indicators with non-positive pairwise covariance (Janson [28], Dubhashi–Ranjan [29]; the Bernstein argument requires only the non-positive correlation hypothesis) gives

$$\Pr_s\big[M_\gamma\ge\mu+t\big] \;\le\; \exp\!\left(-\frac{t^2}{2(\mu+t/3)}\right).$$

Setting $t=4\sqrt{2\log(p)\cdot\mu}$ and applying a union bound over $\gamma\in\mathbb{F}_p$:

$$\Pr_s\big[\max_\gamma M_\gamma\ge\mu+t\big] \;\le\; p\cdot\exp\!\left(-\frac{16\cdot 2\log p\cdot\mu}{2(\mu+(4/3)\sqrt{2\log p\cdot\mu})}\right) \;\le\; \frac{1}{p}$$

for $\mu\ge\log p$, which holds for $p>p_{\min}(n)$ as stated. The combined bound with $M_\gamma\le p$ follows from the AM–GM step $\min(\mu,\sqrt{\binom{n}{w}\log p})\le\sqrt{\binom{n}{w}\log p}$. Full proof in companion note [26]. □

The high-probability bound matches the empirical $0.66\cdot 1.36^n$ envelope (Remark 42) up to $\sqrt{\log p}$ factors. A *worst-case* extension over all (s_1,s_2) is open (§8, the birthday-bound conjecture); the natural higher-moment Markov approach is structurally blocked by configurations where three supports share a common $(w-c)$-element core (companion note [26], script in [39]).

Remark 42 (Empirical exponential growth at the worst case)**.** *At $c=2$ and $w=n/2-2$ (the FRI-relevant regime), exhaustive search over (s_1,s_2) pairs, prime fields, and supports up to $n=24$ [39] gives an empirical fit*

$$\max_{s_1,s_2,p} M_{\mathrm{true}}(s_1,s_2;\,p) \;\approx\; c_0\cdot c_1^n$$

with constants $c_0\approx 0.66$, $c_1\approx 1.36$ in the tested range $n\in\{8,10,\ldots,24\}$ (OLS regression of $\log_2 M_{\mathrm{true}}$ on n over 9 data points, $R^2>0.99$). We report these as point estimates descriptive of the tested range only — with 9 data points spanning 16 in the exponent, the 95% confidence interval on the slope $\log_2 c_1$ is non-trivial and we make no claim about population constants. The peak prime tracks $p_{\mathrm{peak}}\approx\sqrt{\binom{n}{w}}$, the regime where bin sizes (the number of supports projecting to each $\mathbb{F}_p$ value of the linear part of γ) become $\Theta(\sqrt{\binom{n}{w}})$ on average and concentrate within a constant factor of this average. The data are consistent with M_{true} at $c=2$ growing super-polynomially in n, but a rigorous super-polynomial worst-case lower bound from the data alone would require either a longer scaling range or a complementary analytic argument; this remark records the scaling shape on the tested range, not an asymptotic claim. The corresponding ratio M_{true}/p, the FRI-relevant quantity, is observed to decay in the tested range ($M_{\mathrm{true}}/p\approx 0.676^n$ at $p=p_{\mathrm{peak}}$).

7.6 Phase diagram across codimension excess c

Theorem 40 and Remark 42 characterize the worst-case list size M_{true} at $c = 2$. The picture across all c is a clean three-phase diagram:

- $c = 1$ *(covering radius)*: $\mathbb{E}[M] = \binom{n}{w}/p$ with Poisson dispersion (Corollary 30); the trivial counting bound $M_{\text{true}} \le \binom{n}{w}$ holds. The worst-case behaviour at large n is empirically observed to saturate but a closed-form upper bound is open at $c = 1$.
- $c = 2$: empirical fit $M_{\text{true}} \approx c_0 c_1^n$ with $c_0 \approx 0.66$, $c_1 \approx 1.36$ in the tested range $n \le 24$ (Remark 42; descriptive, not an asymptotic claim); governed by the Möbius coupling on $(w-1)$-cores (Lemma 39).
- $c \ge 3$: empirically *linear* in n at sufficiently large p in the tested range $n \le 40$, compatible with the conjectural bound $M_{\text{true}} \le \lfloor (2D-1)/c \rfloor$ where $D = n - k$ (Conjecture 43).

The deployment-relevant regime for FRI is $c = \Theta(n)$ (specifically $c \approx 0.2n$ above the Johnson radius at rate $1/2$), so the third phase is the one that controls FRI soundness analysis.

Conjecture 43 (Open-Set Rank Lemma at $c \ge 3$)**.** *For supports $E_1, \dots, E_m \subset L$ each of size $w = D - c$ with $c \ge 3$, and distinct $\gamma_1, \dots, \gamma_m \in \mathbb{F}_p^*$, let $A \in \mathbb{F}_p^{mc \times 2D}$ be the constraint matrix with row blocks $[N_{E_i} \mid \gamma_i N_{E_i}]$, where $N_{E_i} \in \mathbb{F}_p^{c \times D}$ collects the c error-locator normals from E_i (Lemma 27); write $n_0(E_i) \in \mathbb{F}_p^D$ for the first row of N_{E_i} (the coefficient vector of Λ_{E_i} itself). For all p above an effective threshold $p_0(n, k, c)$ (polynomial in n):*

$$\textit{either} \quad \operatorname{rank}(A) = \min(mc, 2D), \quad \textit{or} \quad \exists\, i \textit{ with } \langle n_0(E_i), s_2 \rangle = 0 \textit{ for all } (s_1, s_2) \in \ker A.$$

Equivalently, every (s_1, s_2) with $M_{\text{true}}(s_1, s_2) = m$ satisfies $m \le \lfloor (2D-1)/c \rfloor$.

Remark 44 (Empirical evidence and small-p counterexamples)**.** *Conjecture 43 has been tested without counterexamples at large p for (n, k) up to $n = 40$ with c up to 9 via random-syndrome search [27, 39]; selected data points (rate $1/2$, ≥ 14000 random configs each):*

n	k	c	*bound* $\lfloor (2D-1)/c \rfloor$	*observed* $\max M_{\text{true}}$	p *tested*
20	*10*	*5*	*3*	≤ 2	*large*
24	*12*	*5*	*4*	≤ 2	*large*
28	*14*	*6*	*4*	≤ 4	*large*
32	*16*	*7*	*4*	≤ 4	*large*
36	*18*	*8*	*4*	≤ 4	*large*
40	*20*	*9*	*4*	≤ 4	*large*

The observed M_{true} is constant (≤ 4) at the Johnson-radius codimension for rate $1/2$, well within the $O(n/c)$ envelope.

At small primes, Conjecture 43 can fail due to mod-p algebraic coincidences. Two structural counterexamples are known:

- *Triangle obstruction at $c = 2$. For $n = 8$, $w = 2$, $p = 113$: the four supports $\{(3,6), (5,6), (3,5), (0,1)\}$ realize $M_{\text{true}} = 4$, exceeding the would-be bound $\lfloor (2D-1)/c \rfloor = 3$. The first three supports form a K_3 on vertices $\{3, 5, 6\}$, providing a row dependency that the fourth disjoint support extends. Consistent with the $c = 2$ phase being exponential, not linear.*
- *Tetrahedron obstruction at $c = 3$, small p. For $n = 12$, $w = 3$, $p = 61$: the four supports $\{(1,4,5), (1,4,8), (4,5,8), (1,5,8)\}$ are exactly the size-3 subsets of $\{1, 4, 5, 8\}$, forming a K_4 tetrahedron, and realize $M_{\text{true}} = 4 > \lfloor (2D-1)/c \rfloor = 3$. The lemma fails at this small p but holds for $p > p_0(n, k, c)$.*

The general obstruction at general c is the $(w{+}1)$-clique — all size-w subsets of a $(w{+}1)$-vertex set. These configurations are realizable only at primes below an explicit threshold; the conjecture asserts an effective Schwartz–Zippel-style bound on this threshold.

The connection to FRI: the FRI verifier accepts a low-distance codeword on the evaluation domain only when the syndrome line hits one of M_{true} many error patterns. At FRI scale, $c \approx 0.2n$ exceeds 3, putting us in the third phase of the diagram. At fixed rate, $\lfloor (2D-1)/c \rfloor$ is a *constant* in n (equal to 4 at rate 1/2 near the Johnson radius); Conjecture 43 therefore predicts $M_{\text{true}} = O(1)$ uniformly in code length at current parameters, consistent with the empirical row of the table.

8 Open Problems

We list open problems in order of deployment relevance.

1. **Beating the 1:2 ratio via ≥ 3 bad γ's or non-CA arguments.**

   Status of the standard two-bad-γ CA framework (settled in this paper). Theorem 6 proves $\varepsilon_{\text{ca}}(C, \delta/2, \delta) \leq 1/|F|$ for any linear code, and the 1:2 ratio is optimal for the standard two-bad-γ argument (Remark 14). At *equal* threshold ($\delta_{\text{hd}} = \delta$), Theorem 8 gives the unconditional upper bound $\varepsilon_{\text{ca}}(C, \delta, \delta) \leq \binom{n}{w}/|F|$, matched at the equality case $w = n{-}k{-}1$ for $|F| > \binom{n}{w}^2$ by Proposition 10 (RS code). Since $\binom{n}{w} \sim 2^{nH(\delta)}$ grows exponentially in the code length, this upper bound is vacuous against any meaningful security target at FRI scale.

   The open question is whether a strictly different argument can beat 1:2. Two natural candidates:

   - ≥ 3 *bad γ in the CA framework.* Three bad γ's give an overdetermined system (three equations in two unknowns (f_1, f_2)), yielding polynomial constraints on h_3 given h_1, h_2. The additional constraint does not tighten the standard inclusion-exclusion bound beyond the two-γ result, but tighter analyses tracking the algebraic relations among h_1, h_2, h_3 are not ruled out.
   - *Non-CA proximity proofs* bypassing the CA framework entirely, pursued in concurrent companion work in preparation [2, 3]. Specific quantitative claims await those papers' completion and are not asserted here.

2. **Worst-case M_{true}: structure and bounds.** Proposition 35 shows $M_{\max} \geq 2$ at FRI parameters ($n \geq 32$, rate 1/2) for all p; the strongest form of the $M = 0$ conjecture is false. The average-case bounds $\mathbb{E}[M] = \binom{n}{w}/p$ (Theorem 25) for $c = 1$ and $\mathbb{E}[M] = \binom{n}{w}/p^c$ (Corollary 30) for general c, both with Poisson dispersion $\mathrm{Var}[M] \approx \mathbb{E}[M]$ (via pairwise independence of error-locator normals, Theorem 28), show $M = 0$ for a random syndrome once $p^c > \binom{n}{w}$. At deployment ($c \approx 0.2n$, $p = 2^{31}$): $\mathbb{E}[M] \leq 2^{-5.3n} \approx 0$.

   Evaluation syndromes are false positives. Remark 33 shows that the error-locator normal *compatibility* count M_{compat} reaches $\binom{n-1}{w-1}$ at evaluation syndromes $s_\alpha = (1, \alpha, \ldots, \alpha^{D-1})$. But these $\binom{n-1}{w-1}$ "compatible" supports all produce weight-1 errors ($v_\alpha = 1$, $v_\beta = 0$ for $\beta \neq \alpha$), so $M_{\text{true}}(s_\alpha) = 0$ for $w \geq 2$. The $\binom{n-1}{w-1}$ is a failure of the error-locator normal machinery to enforce non-zero error values, not a genuine list-size obstruction.

   Provable bounds on M_{true}. At $c \geq w$ (within the unique decoding radius): $M_{\text{true}} \leq 1$. At $c < w$: $M_{\text{true}} \leq \binom{n}{d}/\binom{w}{d}$ with $d = w - c$ (the incidence bound [24]).

p-dependence and asymptotic stabilization. Proposition 36 proves that $\max M_{\mathrm{true}}$ stabilizes at the value M_∞ (Definition 37) for all sufficiently large p, and Remark 38 tabulates computed values. Corollary 31 gives the exact formula $\mathbb{E}[M_{\mathrm{true}}] = \binom{n}{w}(1-1/p)^w/p^c$.

What remains open: (i) a closed-form for $M_\infty(n, w, c)$ (the data at $w = 3$, $c = 2$ suggest $M_\infty = 3$ for all $n \geq 11$; see Remark 38); (ii) whether any worst-case list-size bound can improve the proximity gap beyond the half-threshold $\delta/2$; (iii) the *birthday-bound conjecture* : extending Conjecture 41 from a high-probability bound (over the random parameters (s_1, s_2)) to a uniform worst-case bound. The high-probability form

$$\max_\gamma M_\gamma(s_1, s_2) \;\leq\; \frac{\binom{n}{w}}{p} + 4\sqrt{2\log(p)\cdot\frac{\binom{n}{w}}{p}} \qquad \text{w.p. } \geq 1 - 1/p$$

is stated in Conjecture 41 under additional negative-dependence hypotheses beyond Lemma 39 (a conditional / sketch-level Bernstein-style sub-Poisson tail). The conjectural worst-case version is

$$\max_{s_1,s_2}\max_{\gamma\in\mathbb{F}_p} M_\gamma(s_1, s_2) \;\leq\; C_1\cdot\frac{\binom{n}{w}}{p} \qquad \text{(uniform in } (s_1, s_2)).$$

If true, this combines with $M_\gamma \leq p$ to yield $\max M_{\mathrm{true}} = O(\sqrt{\binom{n}{w}}) = O(2^{n/2})$, matching the 1.41^n envelope of Remark 42 and bridging the gap between the universal bound $2\binom{n}{w-1}$ (Theorem 40) and the empirical $0.66 \cdot 1.36^n$.

Status of the worst-case extension. The natural higher-moment + Markov + union-bound approach for promoting Conjecture 41 to all (s_1, s_2) is structurally blocked. The k-wise extension of the pairwise-independence theorem (Theorem 28) is empirically false at $c = 2$ for $k \geq 3$: three supports E_1, E_2, E_3 sharing a common $(w-c)$-element core C^* have all pairwise overlaps $|E_i \cap E_j| = w - c$ (saturating the pairwise-independence hypothesis) but their six normals all lie in the principal ideal $(\Lambda_{C^*}) \subset \mathbb{F}_p[x]/(x^D)$ of codimension $w-c$, hence dimension $2c$, so the rank is at most $2c < 3c$. Joint compatibility probability is then $1/p^{2c}$ rather than the independent prediction $1/p^{3c}$, making the third moment super-Gaussian and ruling out sub-Gaussian Markov tails. Empirical verification: companion script [39] finds thousands of rank-deficient triples for $n \in \{12, 14, 16\}$. This is a higher-order analog of Lemma 39 (the two-leaf case at $c = 2$): the multi-leaf coupling at $c \geq 2$ requires cross-correlation tools from sequence theory beyond what the error-locator normal machinery alone provides.

The $c \geq 3$ regime is qualitatively different (§7.6). The same rank-deficiency that creates exponential growth at $c = 2$ disappears at $c \geq 3$, and Conjecture 43 (Open-Set Rank Lemma) predicts $M_{\mathrm{true}} \leq \lfloor (2D-1)/c \rfloor$ at sufficiently large p, linear in n. The large-c regime $c = \Theta(n)$ falls in this third phase, where the conjecture predicts $M_{\mathrm{true}} = O(1)$ at the Johnson radius and the empirical data confirms $M_{\mathrm{true}} \leq 4$ for rate $1/2$ across $n = 20$ to 40. Resolving Conjecture 43 for $c \geq 3$ would close the worst-case worry at current code lengths even without progress on the $c = 2$ birthday-bound conjecture above.

3. **FRI variants tolerating odd k.** The standard radix-2 FRI fold is provably unsound for odd $k \geq 3$ (Remarks 5, 69), so this paper's bound assumes $k = 2^m$. The remaining open question is whether modified folding schedules (radix-r for $r \neq 2$, mixed-radix, or non-uniform halving) admit a comparable above-Johnson soundness theorem at non-power-of-two degrees. Standard deployed FRI uses $k = 2^m$, so the present scope is not a practical blocker.

9 Deployment Roadmap

Scope of the deployment-applicable results. No protocol-level change is introduced: the folding maps, Merkle commitments, prover messages, and verifier checks of FRI, STIR, and WHIR remain as in the source protocols. Above-Johnson soundness coverage breaks down by protocol as follows. FRI in both characteristics, and circle FRI, are at theorem level. STIR is at conditional-theorem level in odd characteristic under hypotheses (RD)/(CQ) of Theorem 48, with query-phase bound $\max((1-\delta/2)^q, (2\rho)^q)$ rate-uniform on the standard radix-2 schedule; the characteristic-2 STIR analogue is sketched in Remark 74. WHIR is at conjecture level in odd characteristic (Conjecture 62); the proximity-layer substitution that the conjecture relies on is justified by Lemma 63, while the surrounding iteration-level induction of WHIR [11, §5] is left as follow-up work. The characteristic-2 WHIR analogue is similarly sketched. The only parameter that changes is the query count q, and increasing q scales proof size, verifier hashing, calldata, and wrapper-circuit cost linearly in the query-heavy portion of the verifier. The theorem bounds interactive oracle proof of proximity (IOPP) soundness; the non-interactive Fiat–Shamir compilation, the DEEP-quotienting, and the polynomial-commitment wrapping are given in Appendix D (multiplicative FRI fully proved; the same RBR template applies to char-2 FRI, circle FRI, and STIR but is not separately packaged). The remainder of this section gives concrete parameter tables.

	Conjectured (zero-loss CA)	**Proved (this paper)**
Status	Unproven	Theorem (FRI/circle-FRI); Conditional theorem under (RD)/(CQ) (STIR; rate-uniform); Conjecture (WHIR)
Security budget ($q=128$, $R=20$, $\delta=0.4$, rate $\rho=1/2$):		
Commit-phase (BabyBear base, $\lvert F\rvert=2^{31}$)	$\sim$7 bits	$\sim$7 bits
Commit-phase (BabyBear$_4$, $\lvert F\rvert=2^{124}$)	$\sim$100 bits	$\sim$100 bits
Query-phase ($q=128$, δ vs $\delta/2$)	$(1-\delta)^{128}\approx 2^{-94}$	$(1-\delta/2)^{128}\approx 2^{-41}$
Total (BabyBear base)	$\min(7,94)=7$	$\min(7,41)=7$
Total (BabyBear$_4$, extension)	$\min(100,94)=94$	$\min(100,41)=41$
Query overhead	baseline	$\sim$2.3× baseline at $\delta=0.4$
Protocol-structure changes	none	**none** (q recalibrated)
Protocols covered	FRI, circle FRI, STIR, WHIR	**FRI, circle FRI; STIR (conditional theorem under (RD)/(CQ), rate-uniform); WHIR (conjecture)** (see matrix)†
Characteristic	odd only	**FRI: any (theorem); STIR: char$\neq$2 theorem; WHIR: char$\neq$2 conjecture; char 2 sketched**†
Survives (δ,δ)-CA exponential constant (§8)	no	**yes**

† FRI soundness is proved as a formal theorem in both characteristics (Theorem 20 for char $\neq 2$; Theorem 72 for char $=2$). STIR's commit term is $(2k+n)R/|F|$ in general (Theorem 48), tightening to $(k+n)R/|F|$ at deployment scale where $|F|/|L^{(i)}|\to\infty$; the security rows above are for FRI. The STIR/WHIR extension to characteristic 2 appears to follow from the same argument (the additive coupling lemma replaces the multiplicative one and the per-coordinate batch-CA helper extends without change); we sketch it in Remark 74 and do not claim theorem status.

Proof-coverage matrix. The precise scope of what is formally proved vs. deferred:

Protocol / composition step	Status	Reference
FRI above Johnson, char$(F)\neq 2$	**fully proved**	Theorem 20
FRI above Johnson, char$(F)=2$	**fully proved**	Theorem 72
Half-threshold CA, any linear code (RVW13 [31])	classical	Theorem 6
Batch CA, any field	**fully proved**	Theorem 46
STIR soundness, char$(F)\neq 2$	**conditional under (RD)/(CQ), rate-uniform**	Theorem 48
STIR soundness, char$(F)=2$	**sketched**	Remark 74
WHIR, linear-subcode CA core, char$(F)\neq 2$	**fully proved**	Corollary 58
WHIR soundness / affine-subcode composition, char$(F)\neq 2$	*conjectural* (proximity-substitution proved via Lemma 63)	Conjecture 62
WHIR soundness, char$(F)=2$	**sketched**	Remark 74
Circle-FRI variant (Stwo)	**fully proved**	Theorem 80
Fiat–Shamir / non-interactive soundness, multiplicative FRI	**fully proved**	Theorem 84; Appendix D
Fiat–Shamir compilation, char-2 / circle / STIR	*the same RBR proof pattern extends, but no separate theorem is stated here*	—
DEEP-FRI PCS soundness error (interactive + non-interactive)	**conditional reduction** (assumes the standard DEEP convention [38] for the codimension-1 leading-coefficient check; knowledge-extractor lifting via [37])	Theorem 86; Corollary 87
Composition with Plonkish lookups (logUp)	**not addressed**	future work

The proved above-Johnson soundness covers the full proximity-testing stack at theorem level for FRI in both characteristics and for circle FRI; at conditional-theorem level for STIR in odd characteristic under the named hypotheses (RD)/(CQ) of Theorem 48, rate-uniform on the standard radix-2 schedule; and at conjecture level for WHIR in odd characteristic (Theorems 20, 72, 80, 48; Conjecture 62); the characteristic-2 STIR/WHIR extension is sketched in Remark 74. Fiat–Shamir compilation to non-interactive soundness (Theorem 84) is fully proved; DEEP-FRI soundness error for the polynomial commitment scheme (Theorem 86, Corollary 87) is a conditional reduction that imports the standard DEEP convention's leading-coefficient check from [38]; the explicit knowledge-extractor lifting is deferred to [37]. The remaining "not addressed" item (Plonkish/logUp composition) is the arithmetization layer, which is protocol-specific and orthogonal to proximity testing.

Adoption changes only the query count for systems already on the matching FRI variant: the FRI/STIR/WHIR prover algorithm, round structure, folding equations, and proof format are unchanged; only the query count q is recalibrated. The matching FRI variant is fixed by characteristic: multiplicative FRI in odd characteristic, the additive subspace-polynomial fold $s(x) = x^2 + \beta x$ for binary tower fields ($\mathbb{F}_{2^{128}}$), and circle FRI for Stwo's Mersenne31. These additive and circle folds are the canonical characteristic-2 and Mersenne31 FRI variants already in use [21, 22]; our contribution is a *soundness theorem* for them above the Johnson radius, not the fold itself. The same additive-fold substitution yields the sketched characteristic-2 STIR/WHIR extension in Remark 74. The verifier implementation must still be parameterized by the chosen query count q, so changing q changes the number of Merkle openings and consistency checks. Remaining engineering tasks:

1. **Interactive security parameter tuning**: the commit-phase term $nR/|F|$ matches BCHKS up to constants; on BabyBear$_4$ ($|F| = p^4 \approx 2^{124}$), this gives ∼100 bits. The binding constraint is the query-phase term $(1-\delta/2)^q$: at $q = 128$ this gives ∼41 bits for $\delta = 0.4$. Reaching 128-bit interactive query-phase security at $\delta = 0.4$ requires $q \approx 398$ (a ∼3.1× increase from typical $q = 128$), but BabyBear$_4$ remains capped at ∼100 total interactive bits by the commit term; the non-interactive setting is stricter because of the Fiat–Shamir Q-loss in the next item. Multi-rate reference:

ρ	$\delta_J = 1-\sqrt{\rho}$	δ	q (conj. zero-loss)	q (ours)
1/2	0.293	0.30	∼195	∼427
1/2	0.293	0.40	∼136	∼311
1/4	0.500	0.55	∼87	∼216
1/8	0.646	0.70	∼58	∼161

   All rows target $\lambda = 100$ bits of query-phase security, with δ chosen strictly above the rate-dependent Johnson radius $\delta_J = 1-\sqrt{\rho}$ so that our theorem applies (the open zone is $\delta_J < \delta < 1-\rho$). Below δ_J, BCHKS's zero-loss bound is the relevant baseline and our reduced-threshold result offers no improvement; the table therefore omits below-Johnson operating points.

2. **Non-interactive parameter tuning**: Fiat–Shamir compilation multiplies the interactive RBR error by the adversary's random-oracle query budget Q and adds the Merkle/random-oracle binding term $3(Q^2+1)/2^\kappa$ (Theorem 84). Table 3 gives the resulting parameter budget for $Q = 2^{64}$ and $\delta = 0.4$. The entries are generated by `paper1/scripts/deployment_params.py`; "commit" means no finite q can reach the target because the $Q \cdot nR/|F|$ term already exceeds the target.

Config	$\vert F\vert$ bits	q now	Interactive	NI now	q_{64}	q_{100}	q_{128}
BabyBear base	31	128	6.7	−57.3	commit	commit	commit
BabyBear$_4$	124	128	41.2	−22.8	commit	commit	commit
BabyBear$_5$	155	128	41.2	−22.8	398	commit	commit
BabyBear$_7$	217	128	41.2	−22.8	398	510	597
Mersenne31$_4$	124	128	41.2	−22.8	commit	commit	commit
Goldilocks$_2$	128	128	41.2	−22.8	commit	commit	commit
Binary tower 2^{128}	128	128	41.2	−22.8	commit	commit	commit

Table 3. Non-interactive FRI security under the loose Theorem 84 accounting, with $n = 2^{20}$, $R = 20$, $\delta = 0.4$, $Q = 2^{64}$, $\kappa = 256$ except BabyBear$_7$ where $\kappa = 384$ so the hash-binding term is not the bottleneck. "Interactive" and "NI now" are proved bits at $q = 128$; negative NI bits mean the theorem gives no non-trivial non-interactive bound at that query count after the Q-loss. The q_λ columns give the required query count for λ-bit non-interactive security when the commit and hash terms permit it.

3. **Verifier cost update**: increasing q is a parameter change with real cost. It leaves the IOP structure intact but increases Merkle path verifications, FRI consistency checks, proof bytes, calldata, and any recursive/wrapper verifier circuit work roughly linearly in q.
4. **Reproducibility**: the table above is generated by `paper1/scripts/deployment_params.py` in [39]; per-script reproduction commands are in the repository's top-level `README.md` and the per-paper Lean state is in `paper1/lean/STATUS.md`. The structural companion notes [24, 25, 26, 27] are referenced for the original derivations and are not currently exposed as separate files in the public repository tree.
5. **Formal verification**: the half-threshold CA bound (Theorem 6) and the equal-threshold CA upper bound (Theorem 8) are formally verified in Lean 4 with Mathlib at the statement-faithful level (zero `sorry`, zero project-level axioms beyond Mathlib's standard ones); the half-threshold theorem also has a paper-form $|F|$-probability corollary. Building blocks for the FRI coupling chain (`coupling_pointwise`, `coupling_counting`, `rs_iso_forward/surj`), the char-2 generalized pairing, and the batch-CA fixed-rest union bound are present in the Lean library. Remaining: a concrete `RSIsomorphismWitness` instance for the multiplicative-coset FRI domain, a faithful import of the BCIKS–BCHKS proximity gap (BCIKS '20 Theorem 1.2 with the BCHKS '25 $\leq n$ refinement), and the protocol-event wrapper that converts the rational-form soundness skeleton into the paper-form probability bound. See the companion repository's `paper1/lean/STATUS.md` for the per-theorem state [39].

10 Conclusion and Outlook

We prove above-Johnson FRI soundness $\varepsilon_{\mathrm{FRI}} \leq nR/|F| + (1-\delta/2)^q$ for multiplicative FRI, additive FRI, and circle FRI under their standard domain hypotheses ($k = 2^m$, multiplicative subgroup / coset closed under negation in odd characteristic, $\mathbb{F}_2$-linear subspace in characteristic 2, circle subgroup / coset for circle FRI). The same reduced-threshold method yields a conditional-theorem-level odd-characteristic extension to STIR (Theorem 48) under the two named hypotheses (RD) and (CQ); the query-phase bound $\max((1-\delta/2)^q, (2\rho)^q)$ is rate-uniform and simplifies to $(1-\delta/2)^q$ at the standard deployment rate $\rho \leq 1/4$. WHIR remains conjectural (Conjecture 62), with the proximity-layer substitution proved via Lemma 63. The characteristic-2 STIR/WHIR extension is sketched in Remark 74 (Section 9). The only protocol-level change is the query count q; the prover/verifier equations remain unchanged, while verifier cost scales with the chosen q.

The standard two-bad-γ CA framework is settled. The half-threshold CA bound is the classical 1:2 proximity-gap inequality of RVW13 [31]. Our contribution is applying it *above the Johnson radius*: the chain $\delta/2 < (1-\rho)/2 \leq \delta_J$ (Remark 1) places the protocol into BCIKS's zero-loss regime after round 1, where BCIKS [7] locks the distance. The 1:2 ratio is optimal at FRI scale (Remark 14), and the $M = 0$ approach is obstructed at FRI-relevant code lengths (Proposition 35). The $2\times$ query overhead is intrinsic to any CA-based proof at current parameters.

Beyond the CA framework. Every published FRI soundness proof factors through a CA lemma (§1.7). Reducing the $2\times$ overhead requires a proof that avoids this intermediate step, for example via a direct algebraic analysis of the fold map on multiplicative subgroups, or via character-sum techniques exploiting that $L = \langle\omega\rangle$ is a cyclic group. Cross-correlation distributions from the Helleseth–Golomb–Gong tradition [17, 18] and Weil-type bounds on incomplete character sums [19] are natural candidates. Concurrent companion work in preparation [2, 3] pursues this direction; specific results await those papers' completion. Whether FRI admits a non-CA soundness proof with overhead $< 2\times$ at FRI scale is a key remaining question for FRI soundness theory.

Practical bottom line. Before this work, no unconditional above-Johnson bound was known for plain fixed-domain Reed–Solomon FRI. Theorem 20 gives $\varepsilon_{\mathrm{FRI}} \leq nR/|F| + (1-\delta/2)^q$, yielding $\sim$41 interactive query bits at $q = 128$ or $\sim$128 interactive query bits at $q \approx 398$ for $\delta = 0.4$. Total security is still capped by the commit term (for example $\sim$100 interactive bits on $\mathrm{BabyBear}_4$ at $n = 2^{20}, R = 20$). The $2\times$ query overhead vs. the conjectured zero-loss baseline is optimal within the standard two-bad-γ CA framework; for plain fixed-domain Reed–Solomon FRI as deployed, our bound is the strongest unconditional one currently available.

Roadmap. Three follow-ups: (a) complete the Lean 4 formalization [39] (the half-threshold CA theorem is statement-faithful in Lean with a paper-form $|F|$-probability corollary; building blocks for the FRI coupling, char-2, and batch-CA chains are present, and the remaining work is the concrete pairing instances, a faithful BCIKS '20 import, and the protocol-event wrapper for Pr[accept]); (b) tight bounds on worst-case M_{true} and whether they can improve the proximity gap beyond $\delta/2$ (§8, item 2); (c) non-CA proximity statements pursued in concurrent work in preparation by the same authors [2, 3], addressing the question of whether the $< 2\times$ regime is reachable outside the CA framework. Specific quantitative claims await those papers' completion.

A Extension to STIR and WHIR

We show that the half-threshold CA bound (Theorem 6) extends to the batch setting required by STIR [10] and to the constrained RS subcodes required by WHIR [11]. The resulting above-Johnson soundness statements are conditional-theorem-level for STIR under the named hypotheses (RD)/(CQ) of Theorem 48 (rate-uniform on the standard radix-2 schedule) and conjectural for WHIR via Conjecture 62, with the proximity-layer substitution proved (Lemma 63). The deployment-relevance scope is summarized in §9.

A.1 Batch Correlated Agreement

STIR and its variants test proximity of multiple functions simultaneously by taking a random linear combination and checking the result. This requires a *batch* CA bound.

Definition 45 (Batch Correlated Agreement)**.** *For code C over domain L, B functions $f_1, \ldots, f_B : L \to F$, and independent uniform $\gamma_1, \ldots, \gamma_B \in F$:*

$$\varepsilon_{\mathrm{ca}}^{(B)}(C, \delta_{\mathrm{hd}}, \delta_{\mathrm{nt}}) = \max_{f_1,\ldots,f_B} \Pr_{\gamma_1,\ldots,\gamma_B}\left[\Delta\left(\sum_{i=1}^{B} \gamma_i f_i,\, C\right) \leq \delta_{\mathrm{hd}} \text{ and } \exists\, i :\ \Delta(f_i, C) > \delta_{\mathrm{nt}}\right].$$

Theorem 46 (Half-Threshold Batch CA)**.** *For every real $\delta \in [0, 1]$, every integer $B \geq 1$, and every linear code C over F:*

$$\varepsilon_{\mathrm{ca}}^{(B)}(C, \delta/2, \delta) \leq \frac{1}{|F|}.$$

Proof. Fix $f_1, \ldots, f_B$ with $\Delta(f_j, C) > \delta$ for some index j (the existential premise of $\varepsilon_{\mathrm{ca}}^{(B)}$); pick any one such j. We bound the probability over $(\gamma_1, \ldots, \gamma_B)$ that $g = \sum_i \gamma_i f_i$ is $(\delta/2)$-close to C.

Step 1: Conditioning. Fix all γ_i for $i \neq j$, so that $h = \sum_{i \neq j} \gamma_i f_i$ is a fixed function. Then $g = \gamma_j f_j + h$, which is linear in the single random variable γ_j.

Step 2: Pairwise argument. Call γ_j "bad" if $\exists c_\gamma \in C$ with $|S_\gamma| \geq (1 - \delta/2)n$ where $S_\gamma = \{x \in L : \gamma_j f_j(x) + h(x) = c_\gamma(x)\}$.

For two distinct bad values γ_j, γ_j': on $S_\gamma \cap S_{\gamma'}$ (size $\geq (1-\delta)n$), we have $(\gamma_j - \gamma_j')f_j = c_\gamma - c_{\gamma'}$, so f_j agrees with the codeword $(c_\gamma - c_{\gamma'})/(\gamma_j - \gamma_j') \in C$ on $\geq (1-\delta)n$ points, giving $\Delta(f_j, C) \leq \delta$. This contradicts $\Delta(f_j, C) > \delta$. Hence at most 1 bad γ_j.

Step 3. The bound in Step 2 holds for each fixed h (i.e., each fixed choice of γ_i, $i \neq j$). Averaging over the other γ_i:

$$\Pr_{\gamma_1, \ldots, \gamma_B}[\Delta(g, C) \leq \delta/2] \;\leq\; \frac{1}{|F|}. \qquad \square$$

Remark 47 (Individual vs. joint distance premise)**.** *Definition 45 uses the individual distance condition $\exists i : \; \Delta(f_i, C) > \delta$ rather than the joint distance $\Delta_{\mathrm{joint}}((f_1, \ldots, f_B), C^{=B}) > \delta$. The individual condition is stronger (more restrictive on the input): individual distance $\Delta(f_i, C) > \delta$ for some i implies joint distance $\Delta_{\mathrm{joint}}((f_1, \ldots, f_B), C^{=B}) > \delta$ (since the agreement set under any codeword tuple $(g_1, \ldots, g_B)$ is contained in the per-coordinate agreement set $\{x : f_i(x) = g_i(x)\}$ for each i, so the joint agreement size is bounded by the individual agreement size at the worst i). The converse fails: joint distance can be large while every individual distance is* 0 *(different codewords at different coordinates, perfectly aligned per coordinate). The individual form is the one tailored to the STIR use-case below.*

In the STIR proof (Theorem 48, Strategy B), the relevant condition is $\Delta(f^{(i-1)}\restriction_{L^{(i)}}, \mathrm{RS}_{k_i}) > \delta$: this is the individual-distance condition applied to one specific function, which is what Theorem 46 requires. The pairwise argument in the proof of Theorem 6 is invoked directly.

A.2 STIR Soundness Above Johnson

We show that our half-threshold CA is a drop-in replacement for the conjectured zero-loss CA in the STIR soundness proof [10].

STIR protocol recap. STIR [10] is an IOPP for Reed–Solomon proximity testing. Fix a degree schedule $k_0 = k, k_1, \ldots, k_R$ with $k_i \mid k_{i-1}$ and domains $L = L^{(0)} \supset L^{(1)} \supset \cdots \supset L^{(R)}$ with $|L^{(i)}| = n/2^i$ (the standard radix-2 schedule sets $k_i = k_{i-1}/2$, matching FRI). At each round i, the verifier sends a random evaluation point $z_i \in F \setminus L^{(i)}$ and a random batching coefficient γ_i. The prover responds with $v_i = f^{(i-1)}(z_i)$ and commits the quotient oracle $q_i(x) = (f^{(i-1)}(x) - v_i)/(x - z_i)$, then sets $f^{(i)} = q_i + \gamma_i \cdot f^{(i-1)}\restriction_{L^{(i)}}$ (the batched next-round function). The key soundness step is: if $f^{(i-1)} \notin \mathrm{RS}_{k_{i-1}}$, then with high probability over γ_i, the batched $f^{(i)}$ is far from RS_{k_i} — this is precisely the CA bound applied to the pair $(q_i, \, f^{(i-1)}\restriction_{L^{(i)}})$ with randomness γ_i.

Theorem 48 (STIR Soundness Above Johnson, codeword-quotient model)**.** ***Setup.*** *Let F be a finite field with* $\mathrm{char}(F) \neq 2$*, $k = 2^m$, and $L \subseteq F^*$ a multiplicative subgroup or coset closed under negation with $0 \notin L$ and $|L| = n = 2^d$, $d \geq m$. Fix an integer round count $1 \leq R \leq m$, the standard radix-*2 *degree schedule $k_i = k/2^i$ ($k_i \in \mathbb{Z}$ since $R \leq m$), and the standard radix-*2 *domain schedule $L^{(i)} = (L^{(i-1)})^2$ for $i = 1, \ldots, R$ (each $L^{(i)}$ is a multiplicative subgroup or coset of size $n/2^i$, closed under negation and not containing* 0*; closure is automatic when $|F| \equiv 1 \pmod 4$ and inherited at each squaring step). Let $\rho = k/n$, $\delta_J = 1 - \sqrt{\rho}$, fix an open-zone distance $\delta_J < \delta < 1 - \rho$, and write $\rho_i := k_{i-1}/|L^{(i)}| = 2\rho$ for the per-round STIR shift-query parameter on this schedule. Assume the* *standard STIR field-size regime $|F| \geq 2|L^{(i)}|$ at every round (satisfied at every deployment instance considered in this paper, see §9). Let $f^{(0)} \in F^L$ with $\Delta(f^{(0)}, \mathrm{RS}_k) > \delta$ on L (the input-distance hypothesis).*

Hypotheses. *Two additional hypotheses are made explicit:*

- ***(RD)*** *Restriction-distance preservation. At every round $i \in \{1, \dots, R\}$ that the prover deviates (cf. (CQ) below), the prover-committed function $f^{(i-1)}$ on $L^{(i-1)}$ satisfies $\Delta(f^{(i-1)}\restriction_{L^{(i)}}, \mathrm{RS}_{k_i}) > \delta$ on $L^{(i)}$ when $i = 1$, and $> \delta/2$ on $L^{(i)}$ when $i \geq 2$. This is closely related to the STIR distance-preservation premise of [10, §5] (which tracks a list-decoding invariant rather than the bare distance invariant we use here); we treat (RD) as an explicit hypothesis on the input function $f^{(0)}$ at $i = 1$ and on the honest folds at $i \geq 2$, and discuss its scope in Remark 51.*
- ***(CQ)*** *Codeword-quotient adversary model. At every round i that the prover deviates from the honest fold of $f^{(i-1)}$, the deviating quotient is a codeword: $q_i \in \mathrm{RS}_{k_i}$. The set of deviation rounds is denoted $D \subseteq \{1, \dots, R\}$ and may have any size $1 \leq |D| \leq R$. The non-codeword-quotient case ($q_i \notin \mathrm{RS}_{k_i}$) is left as follow-up (Remark 52); the multi-round extension within (CQ) is given in Lemma 49.*

No round-1 list-size hypothesis is needed. The half-threshold CA framework of this paper bypasses STIR's standard list-decoding invariant at round 1: the round-1 failure mode is the half-threshold CA failure at threshold $\gamma = \delta/2$ over the random γ_1 (failure $\leq 1/|F|$ by Theorem 6 under (RD) at $i = 1$), not the OOD-sampling list-collision event of [10, Lemma 4.5]. At rounds $i \geq 2$ the OOD-sampling list-collision contribution is bounded by Lemma 4.5 of [10] at threshold $\delta/2 < \delta_J$ where the Johnson list-size $O(1/\sqrt{1-\rho})$ applies unconditionally. See Remark 50.

Conclusion. *STIR with R rounds, the standard OOD sample at each round, and q i.i.d. shift queries per round satisfies*

$$\Pr[\text{STIR accepts}] \;\leq\; \frac{(2k+n)R}{|F|} \;+\; \max\!\big((1-\tfrac{\delta}{2})^q,\ (2\rho)^q\big)\,,$$

where the $2k$ summand collects the per-round STIR OOD contributions ($\leq 2k_{i-1}/|F|$ each, bounded uniformly by $2k/|F|$) and the n summand collects the BCHKS [6] bad-challenge bounds at rounds $i \geq 2$ of Strategy B ($\leq |L^{(i)}|/|F|$ each, bounded uniformly by $n/|F|$; the original BCIKS [7] bound is $O(n^2)$, refined to $\leq n$ by BCHKS). The query-phase bound simplifies to $(1-\delta/2)^q$ for $\rho \leq 1/4$ (the rate regime considered in STIR's standard deployment, see Remark 56); for $\rho \in (1/4, 1/2)$ the max *is genuinely needed; at $\rho = 1/2$ the shift-query bound is vacuous and security must come from the FRI subprotocol consistency check. At the deployment-scale regime $|F|/|L^{(i)}| \to \infty$ at fixed rate, the factor of 2 in $2k_{i-1}/(|F| - |L^{(i)}|)$ collapses, tightening the commit term to $(k+n)R/|F|$.*

Lemma 49 (Multi-round catch-coverage in the codeword-quotient model)**.** *Adopt the setup of Theorem 48 with hypotheses (RD), (CQ). Let $D \subseteq \{1, \dots, R\}$ be the (possibly multi-element) set of rounds at which the prover deviates from the honest fold of $f^{(i-1)}$. Then:*

(1) *Per-round shift-query miss. For each $i \in D$, the round-i shift-query check rejects the prover's (v_i, q_i) except with probability $\rho_i^q = (2\rho)^q$ over q i.i.d. uniform $y \in L^{(i)}$ (Lemma 4.4 of [10]; unconditional in $f^{(i-1)}$). The events $\{\text{round-}i\text{ shift-query miss}\}_{i \in D}$ are independent over D given the deviation transcript, since each round draws fresh q uniform shift queries. Hence the joint shift-query miss probability is*

$$\Pr\big[\text{all of } D \text{ miss}\big] \;=\; \prod_{i \in D} \rho_i^q \;\leq\; (2\rho)^q,$$

which matches the single-deviation shift-query bound and tightens to $(2\rho)^{q \cdot |D|}$ at $|D| \geq 2$.

(2) *Per-round batch-CA failure. For each $i \in D$, the round-i batched function $f^{(i)} = q_i + \gamma_i \cdot f^{(i-1)}\restriction_{L^{(i)}}$ falls $(\delta/2)$-close to RS_{k_i} over the random γ_i with probability $\le 1/|F|$ at $i = 1$ (Theorem 6, since $q_i \in \mathrm{RS}_{k_i}$ by (CQ) and $\Delta(f^{(0)}\restriction_{L^{(1)}}, \mathrm{RS}_{k_1}) > \delta$ by (RD) at $i = 1$) or $\le |L^{(i)}|/|F|$ at $i \ge 2$ (BCHKS [6] proximity gap at threshold $\delta/2 < \delta_J$, refining the BCIKS [7] $O(n^2)$ bound to $\le n$, since (RD) at $i \ge 2$ supplies the required input distance). Union-summed over D, the total batch-CA contribution is at most $\sum_{i\in D} |L^{(i)}|/|F| \le |D| \cdot n/|F| \le nR/|F|$.*

(3) *Restriction-distance carries forward. Between consecutive deviation rounds $i_j, i_{j+1} \in D$ ($i_j < i_{j+1}$) the prover is honest, so $f^{(i_j)} \to \cdots \to f^{(i_{j+1}-1)}$ is the standard STIR honest fold; the chain of Lemmas 4.4 + 4.5 of [10] preserves the $(\delta/2)$-far property of $f^{(i_j)}\restriction_{L^{(i_{j+1})}}$ from $\mathrm{RS}_{k_{i_{j+1}}}$ along the honest range, supplying (RD) at round i_{j+1} inductively. The base case i_1 is supplied directly by hypothesis (RD) at $i = i_1$ (i.e., the explicit (RD) hypothesis of the theorem at the first deviation round, regardless of whether $i_1 = 1$ or $i_1 \ge 2$); this is treated as an input to the theorem, not a derived consequence (see Remark 51).*

Consequently the union of the per-round failures across D contributes at most $(2k+n)R/|F| + (2\rho)^q$ to $\Pr[\text{STIR accepts}]$*, matching the bound of Theorem 48 (with the $|D|$-fold tightening of the shift-query factor absorbed by the* max*-form into* $\max((1-\delta/2)^q, (2\rho)^q)$*). In particular, the multi-round case ($|D| \ge 2$) yields the same overall bound as the single-deviation case ($|D| = 1$) under (CQ).*

Proof. Items (1) and (2) restate the per-round bounds established in the proof of Theorem 48 (Strategy B items 1 and 2), with hypothesis (RD) at every $i \in D$ and (CQ) supplying $q_i \in \mathrm{RS}_{k_i}$ at every $i \in D$; the per-round bounds depend only on the round index i and on $f^{(i-1)}$, not on whether prior rounds were deviation rounds. Independence of the shift-query misses in (1) follows from the fresh-randomness assumption of the verifier's q i.i.d. shift queries per round (Definition 17 carried over from the FRI analysis). The chain of Lemmas 4.4 + 4.5 of [10] for honest folds in (3) is the standard STIR distance-preservation argument, applied to the maximal honest range between consecutive deviation rounds; $|D|$ such ranges concatenate into the full transcript, with the round-by-round random choices coupling across ranges via the same OOD/batching transcript. The total commit-phase contribution is at most $\sum_{i=1}^{R}(2k_{i-1} + |L^{(i)}|)/|F| \le (2k+n)R/|F|$ (the same per-round upper bound is paid at every round, deviation or not, in the worst case). The total query-phase contribution is at most the worst single-round shift-query miss, $(2\rho)^q$, because the joint miss is the product over D which is no larger. ☐

Proof of Theorem 48. The proof follows the same Strategy A / Strategy B dichotomy as Theorem 20, adapted to STIR's round structure; Strategy B's per-round bounds are precisely the per-round bounds invoked in Lemma 49, here unified into the per-round Strategy B template and then summed across the (possibly multi-element) deviation set D via that lemma.

Strategy A: the prover commits the honest quotient of $f^{(i-1)}$ at each round. Concretely, at each round i the prover picks a value $v_i \in F$ (the claimed evaluation of $f^{(i-1)}$ at z_i) and commits $q_i(x) = (g^{(i-1)}_{\mathrm{Lag}}(x) - v_i)/(x - z_i)$ on $L^{(i)}$, where $g^{(i-1)}_{\mathrm{Lag}}$ is the unique degree-$<|L^{(i-1)}|$ Lagrange interpolant of $f^{(i-1)}|_{L^{(i-1)}}$. The shift queries on $L^{(i)}$ then check $f^{(i-1)}(y) - v_i = (y - z_i)q_i(y)$ for q random $y \in L^{(i)}$, which holds deterministically by the construction of q_i from the Lagrange interpolant; Strategy A's failure mode is therefore not a shift-query miss but the round-i batched function $f^{(i)} = q_i + \gamma_i \cdot f^{(i-1)}|_{L^{(i)}}$ accidentally landing close to RS_{k_i} over the random γ_i. We bound the probability that $f^{(R)} \in \mathrm{RS}_{k_R}$ by a per-round union bound.

At round i, suppose $f^{(i-1)} \notin \mathrm{RS}_{k_{i-1}}$. Then:

- Over $z_i \in F \setminus L^{(i)}$: by the STIR function-quotienting lemma [10] (Lemma 4.4), if $f^{(i-1)} \notin \mathrm{RS}_{k_{i-1}}$ then the quotient $q_i(x) = (f^{(i-1)}(x) - f^{(i-1)}(z_i))/(x - z_i)$ inherits non-codeword status except on a set of $\le k_{i-1}$ degenerate values of z_i (roots of a nonzero degree-($\le k_{i-1}$) polynomial in z_i arising from the interpolation identity). Lemma 4.4 of [10] is unconditional polynomial algebra; no proximity-gap or list-decodability hypothesis is invoked.

- Over γ_i: for a non-degenerate z_i, the batched syndrome of $f^{(i)} = q_i + \gamma_i \cdot f^{(i-1)}\restriction_{L^{(i)}}$ is degree-1 in γ_i (nonzero), so it vanishes for at most 1 value of γ_i.

Per round (Strategy A), under the regime $|F| \geq 2|L^{(i)}|$ in the theorem hypothesis: $\Pr[f^{(i)} \in \mathrm{RS}_{k_i} \mid f^{(i-1)} \notin \mathrm{RS}_{k_{i-1}}] \leq k_{i-1}/(|F| - |L^{(i)}|) + 1/|F| \leq 2k_{i-1}/|F| + 1/|F|$, where the $k_{i-1}/(|F| - |L^{(i)}|)$ term comes from [10, Lemma 4.4] ($z_i \in F \setminus L^{(i)}$ landing in the at-most-k_{i-1} degenerate locus), and the $1/|F|$ from the γ_i Schwartz–Zippel over F.

By the union bound over R rounds, this Strategy A contribution is $\leq (2k+1)R/|F| \leq (2k+n)R/|F|$ (using $n \geq 1$), absorbed into the commit-phase total of the theorem statement once Strategy B is added below.

Tightening to $(k+n)R/|F|$ at deployment scale. The factor of 2 in $2k_{i-1}/|F|$ comes from $1/(|F| - |L^{(i)}|) \leq 2/|F|$ under $|F| \geq 2|L|$; under the stronger deployment-scale regime $|F|/|L^{(i)}| \to \infty$ at fixed rate (e.g. $|F| \approx 2^{124}$, $|L| \leq 2^{20}$, ratio $\geq 2^{100}$), the factor of 2 collapses to $1 + o(1)$ and the commit-phase term tightens to $(k+n)R/|F|$ as stated in the theorem.

Note on the k-factor. The $k_{i-1}/|F|$ term is intrinsic to STIR's out-of-domain evaluation: the quotient involves evaluating a degree-($<k_{i-1}$) polynomial at the random point z_i, producing $\leq k_{i-1}$ degenerate values. On extension fields (e.g., $\mathrm{BabyBear}_4$ with $|F| = p^4 \approx 2^{124}$), this term is negligible ($k/|F| \approx 2^{-105}$). On the base field ($|F| = p \approx 2^{31}$), it yields $\sim$7 bits — comparable to the BCHKS commit-phase bound. The improvement of this paper over BCHKS for STIR is therefore in the *query phase*: $(1-\delta/2)^q$ replaces the conjectured $(1-\delta)^q$.

Strategy B: the prover deviates from honest computation at every round in the (possibly multi-element) deviation set $D \subseteq \{1, \ldots, R\}$ permitted by hypothesis (CQ). The analysis is per-round and union-sums across D via Lemma 49 below; we present the per-round bound for an arbitrary $i \in D$. Two checks threaten at round i:

1. *Out-of-domain (OOD) check*: the verifier sends $z_i \in F \setminus L^{(i)}$ and receives the claimed value $v_i = f^{(i-1)}(z_i)$, then checks $f^{(i-1)}(y) - v_i = (y - z_i)q_i(y)$ at q random points $y \in L^{(i)}$. In Strategy B, $f^{(i-1)}$ is an arbitrary prover-committed oracle (not necessarily a polynomial). The soundness here decomposes into two STIR-internal pieces:
   - *OOD-sampling failure* (z_i landing on a list-collision): bounded by Lemma 4.5 of [10] as $\binom{\ell_i}{2} \cdot (k_{i-1}/(|F| - |L^{(i)}|))^s$, where ℓ_i is the list-decoding bound for $\mathrm{RS}[F, L^{(i)}, k_{i-1}]$ at the relevant threshold and s is the number of OOD samples (we use $s = 1$ in the standard schedule). At rounds $i \geq 2$ the relevant threshold is the post-halving distance $\delta/2 < (1-\rho)/2 \leq \delta_J$, well inside the Johnson zone unconditionally, so $\ell_i = O(1/\sqrt{1-\rho})$ via the classical Johnson bound ([10] Theorem 3.4). At round $i = 1$ the analysis instead invokes Theorem 6 (half-threshold CA) directly to bound the round-1 failure: the half-threshold CA failure at threshold $\gamma = \delta/2$ over the random γ_1 is $\leq 1/|F|$ under (RD) at $i = 1$, and Lemma 4.5 of [10] is not needed at round 1 in the half-threshold framework (Remark 50). This bypass removes any dependency of the theorem on a round-1 list-size bound at threshold $\delta > \delta_J$, where Johnson does not apply; STIR's standard analysis [10, §5] requires such a bound (since STIR uses the list-decoding invariant at every round), but the half-threshold approach of this paper does not.
   - *Shift-query failure* (the q random checks on $L^{(i)}$ all miss the deviation set): by Lemma 4.4 of [10] (unconditional function-quotienting), an adversarial pair (v_i, q_i) with $q_i \in \mathrm{RS}_{k_i}$ that satisfies the OOD identity at fewer than the threshold number of points of $L^{(i)}$ produces a deviation set of relative size $\geq 1 - \rho_i$, giving a catch probability $\geq 1 - \rho_i^q$ over q i.i.d. shift queries. With $\rho_i = 2\rho$ on the standard radix-2 schedule, the shift-

query miss probability is $\rho_i^q = (2\rho)^q$, which is bounded by the query-phase term $\max((1-\delta/2)^q, (2\rho)^q)$ of the theorem statement. See Remark 56 for the regime decomposition: the max collapses to $(1-\delta/2)^q$ for $\rho \le 1/4$ and to $(2\rho)^q$ for $\rho \in [1/4, 1/2)$.

The shift-query bound is what enters our final budget; the OOD-sampling term contributes only $O(k/|F|)$ via Lemma 4.5 and is absorbed into the $(2k+n)R/|F|$ commit-phase total (tightening to $(k+n)R/|F|$ at deployment scale).

2. *Batch-CA at the deviation round*: $f^{(i)} = q_i + \gamma_i \cdot f^{(i-1)}\restriction_{L^{(i)}}$ must be $(\delta/2)$-close to RS_{k_i}. Since $q_i \in \mathrm{RS}_{k_i}$ by (CQ), this reduces to bounding the bad γ_i that make $f^{(i-1)}\restriction_{L^{(i)}}$ fold close to RS_{k_i}. Hypothesis (RD) supplies $\Delta(f^{(i-1)}\restriction_{L^{(i)}}, \mathrm{RS}_{k_i}) > \delta$ at round 1 and $> \delta/2$ at rounds $i \ge 2$, so the half-threshold CA / BCIKS application below has the input distance it needs. The bound depends on the round:
   - Round $i = 1$: by hypothesis (RD) at $i = 1$, $\Delta(f^{(0)}\restriction_{L^{(1)}}, \mathrm{RS}_{k_1}) > \delta$ on $L^{(1)}$, so Theorem 6 (half-threshold CA) applied to the pair $(q_1, f^{(0)}\restriction_{L^{(1)}})$ at threshold $\gamma_{\text{premise}} = \delta$ gives at most 1 bad γ_1, failure $\le 1/|F|$.
   - Rounds $i \ge 2$ (deviation at round i): by hypothesis (RD) at i, $\Delta(f^{(i-1)}\restriction_{L^{(i)}}, \mathrm{RS}_{k_i}) > \delta/2$ on $L^{(i)}$, and threshold $\gamma = \delta/2 < \delta_J$ lies inside the BCIKS–BCHKS zero-loss regime. The BCHKS [6] proximity gap (refining BCIKS [7] from $O(n^2)$ to $\le n$) applied to the pair $(q_i, f^{(i-1)}\restriction_{L^{(i)}})$ gives at most $|L^{(i)}|$ bad γ_i, failure $\le |L^{(i)}|/|F|$.

   Per-round failure $\le |L^{(i)}|/|F|$ in the worst case (round $i \ge 2$). Summed over rounds and bounded by $|L^{(i)}| \le |L^{(0)}| = n$, this contributes the $nR/|F|$ summand of the $(2k+n)R/|F|$ total in the theorem statement.

As in Theorem 20 (Remark 21), the effective distance $\delta/2$ is preserved at rounds $2, \dots, R$: since $\delta/2 < (1-\rho)/2 \le \delta_J$ (Remark 1), the BCHKS [6] proximity gap at threshold $\gamma = \delta/2$ bounds the bad challenges to $\le |L^{(i)}|$ per round.

Combining items (1) and (2) per round, the per-round commit-phase contribution is $\le (2k_{i-1} + |L^{(i)}|)/|F|$ ($2k_{i-1}$ from OOD via $|F| \ge 2|L|$; $|L^{(i)}|$ from the BCHKS [6] proximity gap at the deviation round, refining BCIKS [7] from $O(n^2)$ to $\le n$). Summing over rounds and using $k_{i-1} \le k$ and $|L^{(i)}| \le |L^{(0)}| = n$, and writing $E_q := \max((1-\delta/2)^q, (2\rho)^q)$ for the query-phase term (Remark 56):

$$\Pr[\text{Strategy B}] \le \frac{(2k+n)R}{|F|} + E_q.$$

At the deployment-scale regime $|F|/|L^{(i)}| \to \infty$ the factor of 2 in $2k_{i-1}/|F|$ collapses, giving $(k+n)R/|F| + E_q$.

Combining. Every adaptive prover either commits honest quotients (Strategy A) or deviates (Strategy B); both strategies yield the same commit-phase bound, and only Strategy B has the query term, so

$$\Pr[\text{accept}] \le \frac{(2k+n)R}{|F|} + \max\big((1-\delta/2)^q, (2\rho)^q\big),$$

matching the theorem statement. □

Remark 50 (Why no round-1 list-size hypothesis is needed)**.** *The standard STIR soundness analysis [10, §5] tracks the list-decoding invariant at every round: at round i, the prover's v_i must avoid landing in a list-collision of $\Lambda_{\mathrm{RS}_{k_{i-1}}}(f^{(i-1)}, \delta_{i-1})$, with failure probability bounded via Lemma 4.5 of [10] by $\binom{\ell_i}{2} \cdot (k_{i-1}/|F|)^s$, where ℓ_i is the list size at threshold δ_{i-1}. At round $i = 1$ with input distance $\delta > \delta_J$ above the Johnson radius, the Johnson list-size bound does not apply, and STIR therefore requires (or implicitly assumes) a separate list-size hypothesis $\ell_1 \le L_{\max}$ at this threshold.*

The half-threshold CA framework of this paper bypasses the round-1 list-collision concern entirely. The reason is structural: our soundness invariant at round 1 is not the list-decoding invariant of STIR's standard analysis ("the prover's intended codeword stays in the list"), but the distance invariant $\Delta(f^{(1)}, \mathrm{RS}_{k_1}) > \delta/2$ on $L^{(1)}$, established via the half-threshold CA bound (Theorem 6) under hypothesis (RD) at $i = 1$. The half-threshold CA argument does not consult the size of the round-1 list-decoding ball; it only uses that some pair $(q_1, f^{(0)}\restriction_{L^{(1)}})$ has joint distance $> \delta$ from $\mathrm{RS}^{=2}_{k_1}$, which (RD) at $i = 1$ supplies. After round 1 the effective distance is $\delta/2 < \delta_J$, and Johnson list-decoding applies unconditionally for the rest of the protocol; the round-2 OOD-sampling list-collision contribution at threshold $\delta/2$ is the standard Johnson-bounded $O(k/|F|)$. No round-1 list-size hypothesis is needed in this framework.

Concretely: a round-1 list-size hypothesis like "$\ell_1 := \max_u |\Lambda_{\mathrm{RS}_k}(u, \delta)| \le L_{\max}$" is not needed in our analysis, because the failure mode it would have controlled (round-1 OOD-sampling list-collision) is replaced by the half-threshold CA failure (failure $\le 1/|F|$, unconditional in ℓ_1). The empirical / conjectural Open-Set Rank Lemma evidence for $\ell_1 = O(1)$ at FRI deployment scale (§7) remains an interesting structural complement, but is not load-bearing for the STIR soundness theorem. The (CQ)-to-arbitrary upgrade is closed under (RD-strong) by Theorem 54 (using Lemma 53); the STIR §5 distance-preservation re-derivation in our framework without either (CQ) or (RD-strong) — i.e., a single unified theorem accommodating both pointwise-adversarial q_i and tower-subgroup-aligned $f^{(0)}$ simultaneously — remains as a genuine follow-up gap (Remark 55).

Remark 51 (Scope of hypothesis (RD); how it is verified)**.** *Hypothesis (RD) is treated as an explicit input to the theorem, not a derived consequence. STIR [10, §5] controls a closely related restriction-distance / list-decoding invariant along honest ranges via its quotient/OOD analysis (Lemmas 4.4 + 4.5 of [10] applied at threshold δ for round 1 and at threshold $\delta/2$ for rounds ≥ 2). We do not re-derive the precise carry-forward in our notation; instead we treat (RD) as the explicit hypothesis of the theorem. This phrasing is honest about the dependency: at $i = 1$, (RD) is a structural assumption on the input function $f^{(0)}$ that does not follow from $\Delta(f^{(0)}, \mathrm{RS}_k) > \delta$ on L alone (the restriction $f^{(0)}\restriction_{L^{(1)}}$ can collapse to a degree-$<k_1$ polynomial on $L^{(1)}$ even when $f^{(0)}$ is far on L); at $i \ge 2$, the post-halving threshold $\delta/2 < \delta_J$ places the analogous list-decoding invariant inside the Johnson zone where the standard chain of Lemmas 4.4 + 4.5 of [10] applies. Under non-codeword-quotient deviations (not covered by (CQ)), the honest-portion guarantee no longer applies after a deviation round, and the (RD) carry-forward at later deviation rounds requires a separate downstream-propagation argument; this is the main remaining gap (Remark 52).*

Remark 52 (Multi-round extension within (CQ); what remains for non-(CQ))**.** *The multi-round extension within hypothesis (CQ) is closed by Lemma 49: the per-round shift-query catch (Lemma 4.4 of [10]) is unconditional in $f^{(i-1)}$, so it applies independently at each deviation round, and the joint miss probability is the product over the deviation set D, bounded by $(2\rho)^q$ regardless of $|D|$. The per-round batch-CA failure is similarly per-round and union-sums to at most $nR/|F|$. The restriction-distance (RD) carries forward across maximal honest ranges between deviation rounds via the standard STIR distance-preservation chain [10, §5]. The multi-round bound therefore matches the single-deviation bound up to the $|D|$-fold tightening of the shift-query factor, which the* max*-form $\max((1-\delta/2)^q, (2\rho)^q)$ already absorbs.*

The non-codeword-quotient case ($q_i \notin \mathrm{RS}_{k_i}$ at some deviation round) is closed under the strengthened input hypothesis (RD-strong) by Theorem 54 below, which uses Lemma 53 (the iterated-restriction proximity-gap chain). In summary: under (RD)/(CQ) the bound is Theorem 48; under (RD-strong) alone (no (CQ)) the bound is Theorem 54, with identical commit-phase rate and the rate-uniform query-phase term restored via STIR's standard final-round consistency check [10, Construction 5.2 Item 2]. See Remark 55 for the discussion of (RD-strong) vs (CQ) as comparable *input/adversary restrictions.*

Lemma 53 (Iterated-restriction carry-forward under arbitrary quotient)**.** *Adopt the setup of Theorem 48 with hypothesis (CQ) dropped (so each q_i is an arbitrary function in $F^{L^{(i)}}$ committed before γ_i). Replace hypothesis (RD) by:*

(RD-strong) *Iterated restriction-distance preservation on the input. For every level $j \in \{1, \ldots, R\}$:*

- *$\Delta(f^{(0)}\restriction_{L^{(j)}}, \mathrm{RS}_{k_j}) > \delta$ on $L^{(j)}$ when $j = 1$,*
- *$\Delta(f^{(0)}\restriction_{L^{(j)}}, \mathrm{RS}_{k_j}) > \delta/2$ on $L^{(j)}$ when $j \geq 2$.*

Then for every adaptive adversary strategy producing the chain $(f^{(0)}, q_1, \ldots, q_R)$ with arbitrary $q_i \in F^{L^{(i)}}$, the per-round restriction distance carries forward. Concretely, define the round-i level-j carry-forward event (for $j \geq i$):

$$E_{i,j} \;:=\; \left\{\Delta\big(f^{(i)}\restriction_{L^{(j)}},\, \mathrm{RS}_{k_j}\big) \leq \delta/2 \text{ on } L^{(j)}\right\}$$

with the convention $f^{(0)} := f^{(0)}$ (so $E_{0,j}$ is excluded by (RD-strong)). Then for every $1 \leq i < j \leq R$:

$$\Pr\big[E_{i,j} \mid \overline{E_{0,j}}, \overline{E_{1,j}}, \ldots, \overline{E_{i-1,j}}\big] \;\leq\; \frac{|L^{(j)}|}{|F|} \quad \text{over } \gamma_i.$$

Consequently:

(1) *Per-round (RD) at round i holds.* $\Pr[E_{i-1,i}] \leq (i-1) \cdot |L^{(i)}|/|F|$*, which supplies the (RD)-at-round-i hypothesis $\Delta(f^{(i-1)}\restriction_{L^{(i)}}, \mathrm{RS}_{k_i}) > \delta/2$ on $L^{(i)}$ at every round. Summed across rounds, total failure probability $\sum_{i=1}^{R}(i-1)|L^{(i)}|/|F| \leq n/|F|$ (geometric sum $\sum_{i\geq 1}(i-1)\cdot 2^{-i} = 1$, so $\sum(i-1)|L^{(i)}| = n\sum(i-1)/2^i \leq n$).*

(2) *Per-round batch-CA failure. Conditional on (RD) at round i (Item 1), the round-i batched function $f^{(i)} = q_i + \gamma_i \cdot f^{(i-1)}\restriction_{L^{(i)}}$ falls $(\delta/2)$-close to RS_{k_i} on $L^{(i)}$ with probability $\leq |L^{(i)}|/|F|$ over γ_i (BCHKS [6] proximity gap on $L^{(i)}$ for the pair $(q_i, f^{(i-1)}\restriction_{L^{(i)}})$, where the joint distance hypothesis is supplied by the individual distance of $f^{(i-1)}\restriction_{L^{(i)}}$ from (RD)). The round-1 contribution tightens to $\leq 1/|F|$ via Theorem 6 (half-threshold CA at threshold $\gamma_{\text{premise}} = \delta$).*

Proof. **Carry-forward step.** Fix $1 \leq i < j \leq R$ and condition on $\overline{E_{l,j}}$ for $l = 0, 1, \ldots, i-1$, i.e., on $\Delta(f^{(l)}\restriction_{L^{(j)}}, \mathrm{RS}_{k_j}) > \delta/2$ on $L^{(j)}$ for every $l < i$. By construction

$$f^{(i)}\restriction_{L^{(j)}} = \; q_i\restriction_{L^{(j)}} \; + \gamma_i \cdot f^{(i-1)}\restriction_{L^{(j)}},$$

where $L^{(j)} \subseteq L^{(i)}$ since $j \geq i$ in the descending tower. Apply the BCHKS [6] proximity gap on the domain $L^{(j)}$ for the code RS_{k_j} to the pair $(q_i\restriction_{L^{(j)}},\, f^{(i-1)}\restriction_{L^{(j)}})$ with random coefficient γ_i. The required joint-distance hypothesis $\Delta_{\text{joint}}\big((q_i\restriction_{L^{(j)}}, f^{(i-1)}\restriction_{L^{(j)}}), \mathrm{RS}_{k_j}^{=2}\big) > \delta/2$ on $L^{(j)}$ is implied by the individual distance $\Delta(f^{(i-1)}\restriction_{L^{(j)}}, \mathrm{RS}_{k_j}) > \delta/2$ on $L^{(j)}$ (which is the conditioning event $\overline{E_{i-1,j}}$): individual distance bounds joint distance for any second function (the second function's choice of nearest codeword cannot enlarge the agreement set beyond what $f^{(i-1)}$ alone allows). Hence the post-halving threshold $\delta/2 < (1-\rho)/2 \leq \delta_J$ places the BCHKS analysis in its zero-loss regime, and the bad-γ_i count is $\leq |L^{(j)}|$, giving $\Pr[E_{i,j}] \leq |L^{(j)}|/|F|$.

The base case $\overline{E_{0,j}}$ for $j = 1, \ldots, R$ is supplied by (RD-strong) directly: $\Delta(f^{(0)}\restriction_{L^{(j)}}, \mathrm{RS}_{k_j}) > \delta/2$ on $L^{(j)}$ (with $> \delta$ at $j = 1$).

Item (1). We bound $\Pr[\bigcup_{l=1}^{i-1} E_{l,i}]$ (which contains $\Pr[E_{i-1,i}]$ as a special case after restricting attention to the last index) by the chain-rule decomposition: writing $A_l := E_{l,i}$ and $B_l := \overline{A_1} \cap$

$\cdots \cap \overline{A_{l-1}}$,

$$\Pr\Big[\bigcup_{l=1}^{i-1} A_l\Big] \;\le\; \sum_{l=1}^{i-1}\Pr[A_l \cap B_l] \;\le\; \sum_{l=1}^{i-1}\Pr[A_l \mid B_l] \;\le\; \sum_{l=1}^{i-1}\frac{|L^{(i)}|}{|F|} \;=\; (i-1)\cdot\frac{|L^{(i)}|}{|F|},$$

where the third inequality is the carry-forward step applied at level $j = i$ to round l (the carry-forward step gives $\Pr[A_l \mid B_l] \le |L^{(i)}|/|F|$ since B_l is exactly the conditioning event $\overline{E_{0,i}}, \overline{E_{1,i}}, \ldots, \overline{E_{l-1,i}}$ required by the carry-forward step's hypothesis). Since $E_{i-1,i} \subseteq \bigcup_{l=1}^{i-1} A_l$, the same bound applies to $\Pr[E_{i-1,i}]$.

Summed across rounds:

$$\sum_{i=1}^{R}(i-1)\cdot\frac{|L^{(i)}|}{|F|} \;=\; \frac{1}{|F|}\sum_{i=1}^{R}(i-1)\cdot\frac{n}{2^i} \;\le\; \frac{n}{|F|}\cdot\sum_{i=1}^{\infty}\frac{i-1}{2^i} \;=\; \frac{n}{|F|},$$

using the identity $\sum_{i\ge 1}(i-1)2^{-i} = 1$.

Item (2). Standard BCHKS [6] application on $L^{(i)}$, exactly as in Strategy B item (2) of the proof of Theorem 48. The round-1 tightening to $\le 1/|F|$ via Theorem 6 carries over verbatim, since (RD-strong) at $j = 1$ supplies $\Delta(f^{(0)}\restriction_{L^{(1)}}, \mathrm{RS}_{k_1}) > \delta$ on $L^{(1)}$ (the half-threshold CA premise threshold). □

Theorem 54 (STIR Soundness Above Johnson, arbitrary-quotient model)**.** *Adopt the setup of Theorem 48 (multiplicative subgroup or coset L closed under negation, $k = 2^m$, radix-2 schedule, $|F| \ge 2|L^{(i)}|$ at every round), with hypothesis (CQ) dropped and hypothesis (RD) replaced by (RD-strong) of Lemma 53. Assume STIR with R rounds, the standard OOD sample at each round, q i.i.d. shift queries per round, and $t_M = q$ final-round consistency queries on $f^{(R)}$ at uniform points of $L^{(R)}$ ([10, Construction 5.2 Item 2]; the standard STIR final check). Then*

$$\Pr[\text{STIR accepts}] \;\le\; \frac{(2k+n)R}{|F|} \;+\; \big(1-\tfrac{\delta}{2}\big)^q,$$

matching the rate-uniform $\rho \le 1/4$ form of Theorem 48's bound while admitting arbitrary (non-codeword) prover quotients q_i.

Proof. The argument follows the Strategy A / Strategy B dichotomy of Theorem 48's proof, with three modifications driven by Lemma 53:

Strategy A (honest fold). Identical commit-phase analysis as Theorem 48: the union bound over per-round OOD ($\le 2k_{i-1}/|F|$ each) and per-round Schwartz–Zippel ($\le 1/|F|$ each, batched into the OOD bound) gives commit-phase contribution $\le (2k+1)R/|F| \le (2k+n)R/|F|$. At the final-round consistency check, the prover's claimed polynomial $\hat{p} \in \mathrm{RS}_{k_R}$ may disagree with the honestly-folded $f^{(R)}$ on $L^{(R)}$; under (RD-strong) at $j = R$, $\Delta(f^{(R)}, \mathrm{RS}_{k_R}) > \delta/2$ on $L^{(R)}$ in Strategy A as well (the honest fold preserves the restriction-distance invariant via the Schwartz–Zippel argument; equivalently, the same Lemma 53 carry-forward applies trivially when each q_i is the honest quotient), so the q random final-check queries catch the disagreement with probability $\ge 1 - (1-\delta/2)^q$, giving the same query-phase term as Strategy B.

Strategy B (deviation, arbitrary q_i). At every round i (deviation or honest), the per-round commit-phase contribution decomposes into:

- OOD evaluation degeneracy, $\le 2k_{i-1}/|F|$ per round, summed to $\le 2kR/|F|$ across rounds — same as in Strategy B of Theorem 48 (Lemma 4.4 of [10] on the at-most-k_{i-1} degenerate values of z_i, plus Schwartz–Zippel over γ_i, both unconditional in $f^{(i-1)}$ and q_i). This bound applies at every round regardless of whether the prover was honest or deviated, because the OOD step is independent of the deviation classification.

- Per-round (RD) carry-forward, from (RD-strong) on the input via Lemma 53 Item (1) — total contribution $\leq n/|F|$.
- Per-round batch-CA failure, $\leq |L^{(i)}|/|F|$ over γ_i, summed to $\sum_{i=1}^{R} |L^{(i)}|/|F| \leq 2n/|F|$ across rounds (geometric sum of $|L^{(i)}| = n/2^i$ is bounded by $2n$) — Lemma 53 Item (2).

The shift-query check provides no catch in this regime: for arbitrary q_i, the prover may pointwise set $q_i(y) := (f^{(i-1)}(y) - v_i)/(y - z_i)$ on every $y \in L^{(i)}$, satisfying the OOD identity deterministically. This loses the per-round $(2\rho)^q$ catch from Lemma 4.4 of [10] that was active under (CQ).

Final-round consistency check supplies the query-phase term. Conditional on the high-probability event from Strategy B (Item (1) and (2)), the final-round function $f^{(R)}$ satisfies $\Delta(f^{(R)}, \mathrm{RS}_{k_R}) > \delta/2$ on $L^{(R)}$. STIR's final-round consistency check ([10, Construction 5.2 Item 2]) at $t_M = q$ uniform points of $L^{(R)}$ then catches with probability $\geq 1 - (1 - \delta/2)^q$ (each query catches a $(\delta/2)$-far point with probability $\geq \delta/2$, and the q queries are independent); this is the standard ε_{fin} bound of [10, Lemma 5.4] adapted to our post-halving threshold $\delta/2$.

Combining. Every adaptive prover either commits honest folds (Strategy A) or deviates somewhere (Strategy B); both yield the same commit-phase total and the same query-phase term, so

$$\Pr[\text{accept}] \;\leq\; \frac{(2k+n)R}{|F|} + \left(1 - \tfrac{\delta}{2}\right)^q.$$

Constants: the OOD contribution sums to $\leq 2kR/|F|$; the Lemma 53 Item (1) carry-forward sums to $\leq n/|F|$ (via the geometric identity, see proof); the Item (2) per-round batch-CA contribution sums to $\sum_{i=1}^{R} |L^{(i)}|/|F| \leq 2n/|F|$ (geometric sum of $|L^{(i)}| = n/2^i$ is bounded by $2n$). Total $\leq 2kR/|F| + 3n/|F| \leq (2k+n)R/|F|$ for every $R \geq 3$ (where $3n \leq nR$); for $R \leq 2$ the same bound holds with the slightly tighter form $(2k+n)R/|F| + (3-R)n/|F|$ on the right-hand side (a constant adjustment of at most $3n/|F|$ in either case, absorbed into the displayed bound). At deployment-scale regime $|F|/|L^{(i)}| \to \infty$ at fixed rate, the factor of 2 in $2k_{i-1}/|F|$ collapses, tightening to $(k+n)R/|F| + (1-\delta/2)^q$. □

Remark 55 ((RD-strong) vs (CQ): comparison of input/adversary restrictions)**.** *Theorems 48 and 54 provide two complementary above-Johnson STIR soundness bounds with the same rate-uniform query-phase term $(1-\delta/2)^q$ at $\rho \leq 1/4$ and the same commit-phase rate $(2k+n)R/|F|$, differing only in their structural hypotheses:*

- *Theorem 48 (**adversary restriction (CQ)**): every prover deviation commits a quotient $q_i \in \mathrm{RS}_{k_i}$, i.e., a low-degree codeword. This restricts the adversary's commit space; it is satisfied by every prover that runs the standard STIR construction honestly except for q_i-degree-class boundary behavior, but excludes pointwise-adversarial q_i's that pass the shift-query check by pointwise construction.*
- *Theorem 54 (**input restriction (RD-strong)**): the prover-committed input $f^{(0)}$ has restriction-distance $> \delta$ (resp. $> \delta/2$) at every level $L^{(j)}$ of the radix-2 tower. This is genuinely stronger than $\Delta(f^{(0)}, \mathrm{RS}_k) > \delta$ on L alone: a sharp counterexample is $f^{(0)}(x) = x^{n/2^j}$ for any $j \geq 2$ with $n/2^j \geq k$, which is $(1-2^{-j})$-far from $\mathrm{RS}_k(L)$ on L but restricts to the constant 1 on $L^{(j)} \subset L$ (since $x^{n/2^j} = 1$ for every $x \in L^{(j)}$ of order dividing $n/2^j$), giving $\Delta(f^{(0)}\restriction_{L^{(j)}}, \mathrm{RS}_{k_j}) = 0$ on $L^{(j)}$.*

Neither hypothesis dominates the other. An adversary that respects (CQ) but violates (RD-strong) (e.g. committing the monomial counterexample above as input and an honest codeword as q_i) is bounded by Theorem 48; an adversary that violates (CQ) but respects (RD-strong) is bounded by Theorem 54. The intersection is bounded by either. The "true" above-Johnson STIR soundness theorem with neither hypothesis, accommodating both pointwise-adversarial q_i and tower-subgroup-aligned $f^{(0)}$ simultaneously, would require a deeper structural argument (presumably re-deriving the *full STIR §5 analysis [10, §5] in the half-threshold framework); we identify this as a follow-up gap. An adversary that violates both (e.g. the monomial counterexample combined with a pointwise-adversarial q_i) is bounded by neither theorem.*

The $(2\rho)^q$ branch of the rate-uniform max in Theorem 48 arose from STIR's Lemma 4.4 shift-query catch under (CQ); under (RD-strong) (Theorem 54) the shift-query check is defeated and the query-phase term comes solely from the final-round consistency check, giving $(1-\delta/2)^q$ alone. This is the same $(1-\delta/2)^q$ that dominates the rate-uniform max at $\rho \le 1/4$ (Remark 56), so for STIR's standard deployment regime the two theorems give numerically identical bounds; the difference is the structural hypothesis swap, not a quantitative one. At $\rho > 1/4$ the max in Theorem 48 can be the larger $(2\rho)^q$ branch (when $2\rho > 1-\delta/2$), so its query-phase factor is then worse than the $(1-\delta/2)^q$ of Theorem 54; the regime where Theorem 48 is quantitatively preferable is precisely $\rho \le 1/4$ (where the two coincide) and never strictly better.

Remark 56 (Query-phase regime decomposition; rate beyond 1/4)**.** *The query-phase bound* $\max((1-\delta/2)^q, (2\rho)^q)$ *admits the following regime decomposition on the standard radix-2 schedule with $\rho_i = 2\rho$:*

- $\rho \le 1/4$*:* $(2\rho)^q \le (1/2)^q \le (1-\delta/2)^q$ *for every $\delta < 1$, so the* max *collapses to* $(1-\delta/2)^q$*. This is the regime considered in STIR's standard deployment ([10], §1) and the regime where the $(1-\delta/2)^q$ improvement over BCHKS [6] is the active one.*
- $\rho \in (1/4, 1/2)$*:* $(2\rho)^q \in ((1/2)^q, 1)$ *may exceed* $(1-\delta/2)^q$*, so the* max *is genuinely needed; the bound retains a non-trivial query-phase factor but the $2\times$ overhead vs. the conjectured baseline shrinks toward the equality case.*
- $\rho = 1/2$*: the per-round STIR shift-query parameter $\rho_i = 1$, so the shift-query bound is vacuous; security at this rate must come from the FRI subprotocol consistency check (i.e., from the underlying base-protocol analysis), which is outside the scope of the present theorem.*

The max*-form bound is rate-uniform on $\rho \in [0, 1/2)$: the underlying Lemma 4.4 [10] catch probability $1 - \rho_i^q$ is what enters, and the choice between $(1-\delta/2)^q$ and $(2\rho)^q$ is purely a question of which side of the inequality $2\rho \lesseqgtr 1-\delta/2$ the parameters land on. At $\rho = 1/2$ the shift-query bound becomes vacuous and the rate restriction is genuine.*

Remark 57 (Comparison with conjectured STIR bound)**.** *The original STIR analysis [10] assumes zero-loss CA ($\varepsilon_{\mathrm{ca}}(C, \delta, \delta) = O(1)/|F|$, conjectured) and achieves an analogous commit term plus $(1-\delta)^q$ in the query phase. The intrinsic OOD contribution $O(k/|F|)$ per round ([10] Lemma 4.4) is independent of which CA bound is invoked. Our proved STIR bound has two differences:*

- *Commit phase: $(2k+n)R/|F|$ unconditionally (and $(k+n)R/|F|$ at deployment scale, see Theorem 48) instead of $O(R)/|F|$. The k-factor comes from the OOD polynomial evaluation at the random point z_i (degree $\le k_{i-1}$, giving k_{i-1} degenerate values). This is intrinsic to STIR's quotient structure and absent in plain FRI. On extension fields ($|F| = p^4$), this term is negligible; on the base field ($|F| = p$), it dominates.*
- *Query phase:* $\max((1-\delta/2)^q, (2\rho)^q)$ *instead of $(1-\delta)^q$. At STIR's typical deployment regime $\rho \le 1/4$ the* max *collapses to $(1-\delta/2)^q$, the same $\sim 2\times$ overhead as for FRI; for $\rho \in (1/4, 1/2)$ the $(2\rho)^q$ term enters, see Remark 56.*

For BabyBear$_4$ ($|F| \approx 2^{124}$, $k = 2^{19}$, $R = 20$): the commit term gives $\approx 2^{-100}$ bits (negligible), and the query term is the bottleneck as for FRI. On the base field BabyBear ($|F| \approx 2^{31}$): the commit term gives only $\sim$7 bits, comparable to BCHKS. The improvement over BCHKS for base-field STIR is entirely in the query phase.

A.3 WHIR Extension

WHIR [11] achieves super-fast verification by working with *constrained* Reed–Solomon codes: linear subcodes $C' \subseteq \mathrm{RS}_k$ cut out by additional linear constraints, and affine constraint sets $\mathrm{RS}_k \cap \{f : \langle a_j, f\rangle = b_j\}$ for the inhomogeneous (OOD-evaluation, multilinear-evaluation) cases. The CA bound (Theorem 6) uses only one property of the code:

Linearity (closure under F-linear combinations).

This is preserved by every *linear* subcode of RS_k over the same domain L. We make the linear-subcode reduction explicit (Corollary 58), then build up to the full WHIR soundness theorem (Conjecture 62) via a per-iteration analysis that handles affine constraints through the sumcheck weight polynomial rather than through CA on affine codes.

Corollary 58 (Linear-subcode CA)**.** *For any integer $B \geq 1$, any threshold δ, and any **linear** subcode $C' \subseteq \mathrm{RS}_k$ over domain L:*

$$\varepsilon_{\mathrm{ca}}(C', \delta/2, \delta) \leq 1/|F|, \qquad \varepsilon_{\mathrm{ca}}^{(B)}(C', \delta/2, \delta) \leq 1/|F|.$$

Proof. The proofs of Theorems 6 and 46 invoke linearity in exactly one place: constructing $g_1, g_2 \in C$ from $h_1, h_2 \in C$ via F-linear combinations. Linear subcodes are closed under such combinations, so the proofs go through verbatim. □

Remark 59 (Why this does not extend to affine subcodes)**.** *For an affine subcode $C' = g_0 + C_0$ with $g_0 \notin C_0$: the construction $g_2 = (c_\gamma - c_{\gamma'})/(\gamma - \gamma')$ in the proof of Theorem 6 produces an element of C_0 (a difference, not a sum), not of C'. Hence we cannot conclude $g_2 \in C'$. WHIR addresses this by enforcing affine constraints at a different layer (the sumcheck weight polynomial), not by extending CA to affine codes. The CA bound is only invoked on the ambient $\mathrm{RS}_{k/2^i}$ at each folding step.*

WHIR protocol structure: what we replace

We adopt the notation of [11] (Construction 5.1). One WHIR iteration consists of:

(i) **OOD sample.** Verifier sends $z \in F \setminus L$; prover responds with a claimed evaluation $y = f(z)$ and continues with the augmented constraint set $\{(z_j, y_j)\}_j \cup \{(z, y)\}$.

(ii) **Sumcheck folding.** Verifier sends folding randomness $\alpha \in F$; the constrained RS code $C^{(i)}$ folds via the sumcheck weight-polynomial machinery (Definition 4.5 of [11]) to a constrained RS code $C^{(i+1)}$ of half the dimension on the half-domain $L^{(i+1)}$.

(iii) **Shift queries.** Verifier samples t i.i.d. points from $L^{(i+1)}$ and checks consistency of the folded function with the prover's commitment.

After M iterations the protocol terminates with q i.i.d. final queries on the smallest folded domain.

Where MCA enters in [11]: only at step (ii), the sumcheck folding. WHIR's analysis (Theorem 4.20 of [11]) bounds the probability that a δ-far function f folds to a δ-close folded function via the MCA error of the constrained RS code at threshold δ. Above the Johnson radius, this requires Conjecture 4.12 of [11] (zero-loss MCA in the open zone). *This is the conjecture we remove from the folding-term / proximity layer* (Lemma 63); the surrounding iteration-level induction of [11, §5] would need to be rerun under the substituted folding term to upgrade Conjecture 62 from a conjecture to a theorem. The OOD sample (i) and shift queries (iii) do not invoke MCA; their soundness is bounded directly by polynomial-identity testing and Lemma 19 respectively.

Substitution: half-threshold CA + proved MCA in the Johnson regime

We replace WHIR's MCA at threshold δ with a two-stage argument:

- *Iteration 1, first folding round only:* apply our half-threshold CA (Theorem 6) on the linear ambient RS_k via Corollary 58. This drops the effective distance from δ (premise) to $\delta/2$ (conclusion), with $\leq 1/|F|$ probability of escape. *This is the only invocation of half-threshold CA in the entire WHIR proof.*
- *All subsequent folding rounds (iteration 1 rounds* $2, \ldots, r_1$ *and all folds of iterations* $2, \ldots, M$*):* the effective distance $\delta/2$ satisfies $\delta/2 < (1-\rho)/2 \leq \delta_J$ (Remark 1), so the protocol enters BCIKS's zero-loss regime (the proved Johnson regime; strictly larger than the unique-decoding ball). WHIR's MCA Corollary 4.11 (proved unconditionally in [11] for $\delta < \delta_J$) applies, giving $\leq n/|F|$ bad challenges per round. We deliberately do *not* re-apply half-threshold CA at later folds: doing so would compound the halving to $\delta/4, \delta/8, \ldots$.

The key structural point: *WHIR's Conjecture 4.12 is invoked only above Johnson; below Johnson, MCA is proved in [11].* Our one-time half-threshold step at iteration 1's first fold brings the effective distance below $(1-\rho)/2 \leq \delta_J$, so all subsequent folds (in iteration 1 and all later iterations) use the proved MCA, never the conjecture.

Affine constraints: handled by sumcheck, not by CA

WHIR's OOD constraints define an affine subcode $C' = g_0 + C_0$ where $C_0 \subseteq \mathrm{RS}_k$ is linear and g_0 encodes the constraint values. The CA bound (Corollary 58) applies only to the linear part, but in WHIR the constraints are not enforced via CA: they are absorbed into the sumcheck weight polynomial $\mathrm{wt}(X)$ (Definition 4.5 of [11]). The folding step (ii) then asks the prover to maintain WHIR's wt-folded sumcheck identity ([11, Construction 5.1, sumcheck step]) over $L^{(i)}$ round by round. This identity is degree-≤ 2 in the sumcheck round variable, with soundness $\leq 2/|F|$ per round (Schwartz–Zippel). The CA bound enters only at the proximity-preserving step that maps $f^{(i)} \to f^{(i+1)}$, where the underlying code is the linear $\mathrm{RS}_{k/2^i}$. Affine vs. linear is therefore handled by separation of concerns: **sumcheck verifies constraint satisfaction, CA verifies proximity**.

Per-iteration error budget

Lemma 60 (Per-iteration commit-phase error, conditional)**.** *Under the above setup and conditional on the cross-iteration distance preservation invariant of Conjecture 62 (which is what currently keeps Theorem 20's analogue a conjecture for WHIR), one WHIR iteration with r folding rounds has commit-phase soundness error (failures from random alpha/sumcheck challenges, not from query sampling)*

$$\varepsilon_{\mathrm{commit}}^{\mathrm{iter}} \leq \frac{c_{\mathrm{OOD}} k}{|F|} + \frac{r\,n}{|F|} + \frac{2r}{|F|},$$

with terms: OOD ($c_{\mathrm{OOD}} k/|F|$, where $c_{\mathrm{OOD}} = 2^{m_i}\ell_{i,0}^2/(2k)$ is the deployment-regime constant from [11, Lemma 4.25], of order 1 *at the WHIR parameter regime of §A.3), folding rounds at $\leq n$ bad challenges each ($rn/|F|$), and sumcheck rounds ($2r/|F|$). Shift queries are handled separately (Remark 61).*

Proof. OOD ($k/|F|$). Imported as a black box from [11, Lemma 4.25]: under the WHIR parameter regime (multilinear-extension degree $m_i \leq \log_2 k$, classical Johnson list-size $\ell_{i,0} = O(1/\sqrt{1-\rho})$ uniform in n in the Johnson zone, $|F| \geq 2|L|$), the OOD-evaluation soundness error is $\varepsilon_i^{\mathrm{out}} \leq 2^{m_i}\ell_{i,0}^2/(2|F|)$, which simplifies to $O(k/|F|)$ at the deployment regime. We write $k/|F|$ for this bounded contribution. The proof of [11, Lemma 4.25] is a Schwartz–Zippel argument applied to a degree-m_i power map on the OOD challenge z_i together with the classical Johnson list-size bound; it does *not* model the prover's oracle as a low-degree polynomial. We do not re-derive this bound here; Lemma 63 (i) verifies that this term has no MCA dependency.

Folding rounds ($rn/|F|$). Iteration 1's first fold uses half-threshold CA: the input is δ-far, Corollary 58 gives at most 1 bad α for $(\delta/2)$-closeness ($\le 1/|F|$, tighter than $n/|F|$). *All other folds* (iteration 1's rounds $2, \dots, r$ and *all* folds of iterations ≥ 2) use the BCHKS [6] proximity gap (refining BCIKS [7] from $O(n^2)$ to $\le n$) at threshold $\delta/2 < (1-\rho)/2 \le \delta_J$ (the input is $(\delta/2)$-far by distance preservation across iterations, Remark 21; we deliberately do *not* re-apply the half-threshold CA at iterations ≥ 2, which would compound the distance halving). BCHKS gives $\le n$ bad challenges per round, $\le rn/|F|$ per iteration. We state $rn/|F|$ as a uniform upper bound covering all r folds (loose by $\frac{n-1}{|F|}$ for iteration 1's first fold).

Sumcheck ($2r/|F|$). Per-round Schwartz–Zippel on the degree-≤ 2 sumcheck identity gives $2/|F|$; total $2r/|F|$.

Union over the three commit-phase failure modes gives the stated bound. ☐

Remark 61 (Shift queries and query-phase max-over-strategies)**.** *The prover commits to a strategy before seeing verifier randomness. Adversarial strategies (mutually exclusive choices) and their query-phase failure probabilities:*

- *Strategy A: honest fold throughout. The cumulative oracle $f^{(M)}$ is $(\delta/2)$-far by distance preservation. Final q queries miss with probability $\le (1-\delta/2)^q$ (Lemma 19).*
- *Strategy B: commit a codeword $f^{(i+1)} \in C^{(i+1)}$ in iteration i instead of the honest fold; then continue honestly. The discrepancy between the committed codeword and the honest fold is $(\delta/2)$-far on $L^{(i+1)}$. Iteration-i shift queries miss this with probability $\le (1-\delta/2)^t$. (Subsequent iterations' shift queries pass automatically: codewords fold to codewords, so consistency is preserved from iteration $i+1$ onward; final queries also pass since $f^{(M)}$ is a codeword.)*
- *Strategy B-multi: deviate at multiple iterations $\{i_1, \dots, i_s\}$ for some $s \ge 1$. The events "shift queries miss in iteration i_j" are independent across iterations (different randomness), so joint miss $\le (1-\delta/2)^{t\cdot s} \le (1-\delta/2)^t$ (using $s \ge 1$).*

The prover picks ONE strategy; the verifier's failure probability is the maximum over strategies, not the sum:

$$\Pr[\textit{query-phase miss}] \le \max\big((1-\delta/2)^q, (1-\delta/2)^t\big) = (1-\delta/2)^q,$$

using $t \ge q$ to make the shift-query bound tighter. No factor of M arises: a single chosen strategy can only escape its single failure mode.

Conjecture 62 (WHIR Soundness Above Johnson, conjectural)**.** *For* $\mathrm{RS}[F, L, k]$ *with $k = 2^m$, L admitting a fixed-point-free involution, $\delta_J < \delta < 1-\rho$, an initial constrained Reed–Solomon code $C^{(0)} \subseteq \mathrm{RS}_k$ defined by linear and affine constraints (multilinear-evaluation and OOD), and the standard WHIR parameter regime of [11, §5] ($|F| \ge 2|L|$ at every iteration, multilinear-extension degree $m_i \le \log_2 k$, with the classical Johnson list-decoding parameter $\ell_{i,0} \le \lceil 1/\sqrt{1-\rho} \rceil$ in the Johnson zone, and per-iteration shift count t_{i-1} chosen as in [11, §5] so that the WHIR-internal absorption $t_{i-1}\ell_{i,0} \le r_i \cdot c_{\mathrm{WHIR}}$ holds for an explicit constant c_{WHIR}): **provided the prover-committed initial oracle** $f^{(0)} : L \to F$ **satisfies** $\Delta(f^{(0)}, C^{(0)}) > \delta$, WHIR [11] (Construction 5.1) with M iterations, r_i folding rounds in iteration i, $t \ge q$ shift queries per iteration, and q final queries satisfies*

$$\Pr[\text{WHIR accepts}] \le \frac{C_{\mathrm{OOD}} Mk + (n+2)R}{|F|} + \left(1 - \frac{\delta}{2}\right)^q,$$

where $R = \sum_{i=1}^{M} r_i$ *is the total number of FRI-folding rounds across all iterations and* $C_{\mathrm{OOD}} := \max_i 2^{m_i} \ell_{i,0}^2/(2k)$ *is the iteration-uniform upper bound on the OOD constants from Lemma 60 (of order* 1 *at the WHIR parameter regime above).*

Status. The MCA-orthogonality of the OOD and shift terms in WHIR Theorem 5.2's four-term decomposition is established rigorously in Lemma 63; this licenses replacing the MCA-bearing folding terms $\varepsilon_{0,s}^{\mathrm{fold}} + \varepsilon_{i,s}^{\mathrm{fold}}$ *by our half-threshold-CA + BCIKS chain at the term level, and removes any dependency on WHIR Conjecture 4.12 from the proximity layer. However, completing the proof to a formal theorem requires re-running the iteration-level induction of [11, §5]'s Theorem 5.2 with the substituted folding-term bound across all* M *iterations, which involves checking that the case-(a)/case-(b) recursion and constraint-set accumulation in WHIR's argument remain sound under the substitution. We leave this re-derivation to subsequent work; the conjecture above is the bound the substitution would deliver if the iteration-level analysis goes through unchanged.*

Evidence and partial justification. The conjecture is what one would obtain by substituting our half-threshold-CA + BCIKS chain for the MCA-bearing folding terms in WHIR's Theorem 5.2; the proximity-layer substitution is rigorous (Lemma 63); the surrounding iteration-level analysis from [11, §5] is stated here in sketch form for context but is not re-derived in full. We retain the case-split structure and Merkle-binding analysis that motivate the iteration-level argument.

Case split on ambient vs. constrained distance. Two cases:

Case (a): $\Delta(f^{(0)}, \mathrm{RS}_k) > \delta$*.* The ambient RS distance is $> \delta$, so Corollary 58 applies directly: half-threshold CA on the linear ambient RS_k at iteration 1's first fold drops the effective distance to $\delta/2$ with $\leq 1/|F|$ escape (subsumed by the $rn/|F|$ uniform bound of Lemma 60).

Case (b): $\Delta(f^{(0)}, \mathrm{RS}_k) \leq \delta$ *but* $\Delta(f^{(0)}, C^{(0)}) > \delta$*.* Some witness $g \in \mathrm{RS}_k$ with $|\{x \in L : f^{(0)}(x) \neq g(x)\}| \leq \delta n$ exists, but every such witness violates at least one constraint defining $C^{(0)} \subsetneq \mathrm{RS}_k$. We must show that, even though the proximity layer cannot reject (the ambient distance is small), the sumcheck/OOD layer rejects.

Pre-query case-determination. The case split is made on the *committed* oracle $f^{(0)} : L \to F$ before any verifier randomness. The Merkle commitment binds $f^{(0)}$ as a function on all of L at commit time (see [20, 11] for the standard binding analysis); the ambient distance $\Delta(f^{(0)}, \mathrm{RS}_k) = \min_{g \in \mathrm{RS}_k} |\{x \in L : f^{(0)}(x) \neq g(x)\}|/n$ is then a property of the committed function, independent of which positions are subsequently queried. The prover cannot retroactively shift $f^{(0)}$ between Case (a) and Case (b) based on query outcomes (modulo the standard collision term $3(Q^2{+}1)/2^{\kappa}$ for κ-bit Merkle binding under Q random-oracle queries, absorbed into the Fiat–Shamir budget).

Black-box reduction to WHIR's sumcheck/OOD soundness. Conditional on Case (b), we invoke *as black boxes* the per-iteration bounds in WHIR's main round-by-round soundness theorem ([11], Theorem 5.2 of Construction 5.1), under the standard WHIR parameter regime of [11, §5] as instantiated in our deployment table §9 (extension fields $|F| = p^d$ with $d \geq 4$, $|F| \gg \mathrm{poly}(k)$, multilinear extension degree $m_i \leq \log_2 k$, Johnson-zone list-decoding parameter $\ell_{i,0} \leq O(1/\sqrt{1-\rho})$):

- *Out-of-domain (OOD) bound* $\varepsilon_i^{\mathrm{out}} \leq 2^{m_i} \ell_{i,0}^2/(2|F|)$, proved in [11] via Lemma 4.25 (Schwartz–Zippel / list-decoding on a degree-m_i power map). At the deployment regime above, $\varepsilon_i^{\mathrm{out}} = O(k/|F|)$; we write $k/|F|$ for this term throughout.
- *Shift / sumcheck-consistency bound* $\varepsilon_i^{\mathrm{shift}} \leq (1{-}\delta_{i-1})^{t_{i-1}} + \ell_{i,0}(t_{i-1}{+}1)/|F|$, proved via Claim 5.4 of [11] (a polynomial identity in the combination randomness γ_i) plus a standard random-point miss bound; we write $2r_i/|F|$ for the commit-phase $\ell_{i,0}(t_{i-1}{+}1)/|F| = O(r_i/|F|)$ contribution at the deployment regime, and absorb the $(1{-}\delta_{i-1})^{t_{i-1}}$ query-phase contribution into our $(1{-}\delta/2)^q$ term (Remark 61).

The constants k and $2r_i$ above are the WHIR-paper figures at the standard deployment regime ([11, §5]); we do not re-derive them. **Crucially, neither bound invokes WHIR Conjecture 4.12 or**

the MCA error $\mathrm{err}^\star$**.** Tracing dependencies through the proof of Theorem 5.2 of [11], the round-i commit-phase escape probability is decomposed into four terms $\varepsilon_{0,s}^{\mathrm{fold}}, \varepsilon_i^{\mathrm{out}}, \varepsilon_i^{\mathrm{shift}}, \varepsilon_{i,s}^{\mathrm{fold}}$ (the equation immediately preceding the proof of [11, Theorem 5.2]; we adopt their term names here). Each term has a derivational lineage in [11] that we trace in Lemma 63 below; the conclusion is that the OOD term $\varepsilon_i^{\mathrm{out}}$ and the shift term $\varepsilon_i^{\mathrm{shift}}$ have no MCA dependency, while $\varepsilon_{0,s}^{\mathrm{fold}}$ and $\varepsilon_{i,s}^{\mathrm{fold}}$ do, and the latter are exactly what the half-threshold CA + BCIKS zero-loss chain of §5 replaces (one halving step at iteration 1's first fold, which lands $\delta/2 < \delta_J$ by Remark 1, then BCIKS distance-locking for all subsequent folds). The OOD and shift/sumcheck layers are imported unchanged.

Lemma 63 (MCA-orthogonality of OOD and shift terms in WHIR Theorem 5.2)**.** *In the proof of [11, Theorem 5.2]:*

(i) $\varepsilon_i^{\mathrm{out}}$ *is bounded in [11, Lemma 4.25] by Schwartz–Zippel applied to a single degree-*m_i *polynomial in the OOD challenge* $z_i \in F$*; its proof invokes only* $|F|$*, the multilinear-extension degree* m_i*, and the list-decoding parameter* $\ell_{i,0}$ *via the Johnson list-size bound ([11, Theorem 3.4], classical Johnson). No MCA conjecture, no Conjecture 4.12, no* $\mathrm{err}^\star$ *ever appears in the chain.*

(ii) $\varepsilon_i^{\mathrm{shift}}$ *is bounded in [11, Claim 5.4] by a polynomial-identity argument in the batching randomness* γ_i *plus an elementary Schwartz–Zippel estimate; its derivation cites [11, Lemma 4.4 (function-quotienting)] (unconditional polynomial algebra) and [11, Theorem 3.4 (Johnson list-size)]. No MCA conjecture appears.*

(iii) $\varepsilon_{0,s}^{\mathrm{fold}}$ *and* $\varepsilon_{i,s}^{\mathrm{fold}}$ *are the proximity-preservation / mutual-CA terms; they are bounded in [11, Theorem 4.20] by the MCA error* $\mathrm{err}^\star$*, which [11] controls in turn via Conjecture 4.12. These are the only two terms in the decomposition that invoke MCA.*

Hence, replacing $\varepsilon_{0,s}^{\mathrm{fold}} + \varepsilon_{i,s}^{\mathrm{fold}}$ *by our half-threshold-CA + BCIKS chain leaves* $\varepsilon_i^{\mathrm{out}}$ *and* $\varepsilon_i^{\mathrm{shift}}$ *unchanged and removes every dependency on Conjecture 4.12.*

Proof of Lemma 63. The lemma is a tracing exercise on the proof of [11, Theorem 5.2]. (i) The chain $\varepsilon_i^{\mathrm{out}} \leftarrow$ [11, Lemma 4.25] $\leftarrow$ [11, Theorem 3.4 (Johnson)] has no node referencing [11, Conjecture 4.12 / Theorem 4.20]; the only inputs are $|F|$, m_i, $\ell_{i,0}$, and the unconditional Johnson list-size bound. (ii) The chain $\varepsilon_i^{\mathrm{shift}} \leftarrow$ [11, Claim 5.4] $\leftarrow$ [11, Lemma 4.4, Theorem 3.4] similarly has no MCA-bearing node. (iii) For the folding terms, $\varepsilon_{0,s}^{\mathrm{fold}}, \varepsilon_{i,s}^{\mathrm{fold}} \leftarrow$ [11, Theorem 4.20] $\leftarrow$ [11, Conjecture 4.12] via the proof of [11, Theorem 4.20], which invokes mutual correlated agreement at the iteration's pre-fold distance. These three derivational chains are disjoint in [11] (no shared lemma other than the Schwartz–Zippel lemma, which is unconditional), so substituting half-threshold CA for the MCA-bearing chain in (iii) leaves (i)–(ii) intact. ☐

Black-box term substitution. Rather than re-derive WHIR's per-iteration recursion from scratch, we apply Lemma 63 as a *term-substitution lemma* on top of WHIR's existing Theorem 5.2 derivation in [11]. Specifically:

(1) Take the round-i commit-phase escape budget of [11, Theorem 5.2], which has the four-term form $\varepsilon_{0,s}^{\mathrm{fold}} + \varepsilon_i^{\mathrm{out}} + \varepsilon_i^{\mathrm{shift}} + \varepsilon_{i,s}^{\mathrm{fold}}$ as recalled above.

(2) By Lemma 63, $\varepsilon_i^{\mathrm{out}}$ and $\varepsilon_i^{\mathrm{shift}}$ have no MCA dependency and we import them unchanged from [11] (with the deployment-regime evaluation $k/|F|$ and $2r_i/|F|$ stated above).

(3) The MCA-bearing folding terms $\varepsilon_{0,s}^{\mathrm{fold}} + \varepsilon_{i,s}^{\mathrm{fold}}$ are bounded by [11, Theorem 4.20] via $\mathrm{err}^\star$, which [11] controls via Conjecture 4.12. We *replace* this MCA chain with the half-threshold-CA chain of §5: at iteration 1's first fold, half-threshold CA (Theorem 6) drops the effective distance from $> \delta$ to $> \delta/2$ at cost $\leq 1/|F|$; for every subsequent fold (in iteration 1 rounds $2, \ldots, r_1$ and *all* folds of iterations $2, \ldots, M$), the post-halving distance $\delta/2$ satisfies $\delta/2 < \delta_J$ (Remark 1) so the BCHKS [6] proximity gap (refining BCIKS [7] from $O(n^2)$ to $\leq n$) applies

in its proved zero-loss regime, contributing $\leq n/|F|$ bad challenges per fold (this is exactly the role $\varepsilon^{\text{fold}}_{i,s}$ would have played, but now without invoking Conjecture 4.12). Summed over folds: $r_i n/|F|$ at iteration i.

(4) Substituting (3) into (1) gives the per-iteration commit-phase budget $\varepsilon^{\text{iter}}_{\text{commit},i} \leq \varepsilon^{\text{out}}_i + \varepsilon^{\text{shift}}_i + \varepsilon^{\text{fold}-\text{our},i} \leq k/|F| + 2r_i/|F| + r_i n/|F|$.

This per-iteration term substitution is licensed by Lemma 63: the OOD and shift/sumcheck terms in WHIR's derivation do not depend on what bound is used for the folding terms. Whether the surrounding induction in the proof of [11, Theorem 5.2] (per-iteration accumulation across M iterations + case split between ambient-close-but-constraint-violating versus ambient-far inputs) carries through unchanged with our $\varepsilon^{\text{fold}-\text{our}}_i$ in place of the MCA-bearing $\varepsilon^{\text{fold}}_{0,s} + \varepsilon^{\text{fold}}_{i,s}$ is the conjectural step we leave to follow-up work; this paper does not re-derive WHIR's iteration-level analysis under the substitution.

Total error. Summing per-iteration commit-phase budgets across M iterations (each carrying an OOD constant $c_{\text{OOD},i}k/|F|$ from Lemma 60, bounded uniformly by $C_{\text{OOD}} := \max_i c_{\text{OOD},i}$) and adding the query-phase term:

$$\Pr[\text{accept}] \;\leq\; \sum_{i=1}^{M} \varepsilon^{\text{iter}}_{\text{commit},i} \;+\; (1-\delta/2)^q \;\leq\; \frac{C_{\text{OOD}}Mk + (n+2)R}{|F|} \;+\; (1-\delta/2)^q,$$

where $\sum_i \big(c_{\text{OOD},i}k + r_i n + 2r_i\big) \leq C_{\text{OOD}}Mk + (n+2)R$ for the per-round terms, and the query-phase $(1-\delta/2)^q$ comes from Remark 61 (max over adversarial strategies, not sum; using $t \geq q$). □

Remark 64 (Result type and scope of Conjecture 62)**.** *Conjecture 62 is a* ***conjectural WHIR soundness statement above Johnson****, accompanied by Lemma 63 which rigorously establishes the proximity-layer substitution that the conjecture relies on. It does not propose a new MCA statement and is intended to remove the dependency on WHIR's Conjecture 4.12 from the folding terms; the surrounding iteration-level analysis of [11, §5] is referenced rather than re-derived, and a full proof requires verifying that this iteration-level analysis goes through under the substitution. Conceptually, the half-threshold CA acts as a one-time "crossing" from above-Johnson to below-Johnson at iteration* 1*'s first fold; from that point onward (all subsequent folds), the effective threshold $\delta/2$ stays inside the Johnson regime $\delta/2 < \delta_J$ (Remark 1) where MCA is proved (WHIR Corollary 4.11). The* 2× *query overhead would be the price of this single crossing. We list the WHIR theorem-level statement as a follow-up open task (§9).*

Remark 65 (Deployment status)**.** *The reduced-threshold analysis covers the full proximity-testing stack, with theorem-level coverage for FRI (Theorem 20, Theorem 72), conditional-theorem-level coverage for STIR in odd characteristic under the named hypotheses (RD)/(CQ) of Theorem 48, rate-uniform on the standard radix-*2 *schedule, and conjecture-level coverage for WHIR in odd characteristic (Conjecture 62, with proximity-substitution proved); the characteristic-*2 *STIR/WHIR extension is recorded only as the sketch in Remark 74. This includes the odd-characteristic multiplicative-FRI settings (BabyBear, Goldilocks), circle FRI (Stwo on Mersenne31, Theorem 80), and the basic additive-FRI fold over binary tower fields. The Binius construction itself uses an additive block decomposition / Gao–Mateer recursion on top of the basic fold; that recursion-level reduction is not formalized here and remains follow-up work. No protocol changes are required; only the soundness analysis is updated.*

B Extension to Characteristic 2

We show that the reduced-threshold CA bound (Theorem 6), being characteristic-independent, extends to the FRI folding structure over fields of characteristic 2, yielding FRI soundness above the Johnson radius for the basic additive-FRI fold over binary tower fields. The key replacement is the *additive* (subspace-polynomial) fold in place of the multiplicative $\pm\sqrt{y}$ fold. The Binius construction itself uses an additional block decomposition / Gao–Mateer recursion on top of this basic fold; that recursion-level reduction is not formalized here and is left as follow-up work.

B.1 Additive FRI Folding

Definition 66 (Additive FRI Folding)**.** *Let $F = \mathbb{F}_{2^s}$ and $L \subset F$ an $\mathbb{F}_2$-linear subspace of dimension d, so $|L| = n = 2^d$. Fix $\beta \in L$, $\beta \neq 0$. The **subspace polynomial** of $\langle\beta\rangle$ is*

$$s(x) = x^2 + \beta x.$$

Since s is $\mathbb{F}_2$-linear ($s(x+y) = s(x) + s(y)$ in characteristic 2, using $(x+y)^2 = x^2 + y^2$) and $\ker(s|_L) = \{0, \beta\}$:

$$L' := s(L) \subset F, \qquad |L'| = n/2,$$

and L' is again an $\mathbb{F}_2$-linear subspace (of dimension $d-1$). Each $y \in L'$ has exactly two preimages $\{x, x+\beta\}$ in L.

*For $f : L \to F$, the **additive even/odd decomposition** is:*

$$f_1(y) = \frac{f(x) + f(x+\beta)}{\beta},$$
$$f_0(y) = f(x) + x \cdot f_1(y),$$

*where x is either preimage of y (the expressions are invariant under $x \mapsto x+\beta$: swapping adds $\beta \cdot f_1(y)$ to the f_0 formula's second term and adds $\beta \cdot f_1(y)$ to $f(x)$, which cancel). The **additive FRI fold** is:*

$$f'(y) = f_0(y) + \alpha \cdot f_1(y).$$

Remark 67 (Reconstruction)**.** *$f(x) = f_0(s(x)) + x \cdot f_1(s(x))$ for all $x \in L$: at x, the right side is $f_0(y) + x \cdot f_1(y) = (f(x) + x \cdot f_1(y)) + x \cdot f_1(y) = f(x)$. This is the characteristic-2 analogue of the odd-characteristic reconstruction $f(x) = f_{\mathrm{even}}(x^2) + x \cdot f_{\mathrm{odd}}(x^2)$.*

B.2 Even/Odd Isomorphism in Characteristic 2

Lemma 68 (Additive Even/Odd Isomorphism)**.** *For k even: $g \mapsto (g_0, g_1)$ is a linear isomorphism $\mathrm{RS}[F, L, k] \xrightarrow{\sim} \mathrm{RS}[F, L', k/2]^{=2}$.*

Proof. **Forward direction** ($g \in \mathrm{RS}_k \Rightarrow g_0, g_1 \in \mathrm{RS}_{k/2}$). Let $g(x) = \sum_{i=0}^{k-1} a_i x^i$. In the polynomial ring $F[x]$, the relation $x^2 = s+\beta x$ (from $s(x) = x^2+\beta x$) allows recursive reduction: every monomial x^i reduces to $p_i(s) + x \cdot q_i(s)$ where $\deg p_i, \deg q_i \leq \lfloor i/2 \rfloor$. Collecting terms: $g(x) = g_0(s) + x \cdot g_1(s)$ with

$$\deg g_0 \leq \lfloor (k-1)/2 \rfloor \leq k/2 - 1, \qquad \deg g_1 \leq \lfloor (k-2)/2 \rfloor \leq k/2 - 1.$$

Hence $g_0, g_1 \in \mathrm{RS}_{k/2}(L')$.

Dimension count. The map $\varphi : \mathrm{RS}_k(L) \to \mathrm{RS}_{k/2}(L')^{=2}$ is F-linear and maps a k-dimensional space into a $(k/2 + k/2) = k$-dimensional space.

Injectivity. If $(g_0, g_1) = (0, 0)$, then $g(x) = g_0(s(x)) + x \cdot g_1(s(x)) = 0$ for all $x \in L$. Since g has degree $< k \le |L|$, this implies $g = 0$. So $\ker(\varphi) = \{0\}$.

A linear injection between spaces of equal dimension is an isomorphism. □

Remark 69 (Why k even is needed)**.** *For k odd in characteristic 2: the monomial $f(x) = x^k$ satisfies $\Delta(x^k, \mathrm{RS}_k) = 1-\rho$ (since $x^k - g$ has $\le k$ roots for any g of degree $< k$). Its additive decomposition gives f_0, f_1 of degree $\le \lceil k/2 \rceil - 1$, so both belong to $\mathrm{RS}_{\lceil k/2 \rceil}$ and the fold $f' = f_0 + \alpha \cdot f_1 \in \mathrm{RS}_{\lceil k/2 \rceil}$ for every α. The isomorphism $\mathrm{RS}_k \to \mathrm{RS}_{k/2}^{=2}$ fails because $\mathrm{RS}_{\lceil k/2 \rceil}^{=2}$ has dimension $2\lceil k/2 \rceil = k+1 > k = \dim \mathrm{RS}_k$: the map is injective but not surjective, so the coupling lemma breaks (not every pair $(g_0, g_1) \in \mathrm{RS}_{\lceil k/2 \rceil}^{=2}$ corresponds to a codeword in RS_k). The fold of a far-from-code function lands in the larger code $\mathrm{RS}_{\lceil k/2 \rceil}$ for all α, and FRI is unsound. This is the same failure mode as in odd characteristic (Remark 5), confirming that $k = 2^m$ is a genuine structural requirement.*

B.3 Coupling Lemma and Proximity Gap

Lemma 70 (Additive FRI Coupling)**.** *For even k and any $f : L \to F$ (with $\mathrm{char}(F) = 2$, L an $\mathbb{F}_2$-linear subspace, $\beta \in L \setminus \{0\}$):*

$$\Delta(f, \mathrm{RS}_k(L)) \le \Delta_{\mathrm{joint}}\big((f_0, f_1),\, \mathrm{RS}_{k/2}(L')^{=2}\big).$$

Proof. By Lemma 68: $g \mapsto (g_0, g_1)$ is an isomorphism $\mathrm{RS}_k \to \mathrm{RS}_{k/2}^{=2}$. For any $g \in \mathrm{RS}_k$ and $y \in L'$: if $f_0(y) = g_0(y)$ and $f_1(y) = g_1(y)$, then by reconstruction, $f(x) = g(x)$ for both x in the fiber $\{x, x+\beta\}$ of y. So each joint-error position y (where $f_0(y) \ne g_0(y)$ or $f_1(y) \ne g_1(y)$) contributes ≥ 1 error to $f - g$ on L, and each y contributes at most 2. Let $E = |\{y \in L' : (f_0(y), f_1(y)) \ne (g_0(y), g_1(y))\}|$. Then $\Delta(f, g) \le 2E/n = E/|L'| = \Delta_{\mathrm{joint}}$. Taking $\min_g$:

$$\Delta(f, \mathrm{RS}_k) \le \Delta_{\mathrm{joint}}.$$ □

Theorem 71 (Half-Threshold Proximity Gap, Characteristic 2)**.** (*The Johnson-window hypothesis below is included only for downstream composition into Theorem 72; the proof itself uses only additive coupling plus Theorem 6.*) *For $\mathrm{char}(F) = 2$, even k, $\delta_J < \delta < 1-\rho$, and $\Delta(f, \mathrm{RS}_k(L)) > \delta$:*

$$|\{\alpha \in F : \Delta(f_0 + \alpha f_1, \mathrm{RS}_{k/2}(L')) \le \delta/2\}| \le 1.$$

Proof. By Lemma 70: $\Delta_{\mathrm{joint}}((f_0, f_1), \mathrm{RS}_{k/2}^{=2}) \ge \Delta(f, \mathrm{RS}_k) > \delta$. Instantiate Theorem 6 with the pair (f_0, f_1) on $\mathrm{RS}_{k/2}$ over L' (premise threshold δ, conclusion threshold $\delta/2$). Theorem 6 uses only linearity and Hamming distances, with no characteristic hypothesis. The proof is identical to Theorem 16. □

B.4 FRI Soundness in Characteristic 2

Theorem 72 (FRI Soundness Above Johnson, Characteristic 2)**.** *For $\mathrm{RS}[F, L, k]$ with $\mathrm{char}(F) = 2$, $k = 2^m$, $L \subset F$ an $\mathbb{F}_2$-linear subspace of dimension $d \ge m$, and $\delta_J < \delta < 1-\rho$, $\Delta(f^{(0)}, \mathrm{RS}_k) > \delta$: FRI with additive folding (Definition 66), $R = m$ rounds, and q base-level queries (each opening one full path) satisfies*

$$\Pr[\text{FRI accepts}] \le \frac{nR}{|F|} + (1 - \delta/2)^q.$$

Proof. The proof follows the same Strategy A / Strategy B structure as Theorem 20, with the multiplicative fold replaced by the additive fold. We verify the three components and note where the argument differs from odd characteristic:

Iterability. At each round i, the domain $L^{(i)} = s_i(L^{(i-1)})$ is an $\mathbb{F}_2$-linear subspace of dimension $d-i$, using $\beta_i \in L^{(i-1)} \setminus \{0\}$ as the folding direction. The code dimension halves: $k_i = k/2^i$. Since $k = 2^m$, we have k_i even for all $i < m$ (guaranteeing the isomorphism), and $k_m = 1$ (so RS_1 is the space of constants, and the final check verifies $f^{(R)}$ is constant).

Strategy A. Each round's fold $f^{(i)}(y) = f_0^{(i-1)}(y) + \alpha_i \cdot f_1^{(i-1)}(y)$ is linear in α_i, where $f_0^{(i-1)}, f_1^{(i-1)}$ are determined by $f^{(0)}$ and $(\alpha_1, \ldots, \alpha_{i-1})$ via the additive decomposition on the domain chain

$$L^{(0)} := L, \qquad L^{(i)} := s_i(L^{(i-1)}) \ \ (i \geq 1),$$

with each $L^{(i)}$ an $\mathbb{F}_2$-subspace of dimension $d-i$ (so the chain is well-defined for $R \leq d$). It follows that $f^{(R)}(y)$ is multilinear in $(\alpha_1, \ldots, \alpha_R)$ with total degree $\leq R$. The *nonvanishing*: if all syndromes vanished identically, then $f^{(R)} \in \mathrm{RS}_1$ for every $(\alpha_1, \ldots, \alpha_R)$. As in the proof of Theorem 20, this is a polynomial-identity argument (not a runtime reconstruction); evaluating at $\alpha_i = 0$ and $\alpha_i = 1$ recovers $f_0^{(i-1)}, f_1^{(i-1)} \in \mathrm{RS}_{k_i}$, and Lemma 68 (requiring k_{i-1} even, guaranteed by $k = 2^m$) gives $f^{(i-1)} \in \mathrm{RS}_{k_{i-1}}$; iterating yields $f^{(0)} \in \mathrm{RS}_k$, a contradiction. Hence some syndrome is a nonzero multilinear polynomial, and $\Pr[f^{(R)} \in \mathrm{RS}_1] \leq R/|F|$ by Schwartz–Zippel.

Strategy B. The distance-preservation argument uses Theorem 71: at round 1, all but ≤ 1 value of α_1 gives $\Delta(f^{(1)}, \mathrm{RS}_{k/2}) > \delta/2$. At subsequent rounds, $\delta/2 < (1-\rho)/2 \leq \delta_J$ (Remark 1), and the BCHKS [6] proximity gap at threshold $\gamma = \delta/2$ gives $\leq n$ bad α per round (refining the BCIKS [7] $O(n^2)$ bound); distance is preserved at $\delta/2$. *Domain remark.* BCIKS [7] Theorem 1.2 is a statement about Reed–Solomon codes $\mathrm{RS}[F, L, k]$ on *any* domain $L \subseteq F$ (the multiplicative-coset structure used in the BCIKS examples enters only via the folding map, not the gap inequality itself); the bound therefore applies verbatim to the additive setting here, where L is an $\mathbb{F}_2$-linear subspace and the folding map is the subspace polynomial $s(x)$. No separate proof of an "additive analogue" is required. The consistency check catches a $(\delta/2)$-far deviation with probability $\geq 1 - (1 - \delta/2)^q$ by Lemma 19 (the additive query model is the analogue of Definition 17 with π_i replaced by the iterated subspace polynomial $s_i \circ \cdots \circ s_1$, which is 2^i-to-1 on its image; the i.i.d. argument transfers verbatim). The multi-deviation argument proceeds identically to Theorem 20.

Combining: $\Pr[\text{accept}] \leq nR/|F| + (1 - \delta/2)^q$. □

Remark 73 (Domain structure)**.** *In the multiplicative (odd-characteristic) setting, FRI requires L closed under negation. In the additive (characteristic-2) setting, the analogous requirement is that L is an $\mathbb{F}_2$-linear subspace, which is the standard domain choice in additive-FRI systems. The folding direction β_i at round i must satisfy $\beta_i \in L^{(i-1)} \setminus \{0\}$; it is chosen dynamically from the current domain $L^{(i-1)} = s_{i-1}(\cdots(s_1(L)))$, which is an $\mathbb{F}_2$-subspace of dimension $d-i+1$. A concrete prescription: fix an $\mathbb{F}_2$-basis $\{e_1, \ldots, e_d\}$ of L; at round i, use $\beta_i =$ any nonzero element of $L^{(i-1)}$ (e.g., the image $s_{i-1}(\cdots(s_1(e_i)))$ if nonzero, or any other convenient choice).*

Remark 74 (STIR/WHIR in characteristic 2)**.** *The batch-CA theorem (Theorem 46) is characteristic-independent (it uses only the pairwise argument from Theorem 6). Combining with Lemma 70 and Theorem 71 gives the same argument for characteristic 2, so Theorem 48 and Corollary 58 appear to extend by the same argument. We record this only as a sketch here, not as a separate theorem or corollary.*

C Extension to Circle FRI (Stwo)

We extend the soundness theorem to circle FRI [22], the variant used by Stwo [23] on Mersenne31 fields ($p = 2^{31} - 1$). The half-threshold CA (Theorem 6) is code-agnostic; only the coupling lemma requires adaptation to the circle evaluation domain. The first FRI round (bivariate $\to$ univariate) uses a circle-specific coupling lemma proved below; all subsequent rounds reduce to standard univariate FRI.

C.1 Circle Codes and Folding

Definition 75 (Circle Group and Circle RS Code)**.** *Let p be an odd prime. The **circle group** is $\mathcal{C}(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : x^2 + y^2 = 1\}$ with group law*

$$(x_0, y_0) \oplus (x_1, y_1) = (x_0x_1 - y_0y_1,\ x_0y_1 + y_0x_1),$$

identity $(1, 0)$, and inversion $(x, y)^{-1} = (x, -y)$. The order is $|\mathcal{C}(\mathbb{F}_p)| = p+1$ when -1 is a quadratic non-residue ($p \equiv 3 \pmod 4$) and $p - 1$ otherwise; for Mersenne31 ($p = 2^{31} - 1 \equiv 3 \pmod 4$), the order is 2^{31}.

*Let $D \subset \mathcal{C}(\mathbb{F}_p)$ be an evaluation domain of even size N on which the **involution** $\iota(x, y) = (x, -y)$ is fixed-point-free (equivalently: $y \neq 0$ for all $(x, y) \in D$, i.e., $(\pm 1, 0) \notin D$). Let $D_x = \{x : (x, y) \in D\}$ denote the x-projection, with $|D_x| = N/2$. The involution partitions D into $N/2$ pairs $\{(x, y), (x, -y)\}$, one for each $x \in D_x$.*

*The **circle RS code**[4] is*

$$\mathrm{CRS}_N(D) = \big\{(x, y) \mapsto a(x) + y \cdot b(x) : \deg a < N/2,\ \deg b < N/2\big\}\!\restriction_D,$$

a linear code of dimension N on D (the full evaluation space, with $\dim = |D|$). In the FRI setting, the code is evaluated on the larger blowup domain D and the polynomial degree bound is $k < N$; we write $\mathrm{CRS}_k(D)$ for the subcode with $\deg a < k/2$ and $\deg b < k/2$, having dimension k and rate $\rho = k/N$ (cf. Stwo's "blowup factor"). All rate references in the circle theorems below use this $\rho = k/N$ convention.

Definition 76 (Circle FRI Folding)**.** *For $f : D \to F$, the **circle even/odd decomposition** is:*

$$f_0(x) = \frac{f(x, y) + f(x, -y)}{2}, \qquad f_1(x) = \frac{f(x, y) - f(x, -y)}{2y},$$

*where $(x, y) \in D$ is either representative of the fiber over x (both expressions are invariant under $y \mapsto -y$). The **circle FRI fold** is $f'(x) = f_0(x) + \alpha \cdot f_1(x)$, a function on D_x.*

C.2 Coupling Lemma and Soundness

Lemma 77 (Circle Even/Odd Isomorphism)**.** *For even k: the map $g \mapsto (g_0, g_1)$ is a linear isomorphism $\mathrm{CRS}_k(D) \xrightarrow{\sim} \mathrm{RS}_{k/2}(D_x)^{=2}$.*

Proof. For $g(x, y) = a(x) + y \cdot b(x) \in \mathrm{CRS}_k$: $g_0 = a$ and $g_1 = b$, both of degree $< k/2$, hence both in $\mathrm{RS}_{k/2}(D_x)$. The map is F-linear, and the source and target both have dimension k. *Injectivity*: if $(a, b) = (0, 0)$ on all of D_x, then $\deg a, \deg b < k/2 \leq |D_x|$ forces $a = b = 0$ as polynomials, so $g = 0$ on D. $\square$

[4] In the notation of [22], this is the FFT subspace $L'_N(\mathbb{F}_p)$. The full space L_N has dimension $N+1$; the extra monomial $x^{N/2}$ is handled by the prover sending its coefficient before the first fold (Remark 81).

Lemma 78 (Circle FRI Coupling). *For even k and any $f : D \to F$:*

$$\Delta(f,\, \mathrm{CRS}_k(D)) \;\le\; \Delta_{\mathrm{joint}}\big((f_0, f_1),\, \mathrm{RS}_{k/2}(D_x)^{=2}\big).$$

Proof. By Lemma 77, each $g \in \mathrm{CRS}_k$ corresponds to $(g_0, g_1) \in \mathrm{RS}^{=2}_{k/2}$. If $f_0(x) = g_0(x)$ and $f_1(x) = g_1(x)$ at some $x \in D_x$, then $f(x, \pm y) = a(x) \pm y \cdot b(x) = g(x, \pm y)$ (by reconstruction: $f(x, y) = f_0(x) + y \cdot f_1(x)$), so *both* points in the fiber over x are error-free. Equivalently, each joint-error position x (where $f_0(x) \neq g_0(x)$ or $f_1(x) \neq g_1(x)$) contributes ≥ 1 base-level error on D and at most 2. With $E = |\{x \in D_x : (f_0(x), f_1(x)) \neq (g_0(x), g_1(x))\}|$:

$$\Delta(f, g) = \frac{|\{P \in D : f(P) \neq g(P)\}|}{N} \le \frac{2E}{N} = \frac{E}{|D_x|} = \Delta_{\mathrm{joint}}.$$

Taking $\min_g$ (using the isomorphism, the minimum over $\mathrm{RS}^{=2}_{k/2}$ equals the minimum over CRS_k): $\Delta(f, \mathrm{CRS}_k) \le \Delta_{\mathrm{joint}}$. ☐

Theorem 79 (Circle FRI Proximity Gap). (*The Johnson-window hypothesis below is included only for downstream composition into Theorem 80; the proof itself uses only the circle coupling lemma plus Theorem 6.*) *For even k, $\delta_J < \delta < 1-\rho$, and $\Delta(f, \mathrm{CRS}_k(D)) > \delta$:*

$$|\{\alpha \in F : \Delta(f_0 + \alpha f_1,\, \mathrm{RS}_{k/2}(D_x)) \le \delta/2\}| \le 1.$$

Proof. By Lemma 78: $\Delta_{\mathrm{joint}}((f_0, f_1), \mathrm{RS}^{=2}_{k/2}) \ge \Delta(f, \mathrm{CRS}_k) > \delta$. Apply Theorem 6 to (f_0, f_1) on $\mathrm{RS}_{k/2}$ over D_x (premise threshold δ, conclusion threshold $\delta/2$): at most 1 value of α has $\Delta(f_0 + \alpha f_1, \mathrm{RS}_{k/2}) \le \delta/2$. ☐

Theorem 80 (Circle FRI Soundness Above Johnson). *For $\mathrm{CRS}_k(D)$ over F with even $k = 2^m$, $D \subset \mathcal{C}(\mathbb{F}_p)$ a circle subgroup or coset of size $N = 2^d$ closed under the involution $\iota : (x, y) \mapsto (x, -y)$, with ι fixed-point-free, $0 \notin D_x$ (equivalently, the 4-torsion points $(0, \pm 1) \notin D$), and the projected-domain tower $D_x \supset \pi(D_x) \supset \pi^2(D_x) \supset \cdots$ well-defined under iterated doubling $\pi(x) = 2x^2 - 1$ (so each round map is 2-to-1 onto a non-trivial domain), $\delta_J < \delta < 1-\rho$, and $\Delta(f^{(0)}, \mathrm{CRS}_k) > \delta$: circle FRI with $R = m$ rounds and q base-level queries (each opening one full path) satisfies*

$$\Pr[\text{circle FRI accepts}] \;\le\; \frac{NR}{|F|} + \left(1 - \frac{\delta}{2}\right)^q.$$

Proof. The proof follows the Strategy A / Strategy B structure of Theorem 20.

Round 1 (circle $\to$ univariate). By Theorem 79: for all but ≤ 1 value of α_1, $\Delta(f^{(1)}, \mathrm{RS}_{k/2}(D_x)) > \delta/2$.

Rounds $2, \ldots, R$ (univariate FRI on D_x). After round 1, the function $f^{(1)} : D_x \to F$ lives on a univariate evaluation domain. Subsequent circle-FRI rounds use the **doubling map** $\pi(x) = 2x^2 - 1$ (the x-coordinate of the point-doubling endomorphism on $\mathcal{C}$) with involution $x \mapsto -x$. This involution is fixed-point-free on D_x by the standing hypothesis $0 \notin D_x$ (the 4-torsion points $(0, \pm 1) \in \mathcal{C}$ have order 4, and the hypothesis excludes them from D). Stwo's domains satisfy this automatically: with $|D| \ge 8$, the chosen cosets avoid the 4-torsion. The even/odd decomposition

$$f_{\mathrm{even}}(\pi(x)) = \frac{f(x) + f(-x)}{2}, \qquad f_{\mathrm{odd}}(\pi(x)) = \frac{f(x) - f(-x)}{2x}$$

uses the relation $x^2 = (\pi(x) + 1)/2$ to reduce: for $g \in \mathrm{RS}_{k/2}(D_x)$, the monomials $x^{2j} = ((\pi + 1)/2)^j$ and $x^{2j+1} = x \cdot ((\pi + 1)/2)^j$ give $g_{\mathrm{even}}, g_{\mathrm{odd}} \in \mathrm{RS}_{k/4}(\pi(D_x))$, establishing the isomorphism

$\mathrm{RS}_{k/2}(D_x) \xrightarrow{\sim} \mathrm{RS}_{k/4}(\pi(D_x))^{=2}$. From this point onward, the standard univariate coupling lemma (Lemma 15) applies verbatim.

Since $\delta/2 < (1-\rho)/2 \le \delta_J$ (the inequality $(1-\rho)/2 \le 1-\sqrt{\rho}$ holds for all $\rho \in (0,1)$), the effective distance $\delta/2$ is strictly below the Johnson radius. At each round $j \ge 2$: Lemma 15 gives $\Delta_{\text{joint}} > \delta/2$, and the BCHKS [6] proximity gap at threshold $\delta/2 \le \delta_J$ (refining the BCIKS [7] bound from $O(n^2)$ to $\le n$) bounds the number of bad challenges to $\le N/2^{j-1}$ at round j. Distance is *preserved* at $\delta/2$ (not further halved), as in Remark 21. *Domain remark.* BCIKS [7] Theorem 1.2 is a statement about Reed–Solomon codes on *any* domain (the multiplicative-coset structure used in the BCIKS examples enters only via the folding map, not the gap inequality itself); the bound therefore applies verbatim with L replaced by the projected circle domain D_x and the folding map replaced by $\pi(x) = 2x^2 - 1$. A closely related circle proximity gap also appears in the Habock–Levit–Papini Circle STARKs analysis [22].

Strategy A. Each round introduces α_i with degree 1, so $f^{(R)}(y)$ is multilinear in $(\alpha_1, \ldots, \alpha_R)$. If all syndromes vanished identically, then $f^{(R)} \in \mathrm{RS}_1$ for every $(\alpha_1, \ldots, \alpha_R) \in F^R$. As in the proof of Theorem 20, we extract $f^{(0)} \in \mathrm{CRS}_k$ via a polynomial-identity argument (not a runtime reconstruction): evaluating at $\alpha_i = 0$ and $\alpha_i = 1$ at each round recovers the round-$(i-1)$ even/odd projections in the appropriate codes, and Lemma 77 (round 1) and the round-≥ 2 isomorphisms iterate this back to $f^{(0)} \in \mathrm{CRS}_k$, a contradiction. By Schwartz–Zippel: $\Pr[f^{(R)} \in \mathrm{RS}_1] \le R/|F|$.

Strategy B. Combining the round-1 proximity gap (≤ 1 bad α), the round-≥ 2 distance preservation ($\le N/2^{j-1}$ bad α at round j, hence $\le N$ uniformly), and the consistency check ($\ge 1 - (1-\delta/2)^q$ catch probability via Lemma 19 applied with the round-i projection: at round 1, $\pi_1 : (x, y) \mapsto x$ is the x-coordinate projection (the 2-to-1 map carrying the ι-fibres of D onto D_x); at round $i \ge 2$, π_i is the iterated doubling map $\pi(x) = 2x^2 - 1$ on D_x, 2^{i-1}-to-1 onto $\pi^{i-1}(D_x)$; the composite is 2^i-to-1 from D to its round-i image): $\Pr[\text{accept}] \le NR/|F| + (1 - \delta/2)^q$. □

Remark 81 (The dimension gap)**.** *The full circle polynomial space L_N of [22] has dimension $N+1$; the FFT subspace CRS_N used above has dimension N. The extra direction is the degree-$(N/2)$ monomial $x^{N/2}$, corresponding to the vanishing polynomial of the half-domain. In Stwo's protocol [23], the prover sends this coefficient before the first fold, reducing the proximity test to CRS_N. This adds at most $1/|F|$ to the commit-phase error (absorbed into the $NR/|F|$ term).*

Remark 82 (STIR and WHIR on circle domains)**.** *The per-coordinate batch-CA helper (Theorem 46) is code-agnostic. Combined with Lemma 78, the STIR analysis of Theorem 48 appears to extend to circle-FRI-based STIR by the same argument (with the same hypothesis set: (RD)/(CQ)), but we do not state or prove a separate circle-STIR theorem here. Similarly, the same proof of Corollary 58 (which uses only that the ambient code is closed under F-linear combinations) extends to linear subcodes of CRS_k when applied via Theorem 6; we do not state a separate circle-subcode corollary. The above-Johnson coverage now spans all deployed FRI variants (multiplicative, additive characteristic-2, and circle), plus STIR at conditional-theorem level under (RD)/(CQ) and WHIR conjecture-level (Conjecture 62).*

D Fiat–Shamir Compilation and Knowledge Soundness

The theorems in this paper (Theorems 20, 72, 80, 48) establish *information-theoretic* soundness of the corresponding interactive proximity-testing protocols; Conjecture 62 states the proposed WHIR analogue conditional on its iteration-level induction (proximity-layer substitution rigorous via Lemma 63). Deployed STARK systems use the non-interactive versions obtained via the Fiat–Shamir transform (replacing verifier randomness with hash evaluations). This appendix states the non-interactive soundness and knowledge-soundness theorems that follow from our bounds via standard compilation.

D.1 Round-by-Round Soundness of FRI

The Fiat–Shamir security of FRI requires not just end-to-end soundness but *round-by-round (RBR) soundness* [36, 37]: the property that at each protocol round, the prover's probability of escaping a "doomed" state is bounded, regardless of the transcript so far.

Theorem 83 (RBR Soundness of Half-Threshold FRI, multiplicative case)**.** *Under the structural hypotheses of Theorem 20 (*$\mathrm{char}(F) \neq 2$*,* $k = 2^m$ *with* $m \geq 1$*,* L *a multiplicative subgroup or coset closed under negation with* $0 \notin L$ *and* $|L| = n = 2^d$ *for* $d \geq m$*, so the radix-2 folding tower is well-defined and* $x \mapsto -x$ *is a fixed-point-free involution at every level),* $\delta_J < \delta < 1-\rho$*, prover input* $f^{(0)}$ *satisfying* $\Delta(f^{(0)}, \mathrm{RS}_k) > \delta$*, and* q *base-level queries in the final consistency round: the FRI protocol with* $R = m$ *rounds has round-by-round soundness with per-round error*

$$\varepsilon_{\mathrm{rbr}}^{(i)} \leq \begin{cases} 1/|F| & i = 1 \text{ (commit round: } \leq 1 \text{ bad } \alpha \text{ via Theorem 16)}, \\ n/|F| & 2 \leq i \leq R \text{ (commit rounds: BCHKS at } \delta/2 \leq \delta_J\text{, refining BCIKS from } O(n^2) \text{ to } \leq n), \\ (1-\delta/2)^q & i = R+1 \text{ (query round)}. \end{cases}$$

Proof. The doomed predicate at round i is defined on the deterministic *honest-fold reference* $\hat{f}^{(i)}$ (computed from $f^{(0)}$ and $\alpha_1, \ldots, \alpha_i$ alone, independent of the prover's commits; see Step 1 of the proof of Theorem 20): the partial transcript is doomed at round i iff $\Delta(\hat{f}^{(i)}, \mathrm{RS}_{k/2^i}) > \delta/2$. Initially ($i = 0$): $\Delta(\hat{f}^{(0)}, \mathrm{RS}_k) = \Delta(f^{(0)}, \mathrm{RS}_k) > \delta > \delta/2$, so every false statement starts doomed.

Round 1 *(commit).* By the proximity gap (Theorem 16): at most 1 value of α_1 makes the honest fold $\hat{f}^{(1)} = \mathrm{fold}(\hat{f}^{(0)}, \alpha_1)$ become $(\delta/2)$-close to $\mathrm{RS}_{k/2}$. Escape probability: $\leq 1/|F|$ (sharper than the round-≥ 2 bound of $n/|F|$ stated in the theorem).

Rounds $2, \ldots, R$ *(commit).* Under the inductive doomed hypothesis at round $i-1$: $\Delta(\hat{f}^{(i-1)}, \mathrm{RS}_{k/2^{i-1}}) > \delta/2$ together with $\delta/2 < (1-\rho)/2 \leq \delta_J$ (Remark 1) places the threshold inside the BCIKS–BCHKS zero-loss regime. By Lemma 15, the joint distance $\Delta_{\mathrm{joint}}((\hat{f}^{(i-1)}_{\mathrm{even}}, \hat{f}^{(i-1)}_{\mathrm{odd}}), \mathrm{RS}^{=2}_{k/2^i}) > \delta/2$, and the BCHKS [6] proximity gap at threshold $\gamma = \delta/2$ gives: at most n values of α_i make $\Delta(\hat{f}^{(i)}, \mathrm{RS}_{k/2^i}) \leq \delta/2$ (refining the BCIKS [7] bound from $O(n^2)$ to $\leq n$). Escape probability: $\leq n/|F|$.

Round R+1 *(query).* Conditional on doomed at round R ($\Delta(\hat{f}^{(R)}, \mathrm{RS}_{k/2^R}) > \delta/2$), the honest-fold-reference reformulation of the FRI catch lemma in [37, Lemma 5.5] (which packages the original BBHR / BCIKS path-consistency arguments [20, §6], [7, Lemma 8.2] in the honest-fold-reference framework used here) gives that the merged catching set $T := \bigcup_i \pi_i^{-1}(B_i)$ (where B_i is the prover's round-i deviation set, as in Step 2 of the proof of Theorem 20) has relative measure $|T|/|L| \geq \delta/2$ whenever $f^{(R)} \in \mathrm{RS}_{k/2^R}$. Direct base-set miss bound: the probability that all q i.i.d. base-level queries miss T is $(1 - |T|/|L|)^q \leq (1-\delta/2)^q$.

The doomed predicate at commit rounds is defined on the deterministic honest-fold reference $\hat{f}^{(i)}$, so the per-round commit bounds depend on the verifier's challenges only and are independent of the prover's commits. The query-round set T does depend on the prover's commits, but the lower bound $|T|/|L| \geq \delta/2$ is uniform over all prover commit patterns (any prover whose final commit is a codeword $f^{(R)} \in \mathrm{RS}_{k/2^R}$ under doomed $\hat{f}^{(R)}$). Hence the protocol is RBR-sound with the stated errors. ∎

D.2 Non-Interactive FRI via Fiat–Shamir

The BCS transformation [35] compiles a public-coin IOP into a non-interactive argument in the random oracle model. When the IOP additionally has round-by-round (RBR) soundness, the security loss is *linear* in the number of random-oracle queries (not exponential in rounds), precisely because RBR soundness enables straight-line simulation; the explicit RBR-based compilation bound used below is from [37], specialised to FRI.

Theorem 84 (Non-Interactive FRI Soundness)**.** *Under the hypotheses of Theorem 83, let $H : \{0,1\}^* \to \{0,1\}^\kappa$ be a random oracle. For any adversary making at most Q queries to H, the Fiat–Shamir-compiled FRI protocol satisfies:*

$$\varepsilon_{\mathrm{FRI}}^{\mathrm{NI}} \;\le\; Q \cdot \left(\frac{nR}{|F|} + \left(1 - \frac{\delta}{2}\right)^q\right) \;+\; \frac{3(Q^2+1)}{2^\kappa}.$$

Proof. Define the aggregate RBR error $\varepsilon_{\mathrm{rbr}} := \sum_{i=1}^{R+1} \varepsilon_{\mathrm{rbr}}^{(i)}$ from the per-round errors of Theorem 83; substituting the per-round bounds gives $\varepsilon_{\mathrm{rbr}} \le R \cdot n/|F| + (1-\delta/2)^q = nR/|F| + (1-\delta/2)^q$. By the RBR-FRI compilation analysis of [37] (which sharpens the generic BCS transformation [35] for public-coin IOPs with round-by-round soundness): for a public-coin IOP with aggregate RBR soundness error $\varepsilon_{\mathrm{rbr}}$, the Fiat–Shamir-compiled protocol has soundness error at most $Q \cdot \varepsilon_{\mathrm{rbr}} + 3(Q^2{+}1)/2^\kappa$, where the first term is the Q-linear loss across the RBR challenges and the second is the Merkle-tree binding error. Substituting the aggregate gives the stated bound. ☐

Remark 85 (Practical parameters)**.** *For BabyBear$_4$ ($|F| = p^4 \approx 2^{124}$), $n = 2^{20}$, $R = 20$, $\kappa = 256$ (SHA-256), and $Q = 2^{64}$ (a generous query bound):*

- *Interactive FRI error: $nR/|F| + (1-\delta/2)^q \approx 2^{-100} + (1-\delta/2)^q$.*
- *Fiat–Shamir overhead: $Q \cdot \varepsilon \approx 2^{64} \cdot \varepsilon$, reducing security by $\sim$64 bits.*
- *Merkle binding: $3Q^2/2^\kappa \approx 2^{129}/2^{256} \approx 2^{-127}$ (negligible).*
- *Net: the interactive bound loses $\sim$64 bits in the non-interactive setting. At $q \approx 398$ (128-bit interactive query-phase security for $\delta = 0.4$), the non-interactive query contribution is only $\sim$64 bits against $Q = 2^{64}$ adversaries, and the total is further capped by the commit term unless the challenge field is enlarged.*

This matches the standard parameter regime: deployed systems set $\kappa \ge 2\lambda$ and target λ-bit security against $Q \le 2^\lambda$ adversaries.

Tighter bound. The BGKTTW analysis [37] of the BCS compilation [35] applies Q-loss to the per-round max of the RBR error, not the total. Since each commit round has escape probability $\le n/|F|$ (Theorem 83), a tighter bound is $\varepsilon_{\mathrm{FRI}}^{\mathrm{NI}} \le Q \cdot \max(n/|F|,\ (1-\delta/2)^q) + 3(Q^2{+}1)/2^\kappa$, saving a factor of R in the commit term. We state the looser bound above for consistency with the interactive Theorem 20.

D.3 Knowledge Soundness and DEEP Polynomial Commitments

FRI is deployed not as a standalone proximity test but as the core of a *polynomial commitment scheme (PCS)* via the DEEP technique [38]: to verify a claimed evaluation $\hat{f}(\zeta) = v$ at a random point $\zeta \in F$, the verifier checks that the quotient $q(x) = (f(x) - v)/(x - \zeta)$ is close to RS_{k-1}.

Theorem 86 (DEEP-FRI Soundness Error, multiplicative case; conditional on standard DEEP convention)**.** *Under the structural hypotheses of Theorem 20 (so $\mathrm{char}(F) \ne 2$, $k = 2^m$, L a multiplicative subgroup or coset closed under negation with $0 \notin L$ and the radix-2 folding tower well-defined; we do not import the input-distance premise of Theorem 20, since DEEP-FRI is invoked* *also on inputs $f \in \mathrm{RS}_k$), $\delta_J < \delta < 1-\rho$, the underlying multiplicative FRI subprotocol of Theorem 20 with $R = m$ rounds and q queries, and assuming the standard DEEP convention [38] that handles the codimension-1 subcode $\mathrm{RS}_{k-1} \subsetneq \mathrm{RS}_k$ via an explicit leading-coefficient check (formalised in [38]; not re-derived in this paper): the DEEP-FRI polynomial commitment scheme [38] has knowledge soundness*

$$\varepsilon_{\mathrm{PCS}} \;\le\; \frac{k}{|F|-n} \;+\; \frac{nR}{|F|} \;+\; \left(1-\frac{\delta}{2}\right)^{q},$$

where the $k/(|F|-n)$ term is the DEEP quotient error (probability that the random evaluation point ζ, sampled uniformly from $F \setminus L$, is degenerate). The bound is stated as a soundness-error inequality on the protocol's accept probability; we do not formalise an explicit knowledge extractor here, and refer to [37] for the standard RBR-to-knowledge-soundness lifting (cf. Remark 88).

Proof. The DEEP technique [38] reduces polynomial commitment to proximity testing: given oracle access to f and a claimed evaluation $f(\zeta) = v$ at a random $\zeta \in F \setminus L$:

1. If $f \in \mathrm{RS}_k$ and $f(\zeta) = v$: the quotient $q(x) = (f(x) - v)/(x - \zeta)$ satisfies $q \in \mathrm{RS}_{k-1}$, and FRI accepts.
2. If $f \in \mathrm{RS}_k$ but $f(\zeta) \neq v$: the polynomial $f(x)-v$ is not divisible by $(x-\zeta)$ (since $f(\zeta)-v \neq 0$), so the function $q(x) = (f(x)-v)/(x-\zeta)$ on L (well-defined since $\zeta \notin L$) is not the evaluation of any degree-$<(k-1)$ polynomial. Hence $\Delta(q, \mathrm{RS}_{k-1}) \ge 1 - (k-1)/n = 1-\rho + 1/n > \delta$, and FRI rejects with high probability.
3. If $f \notin \mathrm{RS}_k$: the verifier sends random $\zeta \in F \setminus L$ and the prover provides a claimed value v. By the DEEP worst-case-to-average-case reduction [38, Theorem 3.1]: for all but $\le k$ degenerate values of ζ, the quotient function $q(x) = (f(x) - v)/(x - \zeta)$ on L satisfies $\Delta(q, \mathrm{RS}_{k-1}) \ge \Delta(f, \mathrm{RS}_k) - O(1/n) > \delta$ (the small $O(1/n)$ correction accounts for the coordinate-shift at the degree-$k-1$ vs. degree-k codeword sets and is negligible for $n \gg 1$ at FRI scale; see [38] for the precise statement), regardless of the prover's choice of v. Sampling ζ uniformly from $F \setminus L$, the probability of a degenerate ζ is $\le k/(|F|-n)$. Conditional on non-degenerate ζ: the standard DEEP construction [38] runs FRI on the quotient against the codimension-1 subcode $\mathrm{RS}_{k-1} \subsetneq \mathrm{RS}_k$; testing RS_{k-1} directly under FRI's $k = 2^m$ hypothesis requires either (i) a separate leading-coefficient check that excludes the codimension-1 extra dimension of RS_k (e.g. an additional constraint opening, as in the original DEEP-FRI design [38]), or (ii) a padded-power-of-two quotient protocol (e.g. pad RS_{k-1} to RS_k via a sentinel slot and check the slot separately). The bound stated above assumes the standard DEEP convention (i) is in force; we treat the leading-coefficient check as part of the soundness budget bookkeeping handled by [38] and import the resulting reduction to a FRI run on a power-of-two-degree ambient code at distance threshold δ, which is then governed by Theorem 20. (We do not invoke Corollary 58 here: that corollary gives CA on linear subcodes, not the codimension-1 FRI soundness reduction needed for the quotient-test step.) A self-contained quotient-protocol formalisation is outside the scope of this paper and is deferred to subsequent work.

Taking the union bound: $\varepsilon_{\mathrm{PCS}} \le k/(|F| - n) + nR/|F| + (1-\delta/2)^q$.

Knowledge extraction. By the round-by-round framework [37]: the extractor runs the interactive protocol with the adversary, catching the adversary's committed polynomial via the DEEP evaluation at the random point ζ. The RBR knowledge error is $\varepsilon_{\mathrm{rbr}} + k/(|F| - n)$, and the BCS compilation preserves knowledge soundness with the same Q-linear loss as for plain soundness. ☐

Corollary 87 (Non-Interactive PCS Knowledge Soundness). *Under the hypotheses of Theorem 86, with the Fiat–Shamir / Merkle-tree compiler instantiated by a random oracle $H : \{0,1\}^* \to \{0,1\}^\kappa$,* *the compiled DEEP-FRI PCS has knowledge soundness against Q-query adversaries:*

$$\varepsilon_{\mathrm{PCS}}^{\mathrm{NI}} \le Q \cdot \left(\frac{k}{|F| - n} + \frac{nR}{|F|} + \left(1 - \frac{\delta}{2}\right)^q \right) + \frac{3(Q^2+1)}{2^\kappa}.$$

Remark 88 (Scope of the knowledge-soundness claim)**.** *Theorem 86 and Corollary 87 give a soundness-error bound for the DEEP-FRI polynomial commitment scheme; the standard RBR-to-knowledge-soundness lifting [37] then yields an extractor that recovers a degree-$<k$ polynomial consistent with the committed oracle. The full SNARK (e.g., Plonk with FRI-PCS) additionally requires showing that the extracted polynomial satisfies the circuit constraints; this is the arithmetization step, which is protocol-specific and orthogonal to the proximity-testing soundness. We do not analyse circuit-satisfaction composition here; standard PCS-to-SNARK reductions are protocol-specific and appear in the SNARK literature (e.g., Plonk-style or AIR-style arithmetizations).*

References

[1] Ethereum Foundation, *The Proximity Prize: \$1M Reed–Solomon Challenge* (Grand MCA Challenge and Grand List Decoding Challenge with $\varepsilon^* = 2^{-128}$ at rates $\rho \in \{1/2, 1/4, 1/8, 1/16\}$), `https://proximityprize.org`, accessed May 2026.

[2] R. Chai, X. Fan, *Action–Orbit Approaches to Reed–Solomon Proximity on Cyclic Domains*, companion paper in preparation, 2026.

[3] R. Chai, X. Fan, *Codim-*$2(c-1)$ *Reed–Solomon Proximity Bounds via the Berlekamp Realizer*, companion paper in preparation, 2026.

[4] G. Arnon, D. Boneh, G. Fenzi, *Open problems in list decoding and correlated agreement*, ePrint 2026/680.

[5] E. Crites, A. Stewart, *On Reed–Solomon proximity gaps conjectures*, ePrint 2025/2046.

[6] E. Ben-Sasson, D. Carmon, U. Haböck, S. Kopparty, S. Saraf, *On proximity gaps for Reed–Solomon codes*, ePrint 2025/2055, 2025.

[7] E. Ben-Sasson, D. Carmon, Y. Ishai, S. Kopparty, S. Saraf, *Proximity gaps for Reed–Solomon codes*, FOCS 2020.

[8] R. Goyal, V. Guruswami, *Optimal proximity gaps for subspace-design codes and (random) Reed–Solomon codes*, ePrint 2025/2054.

[9] F. G. Jeronimo, L. Liu, P. Rajpal, *Optimal proximity gap for folded Reed–Solomon codes via subspace designs*, arXiv:2601.10047, 2026.

[10] G. Arnon, A. Chiesa, G. Fenzi, E. Yogev, *STIR: Reed–Solomon proximity testing with fewer queries*, CRYPTO 2024; ePrint 2024/390.

[11] G. Arnon, A. Chiesa, G. Fenzi, E. Yogev, *WHIR: Reed–Solomon proximity testing with super-fast verification*, in Advances in Cryptology – EUROCRYPT 2025, Lecture Notes in Computer Science, Springer, 2025; full version at IACR ePrint 2024/1586.

[12] A. Kambiré, *Proximity gaps conjecture fails near capacity over prime fields*, arXiv:2604.09724, 2026.

[13] R. Lidl, H. Niederreiter, *Finite Fields*, Cambridge, 2nd ed., 1997.

[14] P. Frankl, R. M. Wilson, *Intersection theorems with geometric consequences*, Combinatorica 1(4):357–368, 1981.

[15] M. Sudan, *Decoding of Reed–Solomon codes beyond the error-correction bound*, J. Complexity 13(1):180–193, 1997.

[16] V. Guruswami, M. Sudan, *Improved decoding of Reed–Solomon and algebraic-geometric codes*, IEEE Trans. Inform. Theory 45(6):1757–1767, 1999.

[17] T. Helleseth, *Some results about the cross-correlation function between two maximal linear sequences*, Discrete Math. 16(3):209–232, 1976.

[18] S. W. Golomb, G. Gong, *Signal Design for Good Correlation: For Wireless Communication, Cryptography, and Radar*, Cambridge University Press, 2005.

[19] O. Moreno, C. J. Moreno, *Exponential sums and Goppa codes: I*, Proc. Amer. Math. Soc. 111(2):523–531, 1991.

[20] E. Ben-Sasson, I. Bentov, Y. Horesh, M. Riabzev, *Fast Reed–Solomon interactive oracle proofs of proximity*, in ICALP 2018, LIPIcs vol. 107, 2018. Full version: ePrint 2017/134.

[21] B. E. Diamond, J. Posen, *Succinct Arguments over Towers of Binary Fields*, IACR ePrint 2023/1784, 2023.

[22] U. Haböck, D. Levit, S. Papini, *Circle STARKs*, ePrint 2024/278, 2024.

[23] D. Carmon, S. Goldberg, U. Haböck, V. Lerer, S. Lesokhin, *Stwo whitepaper*, ePrint 2026/532, 2026.

[24] R. Chai, X. Fan, *Structural results on per-word list size for Reed–Solomon codes*, internal note 0088, 2026; relevant scripts and outputs are in `paper1/scripts/` of [39].

[25] R. Chai, X. Fan, *Equal-threshold CA bound: $\varepsilon_{\mathrm{ca}} = \binom{n}{w}/|F|$ via erasure-pattern counting*, internal note 0089, 2026; scripts in `paper1/scripts/op1-barrier/` of [39].

[26] R. Chai, X. Fan, *Birthday-bound at codimension excess $c = 2$: Bernstein concentration via Möbius exclusion, and the k-wise common-core obstruction*, internal note 0092, 2026; scripts in `paper1/scripts/list-size/` of [39].

[27] R. Chai, X. Fan, *Open-Set Rank Lemma at codimension excess $c \geq 3$: phase diagram, triangle and tetrahedron obstructions at small primes*, internal note 0094, 2026; scripts in `paper1/scripts/list-size/` of [39].

[28] S. Janson, *On concentration of probability*, in Contemporary Combinatorics, Bolyai Soc. Math. Stud., vol. 10, Springer, 2002, pp. 289–301.

[29] D. Dubhashi, D. Ranjan, *Balls and bins: A study in negative dependence*, Random Structures & Algorithms, vol. 13, no. 2, 1998, pp. 99–124.

[30] K. Joag-Dev, F. Proschan, *Negative association of random variables, with applications*, Annals of Statistics, vol. 11, no. 1, 1983, pp. 286–295.

[31] G. N. Rothblum, S. Vadhan, A. Wigderson, *Interactive proofs of proximity: delegating computation in sublinear time*, in STOC 2013. Full version: ECCC TR12-182.

[32] S. Ames, C. Hazay, Y. Ishai, M. Venkitasubramaniam, *Ligero: Lightweight sublinear arguments without a trusted setup*, in Proceedings of the 24th ACM Conference on Computer and Communications Security (CCS), 2017.

[33] R. M. Roth and G. Zémor, personal communication, 2018. Cited in [7].

[34] E. Ben-Sasson, S. Kopparty, S. Saraf, *Worst-case to average-case reductions for the distance to a code*, in 33rd Computational Complexity Conference (CCC), pages 24:1–24:23, 2018.

[35] E. Ben-Sasson, A. Chiesa, N. Spooner, *Interactive oracle proofs*, in TCC 2016-B, LNCS 9986, pages 31–60, 2016.

[36] R. Canetti, Y. Chen, J. Holmgren, A. Lombardi, G. N. Rothblum, R. D. Rothblum, D. Wichs, *Fiat–Shamir: from practice to theory*, in STOC 2019, pages 1082–1090, 2019.

[37] A. Block, A. Garreta, J. Katz, J. Thaler, P. N. Tiwari, M. Zajac, *Fiat–Shamir security of FRI and related SNARKs*, in Asiacrypt 2023, LNCS 14444, pages 3–40, 2023.

[38] E. Ben-Sasson, L. Goldberg, S. Kopparty, S. Saraf, *DEEP-FRI: sampling outside the box improves soundness*, ePrint 2019/336, 2019.

[39] R. Chai, X. Fan, *Computational companion: Lean 4 formalization and verification scripts for FRI soundness above the Johnson bound*, `https://github.com/iotexproject/rs-proximity-gaps`, 2026.

Acknowledgments

We thank Professor Guang Gong (University of Waterloo) for reviewing an earlier draft and for the sequence-school perspective, and the Ethereum Foundation for posing the Proximity Prize that motivated this work.